Local Physics of Disordered Quantum Spin Liquid Systems Individuated by NMR and Inverse Laplace Transform $1/T_1$ Analysis

Local Physics of Disordered Quantum Spin Liquid Systems $Ag_3LiIr_2O_6$, $Zn_xCu_{4-x}(OD)_6FBr$, and $Zn_{0.85}Cu_{3.15}(OD)_6Cl_2$ Individuated by ^{7}Li and ^{19}F NMR, ^{63}Cu NQR, and Inverse Laplace Transform $1/T_1$ Analysis

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A Thesis Submitted to the School of Graduate Studies in Partial Fulfillment of the Requirements for the Degree Doctor of Philosophy of Science

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TITLE:	Local Physics of Disordered Quantum Spin Liquid Systems $Ag_3LiIr_2O_6$, $Zn_xCu_{4-x}(OD)_6FBr$, and $Zn_{0.85}Cu_{3.15}(OD)_6Cl_2$ Individuated by ^{7}Li and ^{19}F NMR, ^{63}Cu NQR, and Inverse Laplace Transform $1/T_1$ Analysis
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Lay Abstract

A quantum spin liquid (QSL) is an exotic state of matter whose magnetism fundamentally differs from those of ordinary materials. At temperatures near absolute zero, the electron spins which make up an ordinary magnet generally freeze in place, whereas in a QSL, they form a highly-entangled quantum superposition. A theoretically attainable QSL state was famously proposed in 1973 by Philip W. Anderson. Since then, several candidate materials have been discovered, and research on QSLs became a major focus in the field of condensed matter physics. The realization of a QSL is predicted to have applications in quantum computing (by hosting more robust quantum bits), and can help us understand the physics of other quantum materials, such as high-temperature superconductors. In this thesis, we report our experimental findings on the QSL candidates Herbertsmithite, Zn-barlowite, and $Ag_3LiIr_2O_6$, where we use nuclear magnetic resonance (NMR) spectroscopy to probe the behaviour of their spins. Hindering past attempts to study these materials is the ever-present influence of disorder, such as chemical and structural imperfections. To combat this, we developed a novel technique for acquiring and analyzing NMR data, known as inverse Laplace transform (ILT) $1/T_1$ analysis, and used it to make unprecedented discoveries about the heterogeneous physics of these disordered materials.

Abstract

One of the main challenges in experimentally identifying a quantum spin liquid (QSL) state is in understanding the influence of disorder. Chemical and structural imperfections exist in many promising QSL candidate materials, and can lead to a spatially inhomogeneous behaviour that obfuscates the interpretation of sample-averaged measurements. This issue highlights the importance of nuclear magnetic resonance (NMR) which can locally probe the intrinsic spin susceptibility χ_{spin} (separate from defect contributions) and low-energy spin fluctuations via the Knight shift K and nuclear spin-lattice relaxation rate $1/T_1$, respectively. The value of $1/T_1$ is typically ascertained by fitting the net nuclear magnetization $M(t_D)$ with an appropriate decay function. However, the $M(t_D)$ measured at a given frequency has contributions from many nuclei, which in a disordered material, can exhibit a broad distribution of $1/T_1$. Analogous to how variations in local χ_{spin} are reflected in the distribution of Knight shifts which make up the inhomogeneously broadened NMR lineshape, the distribution of $1/T_1$ that make up a single $M(t_D)$ curve can represent multiple environments whose local magnetic ground states are qualitatively distinct. We developed a program which computes the inverse Laplace transform (ILT) of our measured $M(t_D)$ data, in order to deduce a probability density function $P(1/T_1)$ representing the $1/T_1$ distribution. Our ILT algorithm primarily employs Tikhonov regularization, which

limits the instability of numerically inverting data with finite noise. This $1/T_1$ analysis method offers significant advantages over the traditional method of fitting $M(t_D)$ against a phenonmenological stretched exponential function, which provides only a crude approximation of the spatial average of the $1/T_1$ distribution. In contrast, our approach of calculating $P(1/T_1)$ using ILT can delineate the behavior of multiple distinct $1/T_1$ components, and hence preserve vital information on the position-by-position distribution of local spin dynamics.

In this thesis, we report on our ^{7}Li NMR measurements of the Kitaev honeycomb iridate $Ag_3LiIr_2O_6$, our ^{63}Cu nuclear quadrupole resonance (NQR) measurements on the Kagome Heisenberg antiferromagnets $ZnCu_3(OD)_6Cl_2$ (herbertsmithite) and $ZnCu_3(OD)_6FBr$ (Zn-barlowite), and further ^{19}F NMR measurements of $ZnCu_3(OD)_6FBr$. Using ILT, we provide crucial insight into both the intrinsic and disorder-induced low-energy spin excitations of these materials. Firstly, we elucidate the effect of stacking faults and unwanted Ag inclusion by comparing the ^{7}Li NMR lineshape and $P(1/T_1^{\mathrm{Li}})$ of $Ag_3LiIr_2O_6$ samples with varying levels of disorder. Next, we observe in $P(1/T_1^{\mathrm{Cu}})$ a fraction of spin singlets with spatially inhomogeneous energy gaps emerging below $\sim$ 30 K within the kagome planes of $ZnCu_3(OD)_6Cl_2$ and $ZnCu_3(OD)_6FBr$. Finally, we develop a novel method using ILT to obtain the two-dimensional correlation map between ^{19}K and $1/T_1^{\mathrm{F}}$ a the ^{19}F sites of $ZnCu_3(OD)_6FBr$, and evidence the existence of spin-polarized domains emerging near interlayer Cu^{2+} defects.

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Chapter 1

Introduction

1.1 Quantum Spin Liquids

The Heisenberg model of magnetism with localized electron spins is described by the Hamiltonian

$$\hat{\mathcal{H}}_H = -J_H \sum_{i,j} \boldsymbol{S}_i \cdot \boldsymbol{S}_j, \tag{1.1.1}$$

where the coupling constant J_H corresponds to the exchange interaction energy between neighboring spins $\boldsymbol{S}$, which can be ferromagnetic ($J_H > 0$) or antiferromagnetic ($J_H < 0$). At high temperatures, these interactions are overwhelmed by thermal fluctuations, resulting in a paramagnetic phase with randomly oriented spins (Fig. 1.1 **(a)**). Below a critical temperature $T_c \sim J_H/k_B$, the spins break rotational symmetry as they stabilize towards the most energetically-favoured configuration (i.e. ground state). In a conventional magnet, this ground state is typically a unique, statically ordered pattern with long-range correlations between spins (Fig. 1.1 **(b)**).

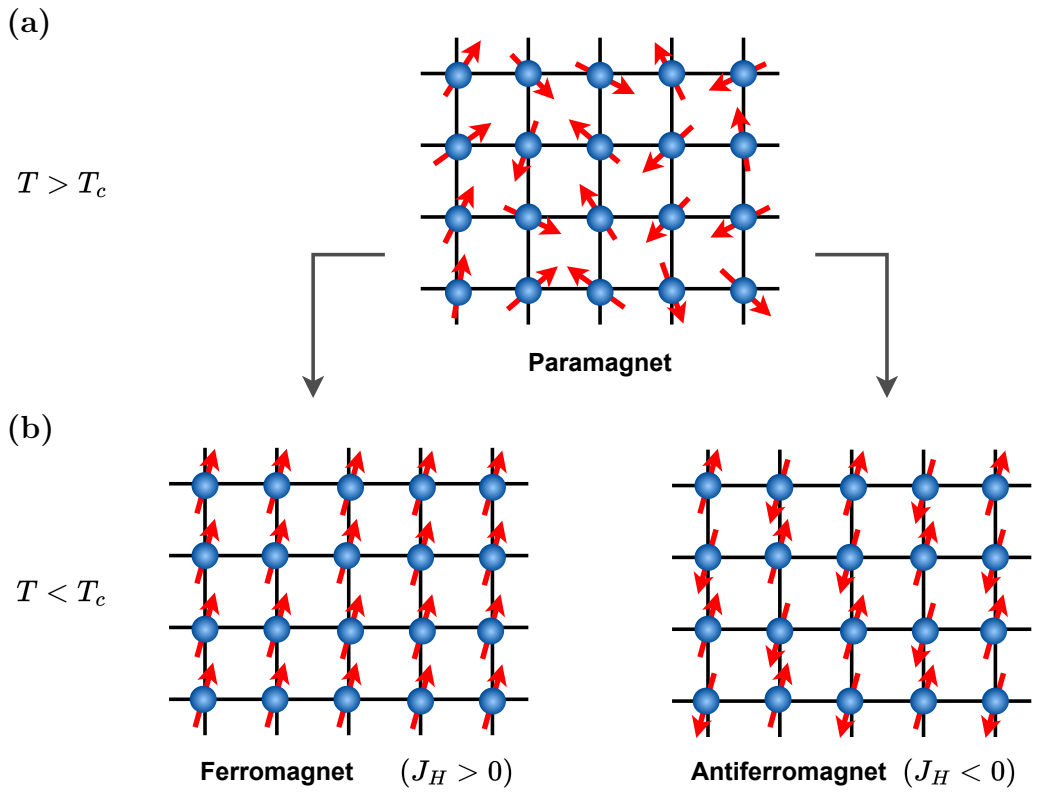


Figure 1.1: Conventional magnet on a square lattice of localized spins. **(a)** High temperature ($T > T_c$) paramagnetic state. **(b)** Low-temperature ($T < T_c$) states with ferromagnetic and antiferromangetic order.

However, this tendency for the spins to order is suppressed when there exists competing interactions that cannot be simultaneously satisfied, such as when the lattice is geometrically frustrated [1]. A simple example of this is three antiferromagnetic Ising spins arranged in a triangle. Since all three spins cannot be simultaneously antiparallel to each other, there is no single, best way for them to align (Fig. 1.2 **(a)**). Extrapolating this onto a two-dimensional (2D) lattice of edge-sharing triangles, the classical ground state becomes highly degenerate, which is to say that there are countless configurations of spins that all equally minimize the total energy (Fig. 1.2 **(b)**).

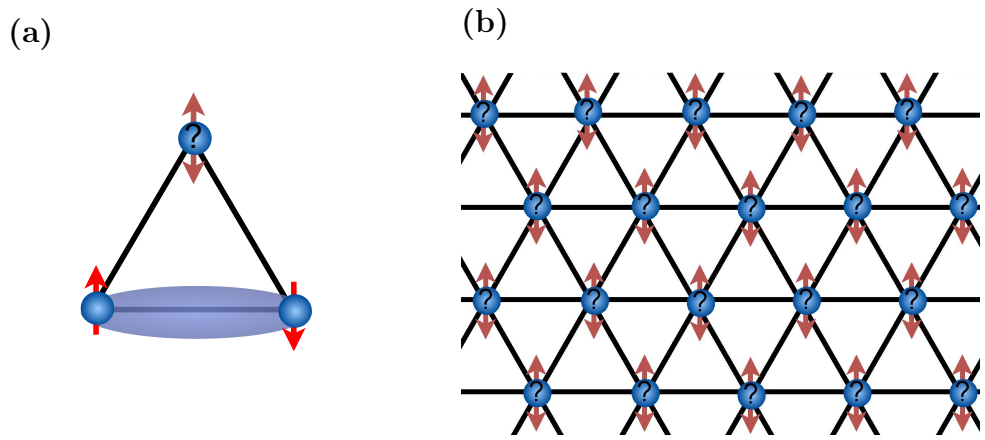


Figure 1.2: Example of geometrical frustration on a **(a)** triangle plaquette of Ising spins. **(b)** Extended to a lattice of edge-sharing triangles, the classical grounds state is highly degenerate.

When strong magnetic interactions coexist alongside a highly degenerate ground state, the spins behave analogously to a"liquid" at low temperatures, in that they remain fluctuating, yet highly-correlated like the molecules in an actual liquid (whereas statically ordered spins are analogous to a solid).

A spin liquid is known as a quantum spin liquid (QSL) when it avoids static order entirely and preserves its symmetry down to $T = 0$ K. This is possible since unlike thermally-driven classical fluctuations, quantum fluctuations exist even at absolute zero due to the Heisenberg uncertainty principle, where the relative size of the zero-point motion of spins is largest for $S = 1/2$ systems. Since these quantum fluctuations can be phase coherent, the spins in a QSL can exhibit long-range quantum entanglement [1].

For a frustrated spin-1/2 Heisenberg antiferromagnet, a QSL can exist in the form of a resonating valence bond (RVB) state, whose theoretical description was first proposed in 1973 by Philip W. Anderson [2]. A valence bond (in the context of magnetism) is a pair of antiferromagnetically-coupled spins that, through quantum

entanglement, form a spin-0 singlet. When all of the spins in a lattice are partitioned into a specific pattern of valence bonds, the result is known as a valence bond solid (VBS), which is not a true QSL as it breaks translational symmetry and lacks long-range entanglement. In contrast, an RVB state only occurs when the valance bonds partitionings can undergo quantum fluctuations, and the ground state becomes a quantum superposition of many different partitionings (shown in Fig. 1.3).

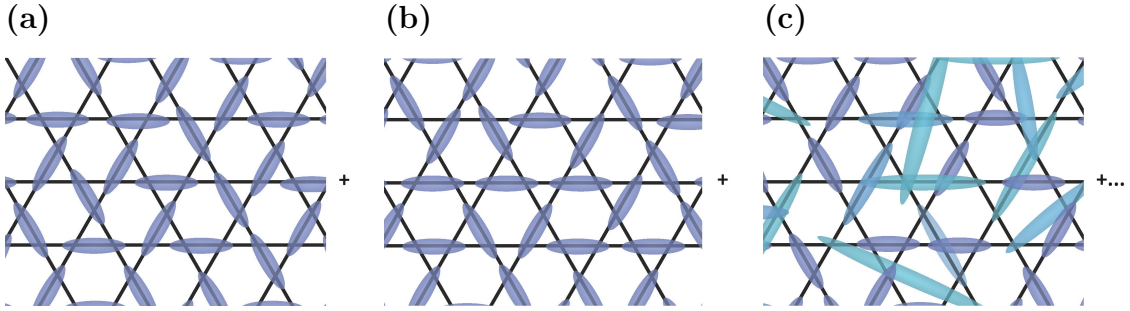


Figure 1.3: Theoretical picture of the RVB state on a kagome lattice (introduced in section 1.3), which is a superposition of many arrangements of **(a)**,**(b)** short-ranged valence bonds (dark blue ellipses) and/or **(c)** long-ranged valence bonds (light blue ellipses).

1.2 Experimental Signatures of QSLs

Finding a QSL generally involves studying materials that exhibit a large $|J_H|$, but whose magnetic order is suppressed by geometric frustration. For a Heisenberg system at high temperatures, the magnetic susceptibility (specifically the electron spin contribution) χ_s typically follows a Curie-Weiss behaviour $\chi_s \approx C/(T - \Theta_{CW})$, where C is the Curie constant and Θ_{CW} is the Curie-Weiss temperature. In the mean-field approximation, these can be expressed as $C = \mu_B^2 g^2 S(S+1)/3k_B$ and $\Theta_{CW} = \frac{2}{3} z S(S+1) k_B J_H$, where μ_B is the Bohr magneton, g is the electron g-factor,

and z is the number of nearest neighbors for each spin. The overall level of magnetic frustration in a material can be represented by a frustration parameter f, defined as $f = |\Theta_{CW}|/T_c$, where T_c is the temperature at which ordering appears [3]. Materials with a high frustration parameter ($f \gtrsim 100$) are good candidates for hosting QSL physics, as their spins remain liquid-like far below the ordering temperature expected for $|J_H|$.

A core issue with experimentally identifying a QSL is that most conventional probes are better suited to ruling out the possibility of a material being a QSL. Apart from measuring the frustration parameter, another test for QSL behaviour is to look for the lack of sharp anomalies (i.e. peaks) in its specific heat capacity, which corresponds to symmetry-breaking phase transitions. More direct evidence of QSL physics can be found by looking for signatures of unconventional elementary excitations with fractional quantum numbers. A well-known example of fractionalized quasiparticle excitations are $S = 1/2$ spinons, which exist in materials containing chains of $S = 1/2$ spins coupled via antiferromagnetic Heisenberg interactions [4].

For context, charge-neutral excitations on normal magnets tend to behave like $S = 1$ magnons. When probing the spin wave dispersion relation via techniques such as inelastic neutron scattering (involving a total change in spin of $\Delta S_z = 1$), magnons appear as sharp bands in the excitation spectrum. In contrast, $\Delta S_z = 1$ neutron interactions with a one-dimensional (1D) spin chain involve the creation/annihilation of two $S = 1/2$ spinons (due to conservation of angular momentum). The extra degree of freedom in the relative momentum of the two spinons results in a continuum in the spin excitation spectrum (such as the one observed in $KCuF_3$ [5]). The antiferromagnetic Heisenberg spin chain completely avoids magnetic ordering [6] and can

be considered as an example of a 1D QSL. However, in systems with higher dimensions, some amount of frustration is needed for achieving a QSL state. For example, an antiferromagnetic three-dimensional (3D) cubic lattice of $S = 1/2$ spins generally favours a Néel-ordered ground state, while in a two-dimensional (2D) square lattice, the Néel state marginally beats out the valence bond state at $T = 0$ [7].

The main platform for realizing a QSL state is in Mott insulator materials, specifically those with localized $S = 1/2$ spins. The original example of this is in copper-oxides containing $S = 1/2$ Cu^{2+} sites coupled via strong antiferromagnetic super-exchange interactions. Popular QSL candidates can be divided into several classes of materials, including those with 2D kagome lattices [8], 3D pyrochlore lattices [9], and organic salts on 2D triangular lattices [10].

However, a highly frustrated material is not necessarily a QSL. Broadly speaking, deviations from the QSL model's theoretical ideal can suppress a QSL state from being realized in a material. This can originate from various factors such as chemical defects, or lattice distortions and asymmetries. These imperfections introduce additional interactions that compete with the QSL model's and can significantly influence the magnetic ground state. Instead of becoming a QSL at low temperatures, spins can undergo magnetic long-range order (LRO) [11], form a VBS [12], or freeze into a spin-glass-like state [13]. Furthermore, there can exist multiple regions in a sample that exhibit qualitatively different behaviours (i.e. spatial inhomogeneity). To accurately investigate a material and determine whether or not it's a QSL requires differentiating its intrinsic physics from the influence of disorder.

1.3 Kagome Heisenberg Antiferromagnets (KHAs)

The kagome lattice is named after a pattern used in Japanese basket weaving consisting of a network of corner-sharing triangles. For achieving a QSL in spin-1/2 Heisenberg antiferromagnets, the kagome lattice is generally preferable to the edge-sharing triangular lattice, which has an ordered classical ground state where spins are oriented 120° relative to each other. In contrast, the kagome lattice is much less prone to this type of order, since a single triangle of 120° canted spins will not constrain the rest of the lattice to favour a specific orientation [14].

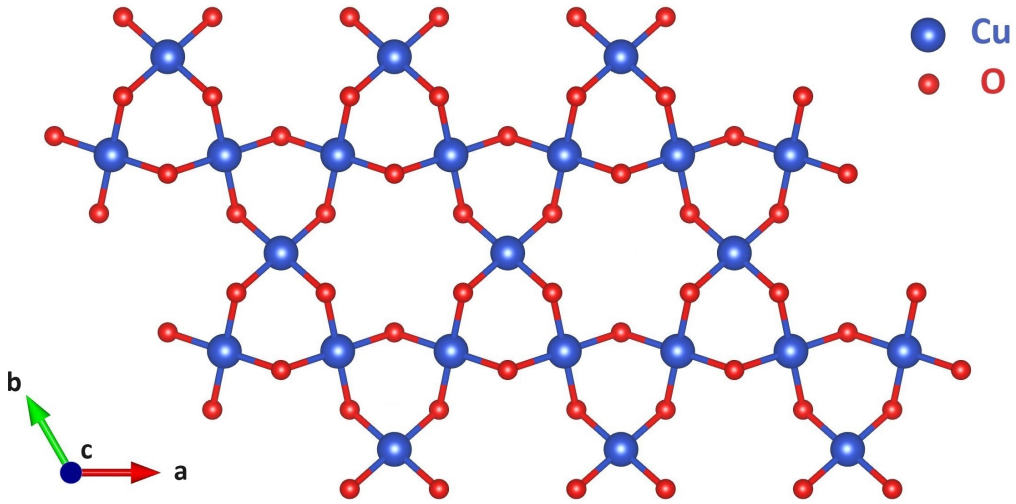


Figure 1.4: Top-down view of kagome planes in Zn-barlowite and herbertsmithite, consisting of a network of spin-1/2 Cu^{2+} sites connected via O^{2-} ligands.

Several naturally-occurring minerals consists of layers of $S = 1/2$ Cu^{2+} sites arranged in the kagome pattern (Fig. 1.4), some of which have been identified as KHA QSL candidates. However, early studies have shown that many of these materials undergo magnetic LRO or spin freezing at low temperatures, precluding a QSL state [8]. Currently, two of the most promising KHA QSL candidates are known

as herbertsmithite ($ZnCu_3(OD)_6Cl_2$) and the more recently discovered Zn-barlowite ($ZnCu_3(OD)_6FBr$), which are the $x = 1$ end members of the series Zn-paratacmite ($Zn_xCu_{4-x}(OD)_6Cl_2$) and Zn-substituted barlowite ($Zn_xCu_{4-x}(OD)_6FBr$), respectively. The crystal structures of herbertsmithite and Zn-barlowite are shown in Fig. 1.5. Note that a key difference between them is in the stacking patterns of their kagome layers [15]. The arrangements of Cu atoms cycle between 3 orientations in herbertsmithite (ABC stacking), while they are arranged identically in Zn-barlowite (AA stacking) along with an alternating pattern of O^{2-} ligands and interlayer atoms.

The successful synthesis of polycrystalline (i.e. powder) [16] and single-crystal [17] samples of herbertsmithite enabled a great deal of experimental research in the past two decades. More recently, powder and single crystal samples of Zn-barlowite were synthesized via Zn^{2+} substitution of its parent compound barlowite ($Cu_4(OD)_6FBr$) [15, 18].

Compared to other KHA materials, the crystal structures of Herbertsmithite and Zn-barlowite have higher symmetries and feature relatively minimal distortions. Both materials show no signs of magnetic order or spin freezing down to $T \sim 50$ mK $\approx 10^{-4}J/k_B$, where $J/k_B \approx 190$ K is the Cu-Cu exchange energy. [18, 19]. Inelastic neutron scattering experiments on herbertsmithite single-crystals also revealed a broad continuum of spin excitations, which are argued to be consistent with deconfined $S = 1/2$ spinons [20].

In contrast to Zn-barlowite, "undoped" barlowite undergoes a Néel transition at $T_N \approx 15$ K into an antiferromagnetically ordered state, as well as a structural transition below ~ 265 K into a lower-symmetry space group ($Pnma$) [11, 21] This suggests that the nonmagnetic Zn^{2+} ions situated in-between the kagome layers of

herbertsmithite and Zn-barlowite are responsible for the suppression of LRO, as they magnetically isolate the layers from one another.

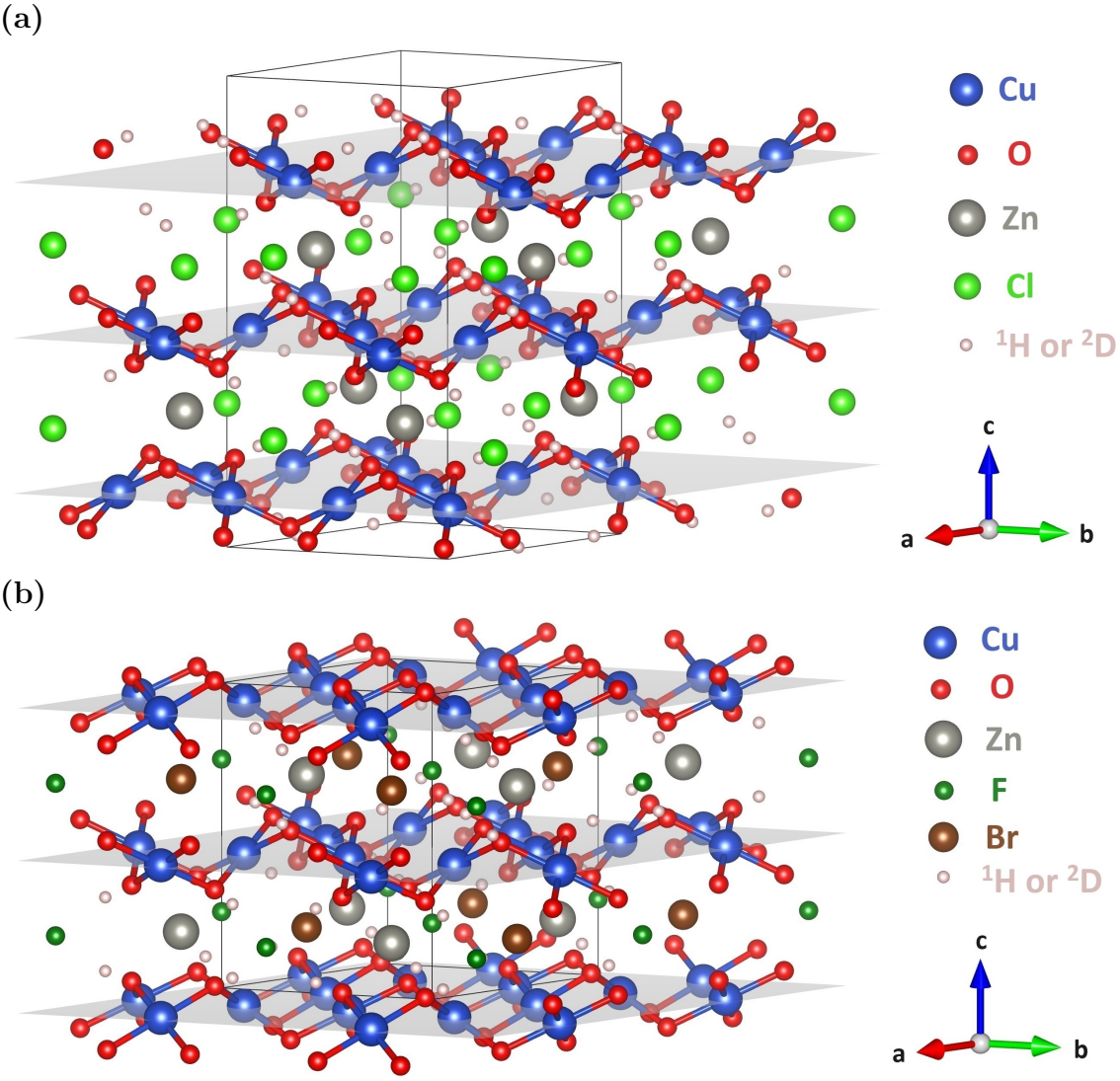


Figure 1.5: **(a)** Atomic lattice of herbertsmithite $ZnCu_3(OD)_6Cl_2$ and **(b)** Zn-barlowite $ZnCu_3(OD)_6FBr$, which can be either protonated (D = ^{1}H) or deuterated (D = ^{2}D). Herbertsmithite and Zn-barlowite are in the space groups $R\bar{3}m$ and $P6_3/mmc$, respectively. The grey horizontal planes indicate the kagome layers, and the solid lines represent the unit cell.

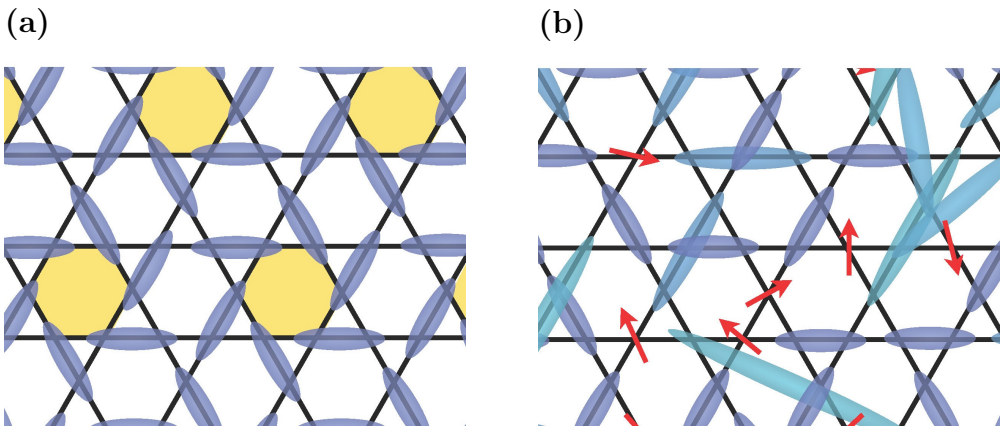


Figure 1.6: Conceptual pictures of other possible states that can be realized in kagome materials. **(a)** A VBS state with a large unit cell, in which singlets are arranged into "pinwheel" formations. **(b)** A"random-singlet" state, consisting of a mixture of singlets (which can be nearly isolated or in resonating clusters) and orphaned paramagnetic spins (red arrows) [22].

There is an ongoing debate about the theoretical ground state of the KHA. Numerous pictures (with similar energies) have been proposed, including a fully gapped QSL [23–25], a gapless critical QSL [26, 27], and a VBS "pinwheel" state (Fig. 1.6 **(a)**) [28] (which has been realized in certain kagome-lattice materials [12]).

More recently, random-bond models employ a distribution of exchange energies J_H in order to account for the quenched disorder common in $S = 1/2$ Heisenberg antiferromagnets [29]. These models describe an inhomogeneous "random-singlet" state (Fig. 1.6 **(b)**), and predict a form of randomness-induced QSL behaviour [22]. It has also been theorized that a gapless valence bond glass phase can arise from a small fraction of missing spins in the kagome layers [30]. Systems composed of localized spin singlets and orphaned paramagnetic spins are proposed to obey a universal scaling relation in their heat capacity (as a function of temperature), which has been observed in numerous $S = 1/2$ quantum magnets with quenched disorder [31].

1.4 Kitaev Honeycomb Model

A new QSL model was proposed in 2006 by Alexei Kitaev, which predicted a QSL ground state to exist in a honeycomb lattice of effective $J_{eff} = 1/2$ spins engaged in bond-dependent Ising interactions [32]. The Kitaev model Hamiltonian is defined as

$$\hat{\mathcal{H}}_K = \sum_{\langle i,j \rangle_\gamma} J_K^\gamma S_i^\gamma S_j^\gamma, \tag{1.4.1}$$

where J_K^γ represents the Ising interaction energy between nearest neighbors i, j, which is aligned along axis $\gamma = x, y, z$, depending on the direction of the bond (Fig. 1.7 **(a)**). Similar to KHAs, the Kitaev model's ground state is also highly degenerate, although it arises from a different form of geometric frustration (Fig. 1.7 **(b)**).

In contrast to the KHA model, the Kitaev model is exactly solvable, so there is no disagreement amongst theorists about its ground state. The exact solution also yields a full spectrum of excitations, which includes fractionalized quasiparticle excitations known as Majorana fermions.

A Kitaev QSL can be realized in Mott insulators containing magnetic transition-metal atoms with strong spin-orbit coupling (e.g. Ir^{4+}, Rh^{4+}, or Ru^{3+}) which act as the $J_{eff} = 1/2$ spins [34]. Early examples of materials recognized as Kitaev QSL candidates include Na_2IrO_3[35], α-Li_2IrO_3 [36], and α-$RuCl_3$ [37].

However, real Kitaev materials differ from the ideal Kitaev model, and exhibits additional interactions such as an isotropic Heisenberg exchange component (which favours ferromagnetic/antiferromagnetic order) and a symmetric off-diagonal exchange

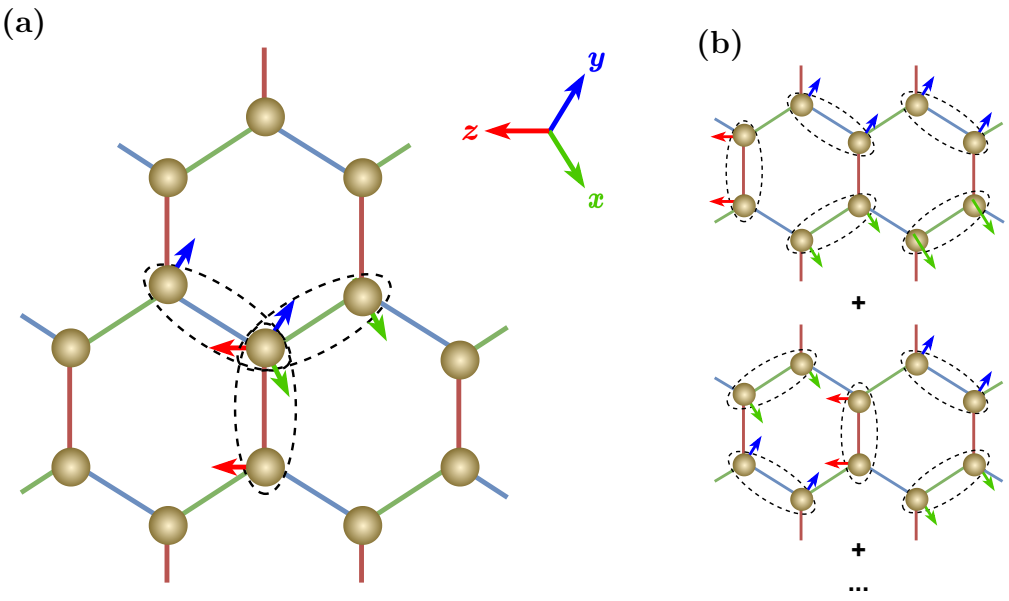


Figure 1.7: Kitaev model on a $J_{eff} = 1/2$ honeycomb lattice. **(a)** Each spin can only align with one of the three perpendicular bonds which can favour the x, y, or z direction. **(b)** The Kitaev QSL state, which is a superposition of many different ways to tessellate the lattice with pairs energetically-satisfied spins [33].

component $\hat{\mathcal{H}}_\Gamma = \sum_{\langle i,j \rangle} \Gamma(S_i^\alpha S_j^\beta + S_i^\beta S_j^\alpha)$. Therefore, the spin system is better described by the Heisenberg-Kitaev Hamiltonian

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_K + \hat{\mathcal{H}}_H + \hat{\mathcal{H}}_\Gamma. \tag{1.4.2}$$

Depending on the relative strength of these interactions, the Heisenberg-Kitaev model's ground state can be one of a variety of different ordered or spin liquid phases [38]. For example, both Na_2IrO_3 and α-Li_2IrO_3 are observed to undergo antiferromagnetic LRO at $T_N \approx 15$ K.

With this in mind, a second generation of metastable Kitaev iridates were synthesized via topotactic exchange methods. These materials demonstrated several

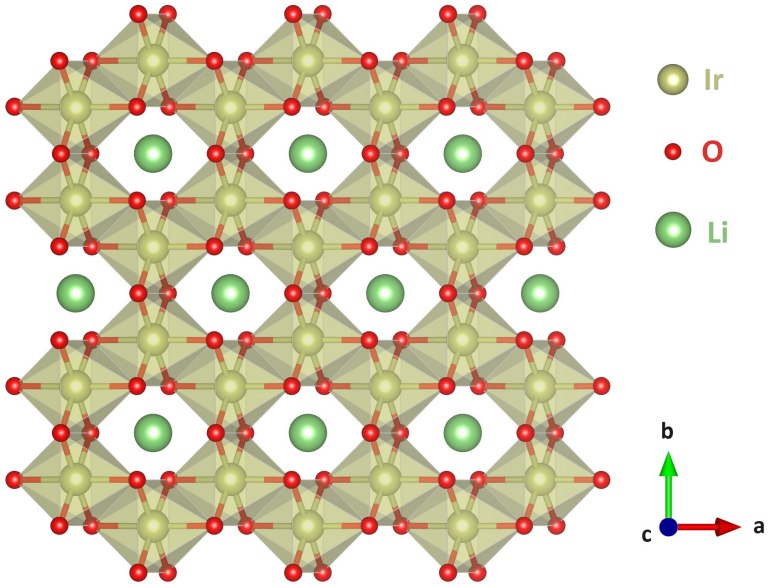


Figure 1.8: Lattice of $Ag_3LiIr_2O_6$. Ir^{4+} sites are located at the center of edge-sharing IrO_6 octahedrons, arranged in a Kiatev honeycomb lattice.

strategies for tuning the Kitaev lattice in order to minimize the non-Kitaev interactions [39]. Cu_2IrO_3 was created by replacing the Na^+ ions in Na_2IrO_3 with Cu^+ [40], where the change in ionic radius modifies the Ir-Ir-Ir and Ir-O-Ir bonds angles and brings them closer to the ideal Kitaev geometry of 120° and 90°, respectively. $H_3LiIr_2O_6$ [41] and $Ag_3LiIr_2O_6$ [42] (shown in Fig. 1.8) were made by replacing the interlayer Li^+ ions in α-Li_2IrO_3 with H^+ and Ag^+, respectively. The intercalation of H^+ introduces random bond disorder in $H_3LiIr_2O_6$ and suppresses magnetic order, which may lead to a disordered QSL-like state [43, 44]. Meanwhile, the interlayer Ag^+ in $Ag_3LiIr_2O_6$ are linearly coordinated with nearby O^{2-} atoms (as opposed to the octahedrally coordinated Li^+ sites in α-Li_2IrO_3). This increases the spacing between the Kitaev layers and modifies the interlayer orbital overlaps [42]. Compared to their parent materials, magnetic order was observed to be greatly suppressed in these new Kitaev iridates.

1.5 Thesis Outline

The remainder of this thesis is organized as follows. In chapter 2, we explain the principles behind nuclear magnetic resonance (NMR) and and Nuclear Quadrupole Resonance (NQR), in order to provide a brief introduction to help readers understand the experimental methods used in our investigations and interpret the results presented in chapters 4-6. In chapter 3, we describe the process of numerical inverse Laplace transform (ILT) $1/T_1$ analysis, a data analysis technique that is crucial to NMR and NQR studies of systems with spatially inhomogenous spin dynamics, such is the case for many disordered QSL candidate materials. At the end of this chapter, we introduce a novel ILT-based technique (used in chapter 6), which generates a 2D correlation map between $1/T_1$ and Knight shift.

In chapter 4, we report our ^{7}Li NMR results on $Ag_3LiIr_2O_6$, and include a reprint of our article published in 2021 in the journal Physical Review B [45]. With the aid of ILT, we report signatures of magnetic long-range order and demonstrate how they can be easily masked by inhomogeneity in the sample.

In chapter 5 we report our ^{63}Cu NQR results on the KHAs herbertsmithite and Zn-barlowite, where ILT was used to detect the gradual emergence (below $\sim$30 K) of spin singlets with spatially varying excitation gaps. We further report ^{19}F NMR results on Zn-barlowite in chapter 6, and compare them to that of (undoped) barlowite, in which all interlayer sites are occupied by Cu^{2+}. Based on our findings on Zn-barlowite, we propose a picture in which spin-polarized domains are formed around interlayer Cu^{2+} defect spins, occupying $\sim$ 60% of the kagome planes at 2 K. The results reported in chapters 5 and 6 were published in Ref. [46] and [47], respectively. Finally, we conclude our research in chapter 7.

Chapter 2

Nuclear Magnetic Resonance and Nuclear Quadrupole Resonance

Nuclear magnetic resonance (NMR), and Nuclear Quadrupole Resonance (NQR) are spectroscopic techniques which manipulate and observe the magnetic moment of atomic nuclei (i.e. nuclear spin) within a sample. They locally probe the structural, magnetic and electronic environment experienced by the nuclei through quantities such as the NMR Knight shift K and NQR resonance frequency ν_Q. Spin-lattice $(1/T_1)$ and spin-spin $(1/T_2)$ relaxation measurements inform about the local spin dynamics, with $1/T_1$ directly embodying the quantity of low-energy excitations at the site of each nuclear spin. These techniques play an important role in the study of disordered quantum materials, where the spatially inhomogeneous physics make sample-averaged measurements difficult to interpret. In this chapter, we provide an overview of the principles behind NMR and NQR, as well as the process of conducting resonance experiments.

2.1 Basic Resonance Theory

An nucleus possesses a magnetic moment $\boldsymbol{\mu}$ parallel to its total angular momentum $\hbar\boldsymbol{I}$. They are related via the nuclear gyromagnetic ratio γ_N, such that $\boldsymbol{\mu} = \gamma_N\hbar\boldsymbol{I}$ [48, 49]. Consider the motion of a free nuclear spin in an externally-applied, static magnetic field $\mathbf{B}_{\text{ext}} = B_{\text{ext}}\hat{\boldsymbol{z}}$ (where $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}}$ are Cartesian unit vectors). In the classical model, $\mathbf{B}_{\text{ext}}$ exerts a torque $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}_{\text{ext}}$, resulting in the equation of motion

$$\frac{d\boldsymbol{\mu}}{dt} = \gamma_N \frac{d\boldsymbol{I}}{dt} = \boldsymbol{\mu} \times \gamma_N \mathbf{B}_{\text{ext}}, \tag{2.1.1}$$

where $\boldsymbol{\mu}$ precesses about $\mathbf{B}_{\text{ext}}$ at the Larmor frequency $\nu_L = \gamma_N B_{\text{ext}}$.

Quantum mechanically, a nucleus' total angular momentum can be described by the dimensionless spin operator $\hat{\boldsymbol{I}}$, which is proportional to the magnetic moment operator $\hat{\boldsymbol{\mu}} = \gamma_N\hbar\hat{\boldsymbol{I}}$. A nucleus' spin quantum number I is inherent to each isotope, and can have integer or half-integer values. The Hamiltonian of the Zeeman interaction between a spin-I nucleus and a static, local magnetic field $\mathbf{B}_{\text{ext}}$ is defined as

$$\hat{\mathcal{H}}_0 = -\hat{\boldsymbol{\mu}} \cdot \mathbf{B}_{\text{ext}} = -\gamma_N\hbar B_{\text{ext}}\hat{I}_z, \tag{2.1.2}$$

where $\hat{I}_\gamma$ ($\gamma = x, y, z$) is the the γ-component of $\hat{\boldsymbol{I}}$. This results in $2I+1$ equally-spaced energy eigenstates with energy levels $E_m = \gamma_N\hbar m B_{\text{ext}}$, indexed by the magnetic quantum number $m = \{-I, -I+1, ..., I-1, I\}$. The energy between adjacent states is then $\Delta E = E_m - E_{m+1} = \gamma_N\hbar B_{\text{ext}}$.

Transitions between eigenstates states $|m\rangle$ can be induced by applying a rotating magnetic field

$$\mathbf{B}_1(t) = B_1 \cos(2\pi f t)\hat{\boldsymbol{x}} - B_1 \sin(2\pi f t)\hat{\boldsymbol{y}}. \tag{2.1.3}$$

perpendicular to $\mathbf{B}_{\text{ext}}$ [49]. In the limit $B_{\text{ext}} \gg B_1$, this acts as a time-dependent perturbation on $\hat{\mathcal{H}}_0$ with frequency f

$$\hat{\mathcal{H}}_{pert} = \gamma_N \hbar B_1 \cos(2\pi f t)\hat{I}_x - \gamma_N \hbar B_1 \sin(2\pi f t)\hat{I}_y, \tag{2.1.4}$$

which has non-zero matrix elements $\langle m'| \hat{\mathcal{H}}_{pert} |m\rangle$ only for $m' = m \pm 1$. Thus, transitions occur between adjacent states, and only when f satisfies the resonance condition

$$f = \Delta E/h = \nu_0, \tag{2.1.5}$$

where ν_0 is known as the resonance frequency.

NMR probes the energy ΔE between the $2I + 1$ nuclear spin states (Figure 2.1 (a)) by inducing transitions via spectral absorption. The resulting precession of nuclear spins, otherwise known as free induction decay (FID) or spin echo, generates a measurable signal (discussed in Sec. 2.3 and 2.4). For typical γ_N and $B_{\text{ext}} \sim 1$ T, ν_0 is in the radio frequency (RF) range, on the order of MHz. Table 2.1 lists the values of γ_N, I, quadrupole moment Q, and natural abundance for the isotopes studied in this thesis.

Isotope	$\gamma_N/2\pi$ (MHz/T)	I	Q (barns)	Natural abundance (%)
^{7}Li	12.089	3/2	-0.03	95.2
^{63}Cu	11.285	3/2	-0.211	69.2
^{65}Cu	12.089	3/2	-0.204	30.8
^{19}F	40.05	1/2		100

Table 2.1: Nuclear gyromagntic ratio γ_N, spin quantum number I, quadrupole moment Q, and natural abundance for the isotopes ^{7}Li, ^{63}Cu, ^{65}Cu, and ^{19}F

In practice, NMR measures the signal intensity as a function of f, resulting in a spectral lineshape which peaks at $f = \nu_0$ (shown in Fig. 2.1 (b)). This peak is homogeneously broadened by sources such as nuclear dipole-dipole interactions (see Sec. 2.5) and the bandwidth of the spectrometer (see Sec. 2.8), and its width is typically on the order of ∼100 kHz in our experiments. In addition to homogeneous broadening, the spatial inhomogeneity of the local hyperfine field (see Sec. 2.2) also determines the width and profile of the NMR lineshape, which makes NMR an ideal probe of this local hyperfine field.

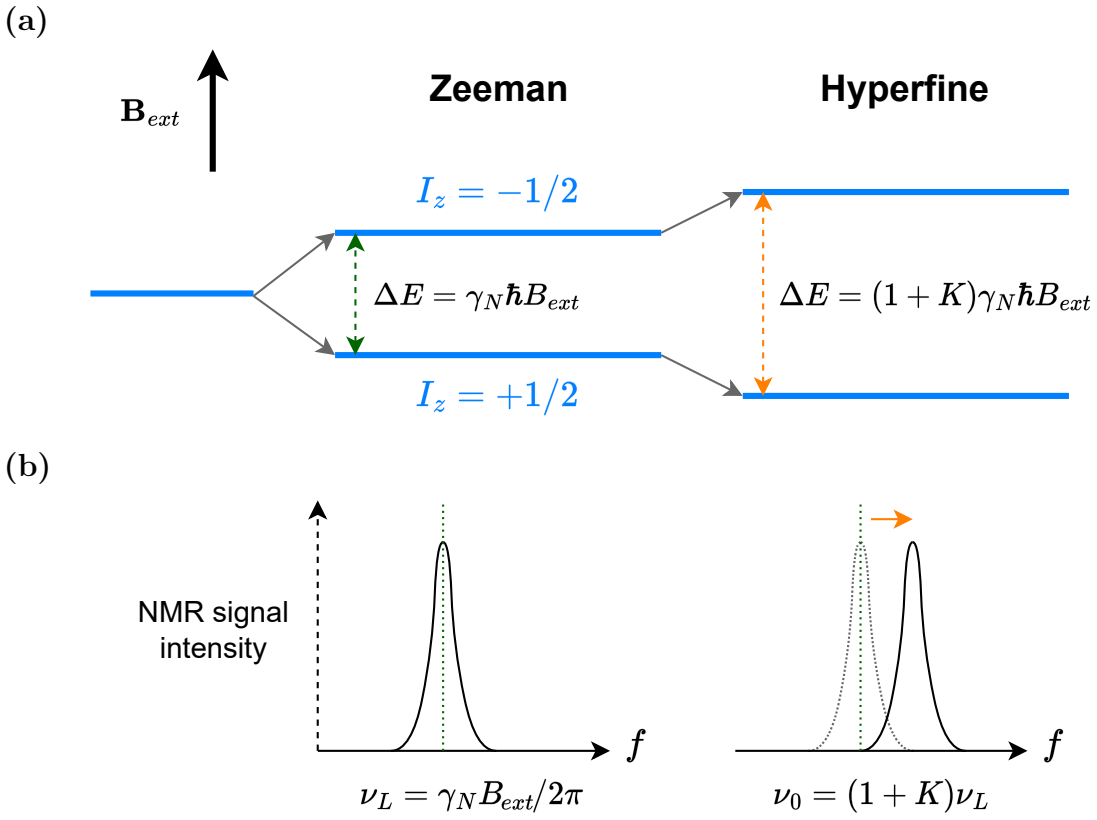


Figure 2.1: **(a)** Energy levels of a spin-1/2 nucleus separated via Zeeman interactions with the external field $\mathbf{B}_{\text{ext}}$ (left), and shifted by the static hyperfine field (right). **(b)** The corresponding NMR lineshapes.

2.2 Knight Shift

The magnetic field experienced by the nuclei inside a material is dependent on its local environment, where the effective field $\mathbf{B}_0$ equals the sum of the external field $\mathbf{B}_{\text{ext}}$ and a local hyperfine field $\mathbf{B}_{\text{hf}}$ generated by nearby electrons. This causes the resonant frequency ν_0 to be shifted from the resonant frequency of a bare nucleus ν_L to $\nu_0 = (1 + K)\nu_L$, where K is known as the Knight shift.

The Knight shift is composed of two separate contributions, the spin shift K_s, and chemical shift K_{chem}, and can be written as

$$K = K_s + K_{chem} = K_s + K_{V.V} + K_{dia}. \tag{2.2.1}$$

K_{chem} stems from interactions between the nucleus and the orbital magnetic moments of electrons, and is the sum of the diamagnetic (K_{dia}) and Van Vleck ($K_{V.V.}$) shifts. K_{dia} corresponds to the diamagnetic susceptibility χ_{dia} arising from the shielding current of electrons, and is temperature-independent [50]. The orbital shift $K_{V.V.}$ is proportional to the Van Vleck paramagnetic susceptibility $\chi_{V.V}$, which originates from the mixing of electron orbitals $|n\rangle$ when perturbed by $\mathbf{B}_{\text{ext}}$ [50, 51]. $\chi_{V.V}$ is shown to be

$$\chi_{V.V} = 2\mu_B^2 \sum_n \frac{|\langle 0|\, \hat{\boldsymbol{L}}_z\, |n\rangle\,|^2}{E_n - E_0}. \tag{2.2.2}$$

where $\hat{\boldsymbol{L}}_z$ is the orbital angular momentum operator along the direction of $\mathbf{B}_{\text{ext}}$. Since the energy splitting produced by the crystal field is on the order of $\sim$ 1 eV ($\sim 10^4$ K) , $\chi_{V.V}$ usually does not have significant temperature dependence in our NMR experiments.

The spin shift K_s comes from the coupling between the nuclear and electron spins

via the Fermi-contact and the dipole-dipole interactions, which we define in section 2.7. The strength of these hyperfine interactions is represented by the hyperfine form factor A_{hf} (see Sec. 2.7), and K_s is proportional to the electron spin susceptibility χ_s, such that

$$K_s = \frac{A_{hf}(\boldsymbol{q}=0)}{N_A \mu_B}\chi_s, \tag{2.2.3}$$

where N_A is the Avogadro constant. In this equation, χ_s is defined in units of [emu/mol], and A_{hf} is in units of $[kOe/\mu_B]$. Typically, χ_s is temperature-dependent for correlated electron systems, and thus K_s is responsible any temperature dependence in the Knight shift.

For a pristine (i.e. spatially uniform) sample, the bulk-averaged magnetic susceptibility χ_{bulk}, measured via superconducting quantum interference device (SQUID), consists of $\chi_{bulk} = \chi_s + \chi_{V.V.} + \chi_{dia}$. However, since the Knight shift is a probe of the local magnetic environment, it can measure the intrinsic spin susceptibility, separate from any defect contributions to χ_{bulk}. More generally, the spatial inhomogeneity of $\mathbf{B}_0$ is quantified by the distribution of resonant frequencies among the nuclei, which composes the NMR lineshape.

If there is only one temperature-dependent contribution in χ_s, A_{hf} can be determined experimentally by measuring K and χ_{bulk}. By plotting K_s and χ_s with temperature as the implicit parameter, we can obtain A_{hf} from the slope of the linear relation

$$\begin{aligned} K_s &= \frac{A_{hf}}{N_A \mu_B}\chi_s \\ \implies A_{hf} &= N_A \mu_B \frac{\Delta K}{\Delta \chi}. \end{aligned} \tag{2.2.4}$$

2.3 Motion of Nuclear Spin

Solving the time-independent Schrödinger equation for $\hat{\mathcal{H}}_0$, it can be shown that the expectation value of the nuclear magnetic moment operator $\langle\hat{\boldsymbol{\mu}}(t)\rangle$ behaves similarly to the classical equation of motion

$$\frac{d\langle\hat{\boldsymbol{\mu}}(t)\rangle}{dt} = \langle\hat{\boldsymbol{\mu}}(t)\rangle \times \gamma_N \mathbf{B}_0. \tag{2.3.1}$$

In fact, this holds true even for a time-dependent magnetic field [48], allowing us to treat $\langle\hat{\boldsymbol{\mu}}(t)\rangle$ as a classical vector going forward.

Changing coordinates $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}} \rightarrow \hat{\boldsymbol{x}}', \hat{\boldsymbol{y}}')$, to a reference frame rotating with angular velocity $\boldsymbol{\Omega} = -\gamma_N \mathbf{B}_0$, Eq. 2.1.1 is rewritten as

$$\frac{d\boldsymbol{\mu}'}{dt} = \boldsymbol{\mu}' \times (\gamma_N \mathbf{B}_0 + \boldsymbol{\Omega}) = 0. \tag{2.3.2}$$

Applying the magnetic field $\mathbf{B}_1(t)$ from Eq. 2.1.4 with $f = \nu_0$, we see that in the rotating frame, $\boldsymbol{\mu}'$ precesses about a fixed $\mathbf{B}_1' = B_1\hat{\boldsymbol{x}}'$ with angular velocity $-\boldsymbol{\omega}' = \gamma_N B_1 \hat{\boldsymbol{x}}'$.

Experimentally, we can generate $\mathbf{B}_1(t)$ within a sample by placing it inside an inductive coil (solenoid) aligned perpendicular to $\mathbf{B}_{\text{ext}}$, and flowing through an alternating electric current (Figure 2.2 **(a)**). The oscillating field generated along the length of the solenoid $\mathbf{B}_x(t) = 2 * B_1 \cos(2\pi\nu_0 t)\hat{\boldsymbol{x}}$ is the sum of 2 counter-rotating fields

$$\begin{aligned} \mathbf{B}_{x,R}(t) &= B_1 \cos(2\pi\nu_0 t)\hat{\boldsymbol{x}} + B_1 \sin(2\pi\nu_0 t)\hat{\boldsymbol{y}}, \text{ and} \\ \mathbf{B}_{x,L}(t) &= B_1 \cos(2\pi\nu_0 t)\hat{\boldsymbol{x}} - B_1 \sin(2\pi\nu_0 t)\hat{\boldsymbol{y}}. \end{aligned} \tag{2.3.3}$$

In the rotating frame, $\mathbf{B}_{x,L}(t)$ is fixed (and equal to $\mathbf{B}_1(t)$ defined in Eq. 2.1.3), while the effect of $\mathbf{B}_{x,R}(t)$ can be ignored since it rotates at a relative frequency $4\pi\nu_0 \gg \gamma_N B_1$ [48].

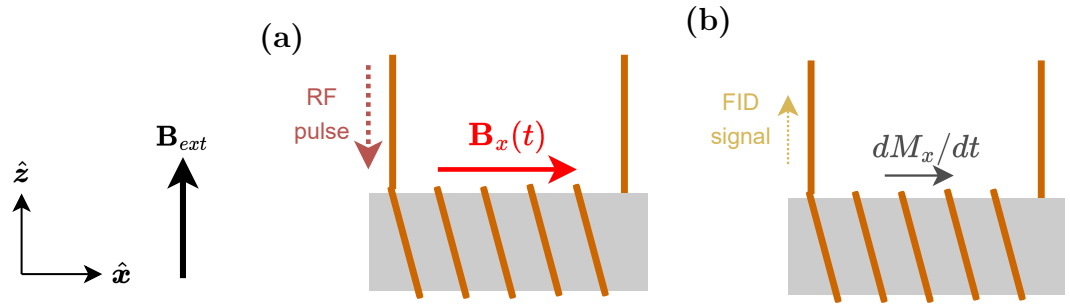


Figure 2.2: Diagrams of solenoid in external field $\mathbf{B}_{\text{ext}}$ showing **(a)** $\mathbf{B}_x(t)$ generated by RF pulse and **(b)** the resulting FID signal.

An RF pulse transmitted into the solenoid can turn on and off $\mathbf{B}_1(t)$. The duration of the pulse required to rotate $\boldsymbol{\mu}$ by 90° and 180° (known as the $\pi/2$-pulse and the π-pulse) are $t_{90} = (\pi/2)/(\gamma'_N B_1)$ and $t_{180} = 2t_{90}$, respectively, where the effective nuclear gyromagnetic ratio γ'_N depends on the nuclear spin I and transitions $(m \leftrightarrow m+1)$ being probed. Regarding the studies in this thesis, $\gamma'_N = \gamma_N$ for ^{19}F NMR ($I = 1/2$), as well as ^{7}Li NMR ($I = 3/2$), in which all transitions are simultaneously probed [49, 52]. For ^{63}Cu ($I = 3/2$) NQR,

$$\gamma'_N = \sqrt{I(I+1) - m(m+1)} * \gamma_N = \sqrt{3}\gamma_N \tag{2.3.4}$$

if $\mathbf{B}_1(t)$ is applied along the direction perpendicular to the principal axis of the electric field gradient $\boldsymbol{Z}$ (see Sec. 2.6). Note that for a randomly oriented powder sample, the optimal t_{90} pulse length is the same as that of an aligned sample with $\mathbf{B}_1(t) \perp \boldsymbol{Z}$, albeit with a lower NMR signal intensity [51].

In any case, t_{90} is typically on the order of around several μs, since B_1 is proportional to the amplitude of the RF pulse, which is generally limited by the breakdown voltage of the gas surrounding the NMR tank circuit.

The collective behavior of nuclear spins across a sample is quantified by the nuclear magnetization $\mathbf{M} = M_x\hat{\boldsymbol{x}} + M_y\hat{\boldsymbol{y}} + M_z\hat{\boldsymbol{z}}$. In additional to applied external perturbations (e.g. $\mathbf{B}_1(t)$), transitions between nuclear spin states are randomly induced from thermal fluctuations within the sample. At thermal equilibrium, $\mathbf{M} = M_0\hat{\boldsymbol{z}}$ is parallel to $\mathbf{B}_0$, where M_0 is the static nuclear magnetization (Eq. 2.3.7).

Starting from $\mathbf{M} = M_0\hat{\boldsymbol{z}}$, applying a $\pi/2$-pulse to a sample rotates $\mathbf{M}$ into the xy-plane. After the pulse is turned off, the precession of $\mathbf{M}$ about $\mathbf{B}_0$ induces an electromotive force inside the solenoid (Faraday's law)

$$\mathcal{E} \propto \frac{dM_x}{dt} \propto \nu_0 M_0 \tag{2.3.5}$$

which we measure as the FID signal (Figure 2.2 **(b)**).

At finite temperatures T, the populations of energy levels in an ensemble of N nuclear spins obey Boltzmann statistics [49]. Since the interaction energy between the nuclei and magnetic fields used in our experiments are typically on the order of

$$\Delta E/k_B = \gamma_N B_0 \hbar/k_B \sim [10^7 \text{ MHz/T}][1 \text{ T}]\hbar/k_B \approx 10^{-4} \text{ K}, \tag{2.3.6}$$

we can approximate M_0 in the $\Delta E \ll k_B T$ limit as

$$M_0 \approx \frac{2\pi N \gamma_N^2 \hbar^2 I(I+1)}{3k_B T} B_0 \propto \frac{\nu_0}{T}. \tag{2.3.7}$$

This means that effectively, only a small fraction ($\propto \nu_0/T$) of the total number of spins are probed via NMR. Combining equations 2.3.5 and 2.3.7, we see that the NMR signal intensity $\mathcal{E}$ is proportional to ν_0^2/T.

2.4 Spin Echo Technique and $1/T_2$

Pulsed NMR techniques are essential to measuring the faint FID signal, which is many orders of magnitude smaller than the RF pulse used to generate $\mathbf{B}_1(t)$. This is partly because nuclear magnetic moments are ~ 1000 times smaller than those of electrons [49]. After an RF pulse is turned off, RF waves persist inside the NMR circuitry for a short time, typically a few μs or longer. This transient noise, known as ringdown, has an intensity comparable to (often much larger than) the FID signal [52].

Furthermore, there is always some spatial inhomogeneity in the static field $\Delta\mathbf{B}_0$, and hence a finite variance in the precession rates of individual nuclear spins within a sample: $\Delta\nu_0 = \gamma_N \Delta\mathbf{B}_0$. After a set of initially aligned nuclear spins are rotated by a $\pi/2$-pulse, their precessions within the xy-plane gradually desynchronize, and the amplitude of the transverse magnetization decays over time. Dipolar interactions between neighboring nuclear spins also result in the dephasing of their precessions, and is characterised by the spin-spin relaxation rate $1/T_2$. In other words, the decay rate of the transverse magnetization $1/T_2^* \approx \gamma_N \Delta\mathbf{B}_0 + 1/T_2$ has contributions from both the field inhomogeneity and nuclear spin-spin interactions, with mainly the latter being reflective of the intrinsic spin dynamics of the material [52].

These two factors motivate the widespread use of an NMR technique known as the spin echo, which refocuses desynchronized spins at a chosen time away from the

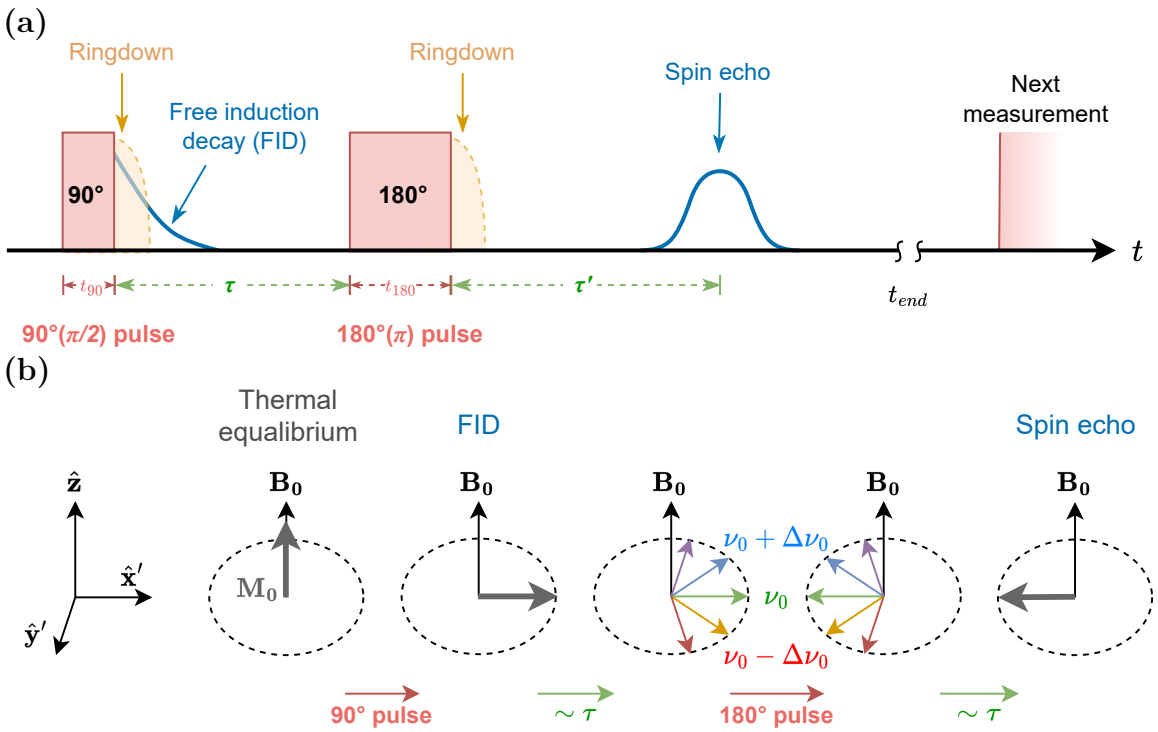


Figure 2.3: **(a)** Spin echo pulse sequence. **(b)** Decoherence and refocusing of nuclear spins during spin echo process.

RF pulses, forming a measurable signal. To perform a spin echo pulse sequence, we apply a $\pi/2$-pulse, followed by a π-pulse after a delay time τ, as illustrated in Figure 2.3 **(a)**. In the time between the RF pulses, spins with faster/slower precession rates will drift counterclockwise/clockwise in the rotating reference frame, away from each other (Figure 2.3 **(b)**). The π-pulse then inverts the spins about the $\hat{\boldsymbol{y}}'$ axis, such that the faster/slower spins are now drifting towards each other, and will realign after time τ (more precisely, the time between the end of the t_{180} pulse and the peak of the spin echo is $\tau' = \tau + t_{180}/\pi$ [48]). The decay rate of ringdown depends on the quality factor Q of the NMR tank circuit [52], and may be comparable to $1/T_2^*$. However, with sufficiently long values of τ, the spin echo signal can be separated from the

ringdown.

The intensity of the NMR lineshape at frequency f is obtained from the integrated amplitude of the spin echo signal's envelope function (see Sec. 2.8). It represents the part of the magnetization M originating from the nuclear spins with resonant frequency $\nu_0 = f$, as the signal resulting from nuclei with different resonant frequencies (outside the bandwidth) integrates to zero [52].

To diminish the relative intensity of random noise, such as electronic noise or external RF interference, we can average the spin echo signal over many repeated measurements. For a set of N scans, the sum of random noise scales with $\sqrt{N}$, which means the ratio of accumulated signal intensity to noise (SNR) is $\propto \sqrt{N}$. To ensure that $\mathbf{M}$ is in thermal equilibrium at the start of each spin echo measurement, a waiting period of t_{end}, in which no RF pulses are applied, is necessary between repeated measurements. Generally, t_{end} should be chosen such that M_z returns to within $\lesssim 1\%$ of M_0, where the speed of this process corresponds to the $1/T_1$ relaxation rate (see Sec. 2.5).

We can measure $1/T_2$ (independent of $\gamma_N \Delta \mathbf{B}_0$ contributions) by repeating a set of ($\sim$ 10-20) spin echo measurements with various delay times (10^{-6} s $\leq \tau \leq 10^{-3}$ s) and plotting the integrated spin echo intensity as a function of τ. The decay of the spin echo intensity depends on the nature of nearby nuclear spins (as well as the $1/T_1$ processes discussed in Sec. 2.5 and 2.7), but typically follows a Lorentzian $M(2\tau) \propto M_0 \exp(2\tau/T_2)$ or Gaussian $M(2\tau) \propto M_0 \exp((2\tau/T_2)^2)$ decay function, or a combination thereof [51].

2.5 $1/T_1$ Relaxation

The spin-lattice relaxation rate $1/T_1$ represents the rate at which the longitudinal component of nuclear magnetization M_z relaxes back to thermal equilibrium M_0. Since the Zeeman interaction energy ($-\boldsymbol{\mu} \cdot \mathbf{B}_0$) is not conserved in this process, it requires energy to be transferred out of the nuclear spins via perturbations from the lattice. As discussed in Sec. 2.7, $1/T_1$ probes the quantity of lattice excitations with energies corresponding to the NMR frequency (ν_0).

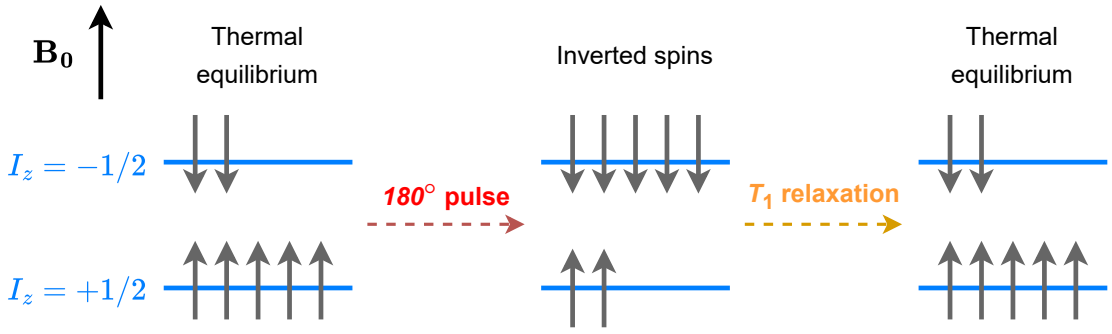


Figure 2.4: $I = 1/2$ nuclear spins inverted by a π-pulse and returning to thermal equilibrium via $1/T_1$ spin-lattice relaxation.

We can measure $1/T_1$ using an inversion recovery sequence. To do this, we initially apply a π-pulse to flip the direction of M_z, which decays toward equilibrium as

$$M_z = M_0[1 - 2\exp(t/T_1)]. \tag{2.5.1}$$

We then perform a standard spin echo sequence after some delay time t_D. If $t_D \gg T_1$, the inverted nuclear spins will have relaxed back to equilibrium by the start of the spin echo sequence (Figure 2.5 **(a)**). On the other hand, if $t_D \ll T_1$, the spin echo pulses are applied while M_z is still inverted, and will instead produce a negative

signal (Figure 2.5 **(b)**). We can therefore determine $1/T_1$ by measuring the spin echo intensity as a function of t_D, which we usually vary over a range of logarithmically-spaced values (10^{-6} s $\leq t_D \leq 10^{1}$ s).

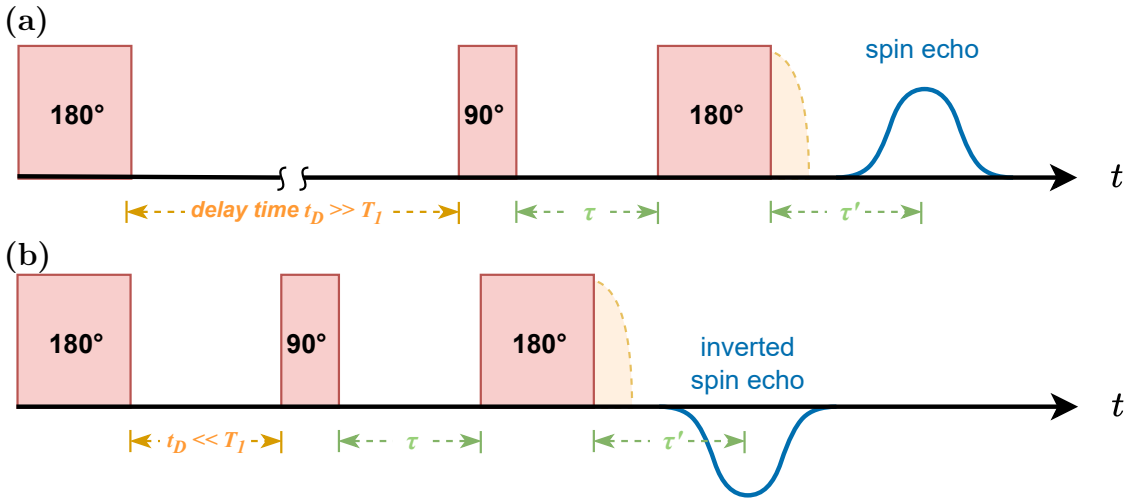


Figure 2.5: $1/T_1$ inversion recovery pulse sequences for delay times **(a)** $t_D \gg T_1$, resulting in a normal spin echo, and **(b)** $t_D \ll T_1$, resulting in an inverted spin echo.

An alternate method of measuring $1/T_1$, known as saturation recovery, involves randomizing the nuclear spins via a series (a.k.a "comb") of $\pi/2$-pulses, then performing a spin echo sequence after t_D. The time t_{comb} between the comb pulses should be chosen such that $T_2^* < t_{comb} \ll T_1$ [52]. This limits the feasibility of saturation recovery, as the nuclear spins require enough time to dephase between comb pulses, but not enough time to relax to equilibrium. On the other hand, saturation recovery has the advantage of not having to wait t_{end} between measurements, as the comb pulses will saturate the magnetization regardless its initial direction. We employed inversion recovery in our ^{7}Li NMR and ^{63}Cu NQR experiments (Chpt. 4 and 5) and saturation recovery in our ^{19}F NMR experiments (Chpt. 6).

As shown in Figure 2.6 **(a)**, the value of $1/T_1$ can be determined by fitting the

integrated spin echo intensity (as a function of t_D) with an exponential function

$$M(t_D) = M_0[1 - \lambda \exp(-t_D/T_1)], \tag{2.5.2}$$

where, under ideal conditions, $\lambda = 2$ for inversion recovery and $\lambda = 1$ for saturation recovery. In practice, λ is fitted as free parameter, since the inversion/saturation of M_z is often incomplete.

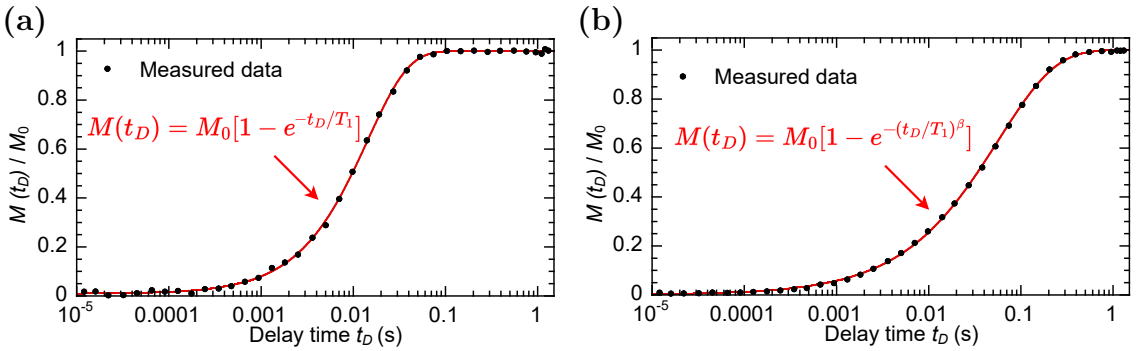


Figure 2.6: **(a)** Normalized $M(t_D)$ data fitted with a single exponential function. **(b)** $M(t_D)$ data resulting from a wide distribution of $1/T_1$ components, fitted with a stretched exponential function ($\beta = 0.7$). Note that the t_D axis is on a log scale.

Recall that an NMR measurement conducted at frequency f excites and probes the response of all nuclei whose resonant frequency ν_0 are close to f. This means that the net nuclear magnetization curve $M(t_D)$ measured at f is composed of all of the individual magnetic moments of nuclei with $\nu_0 \approx f$, which can be situated in environments with different local $1/T_1$. Therefore, $M(t_D)$ is typically not in the form of a single exponential (Eq. 2.5.2), but equals the sum of many exponential functions with different values of $1/T_1$. NMR experimentalists often account for a broad distribution of $1/T_1$ components by fitting $M(t_D)$ with a phenomenological

stretched exponential function

$$M(t_D) = M_0[1 - \lambda \exp\left(-(t_D/T_{1,\mathrm{str}})^{\beta}\right)], \tag{2.5.3}$$

where $1/T_{1,\mathrm{str}}$ is a characteristic relaxation rate meant to represent the sample-averaged $1/T_1$, and $0 \leq \beta \leq 1$ is the stretch parameter. The case of $\beta = 1$ corresponds to a single exponential, while lower values of β correspond to broader (more inhomogeneous) distributions of $1/T_1$, as shown in Fig. 2.6 **(b)**. It's important to keep in mind that the stretch fit method is a working ansatz that assumes a specific form of $M(t_D)$. Alternatively, $M(t_D)$ can be fitted with a predetermined set of exponential functions to extract the value(s) of $1/T_1$. In any case, these methods all require either knowing or assuming the form of the $1/T_1$ distribution, which can produce highly misleading results. We will provide a more detailed discussion of these issues in Chpt. 3, and demonstrate how the limitations of traditional $1/T_1$ analysis can be overcome by instead directly computing the $1/T_1$ distribution from the $M(t_D)$ data.

2.6 Quadrupole Interactions and NQR

The nucleus is composed of a spatial distribution of electric charges, which can be expressed by spherical harmonics of order l, and interacts with electrostatic potential. The monopole ($l = 0$) moment of the nucleus is simply equal to its charge Ze, and the Coulomb energy is independent of its spin state. The electric dipole ($l = 1$) moment, as well as all charge multipoles with odd l, vanish due to the well-defined parity of the nuclear states [48].

The electric quadrupole ($l = 2$) moment of the nucleus is a 3×3 tensor operator

that's dependent on the nuclear spin state, and couples with the electric field gradient (EFG) via the electric quadrupole interaction. The EFG is defined as the second partial derivatives of the electrostatic potential at the position of the nucleus, which is a symmetric, traceless tensor $V_{\alpha\beta} = (\delta^2 V)/(\delta\alpha\delta\beta)$. $V_{\alpha\beta}$ can be diagonalized to be defined along its principal axes $\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z}$, such that

$$V_{ZZ} > V_{YY} > V_{XX} = 0, \qquad \text{and} \qquad V_{ZZ} + V_{YY} + V_{XX} = 0. \tag{2.6.1}$$

The EFG is generally exerted by the nucleus' unpaired orbital electrons and neighboring ions, where the latter can be calculated from lattice charge simulations [53]. Higher order moments ($l > 3$) may also exist, but the interaction energy rapidly falls off $\sim (10^{-5})^l$. Therefore the quadrupole interaction dominates the electrostatic effect on the energy separations between nuclear spin states.

The nuclear quadrupole moment can be defined using a single constant

$$eQ = \frac{1}{2}\int (3z^2 - r^2)\rho(\boldsymbol{r}) d^3\boldsymbol{r} \tag{2.6.2}$$

where $\rho(\boldsymbol{r})$ is the charge density at position $\boldsymbol{r}$ of the nucleus in its maximum magnetic eigenstate [48]. The Hamiltonian of the quadrupole interaction is then defined as

$$\hat{\mathcal{H}}_Q = h\nu_Q^Z[3I_Z^2 - I(I+1) + \frac{1}{2}\eta(I_+^2 + I_-^2)], \tag{2.6.3}$$

where I_Z is the nuclear spin operator along the main principal axis $\boldsymbol{Z}$, and I_+ and I_- are the raising and lowering operators. The quadrupole frequency and the asymmetry

parameter are, respectively,

$$\nu_Q^Z = \frac{V_{ZZ}eQ}{4I(2I-1)h}, \qquad \text{and} \qquad \eta = \left|\frac{V_{YY}-V_{XX}}{V_{ZZ}}\right|. \tag{2.6.4}$$

The quadrupole interaction does not affect the energy of $I = 1/2$ nuclear spin states. While for $I = 3/2$ nuclei in the high-field limit ($\gamma_N B_{\text{ext}} \gg \nu_Q^Z$), $\hat{\mathcal{H}}_Q$ acts as a perturbation which shifts the $m \leftrightarrow m+1$ transition energies, splitting the NMR lineshape into 3 separate peaks [49]. However, in the case of ^{7}Li NMR in $Ag_3LiIr_2O_6$, quadrupole splitting is negligible due to its low quadrupole moment and the high symmetry of charges surrounding the ^{7}Li sites (Figure 2.7 **(a)**).

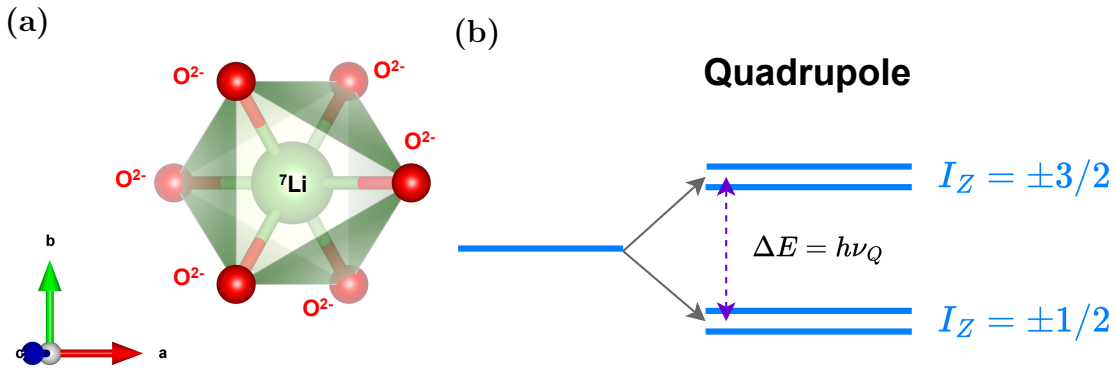


Figure 2.7: **(a)** ^{7}Li site at the center of an octahedron formed by O^{2-} ions, such that $V_{\alpha\beta} \approx 0$ at the position of the ^{7}Li nucleus. **(b)** $m = \pm 1/2$ and $m = \pm 3/2$ states of a $I = 3/2$ nucleus split by the quadrupole interaction.

In contrast to NMR, nuclear quadrupole resonance is performed in $B_{\text{ext}} = 0$, where instead of the Zeeman interaction, the nuclear spin Hamiltonian is dominated by the quadrupole interaction. For $I = 3/2$ nuclei, the NQR resonance frequency equals

$$\nu_Q = \nu_Q^Z(1+\eta^2/3), \tag{2.6.5}$$

and represents the energy separation between the degenerate $m = \pm 1/2$ and $m = \pm 3/2$ states (Figure 2.7 **(b)**). NQR can be greatly advantageous for probing things such as the $1/T_1$ of $I = 3/2$ nuclei in a powder sample (i.e. many randomly oriented crystals), as it avoids the large, orientation-dependent quadrupole splitting that would be present in high-field NMR [49]. Instead, the width of the NQR lineshape (of a given isotope) reflects the variation of local charge environments.

For NQR of $I = 3/2$ nuclei, the $1/T_1$ recovery curve $M(t_D)$ follows the exponential function

$$M(t_D) = M_0[1 - \lambda \exp\left(-(3t_D/T_1)^{\beta}\right)], \tag{2.6.6}$$

where the pre-factor of 3 is due to the rate of transitions between the $m = \pm 1/2 \leftrightarrow \pm 3/2$ spin states, and is calculated in Ref. [54, 55]. In general, the decay curve $M(t_D)$ corresponding to the quadrupole-split $m \longleftrightarrow m + 1$ transitions of a spin-I nucleus has functional form

$$M(t_D) = M_0 \sum_k [1 - p_k \lambda \exp(-q_k t_D/T_1)], \tag{2.6.7}$$

where there exist up to $2I$ normal modes characterised by the coefficients p_k and q_k. For each type of transition that we probe, p_k and q_k can be calculated from the transition probabilities $W_{m \longleftrightarrow m+1}$ between all of the adjacent spin states [54]. In the case of $I = 1/2$ NMR and $I = 3/2$ NQR, $\{p_k, q_k\}$ equals $\{1, 1\}$ and $\{1, 3\}$, respectively.

2.7 Hyperfine Form Factor

The general expression for the hyperfine form factor A_{hf} is the wavevector-dependent $A_{hf,\alpha}(\boldsymbol{q})$, which quantifies the geometry of the hyperfine interactions between nuclear and electron spins. It's defined as

$$A_{hf,\alpha}(\boldsymbol{q}) = \sum_i A_\alpha^i e^{i\boldsymbol{q}\cdot\boldsymbol{r}_i}, \tag{2.7.1}$$

where the summation is over all electron spins i, and A_α^i is the hyperfine coupling strength along α ($\alpha = x, y, z$) between a nucleus and an electron at position $\boldsymbol{r}_i$ relative to the nucleus.

The on-site hyperfine field may be a combination of a variety of contributions [56]

$$A_{hf,\alpha} = A_{contact} + A_{cp} + A_{dip-dip,\alpha} + A_{spin-orbit,\alpha}. \tag{2.7.2}$$

The first term $A_{contact}$ is the isotropic Fermi contact interaction, originating from the probability of electrons in s-like orbitals to be at the position of the nucleus, which would result in a relatively large interaction energy. Its Hamiltonian can be expressed as

$$\hat{\mathcal{H}}_{contact} = \frac{8\pi}{3}\gamma_e\gamma_N\hbar^2\hat{\boldsymbol{I}}\cdot\hat{\boldsymbol{S}}|\Psi(0)|^2, \tag{2.7.3}$$

where, $\hat{\boldsymbol{S}}$ is the electron spin operator, γ_e is the electron gyromagnetic ratio, and $\Psi(0)$ is the value of the electron wavefunction at the position of the nucleus [57]. Polarized outer electrons do not directly contribute to $A_{contact}$ due to their vanishing probability density at the position of the nucleus. However, they can polarize the

inner (core) s-like electrons via direct exchange interactions, leading to the isotropic (typically negative) core-polarization term A_{cp} [56].

The dipole-dipole interaction between nuclear and non-s electron spins is defined by the Hamiltonian

$$\hat{\mathcal{H}}_{dip-dip} = \frac{\gamma_e \gamma_N \hbar^2}{r^3}[2\frac{(\hat{\boldsymbol{I}} \cdot \hat{\boldsymbol{S}})(\hat{\boldsymbol{I}} \cdot \hat{\boldsymbol{S}})}{r^2} - \hat{\boldsymbol{I}} \cdot \hat{\boldsymbol{S}}], \tag{2.7.4}$$

which averages to zero for spherically symmetric orbitals of s-electrons, leading to the anisotropic term $A_{dip-dip,\alpha}$. $A_{spin-orbit,\alpha}$ is generated by the orbital moment of electrons polarized by its spin via spin-orbit coupling.

A nucleus can also experience a transferred hyperfine field $B_{transfer}$ originating from neighboring magnetic sites, due to the mixing of states between the nucleus' s-electrons and the polarized electrons nearby [56]. For example, the hyperfine fields at the non-magnetic Li^+ sites in $Ag_3LiIr_2O_6$, and F^- sites in barlowite are generated primarily by the neighboring 6 spin-1/2 Ir^{4+} ions and 12 spin-1/2 Cu^{2+} ions, respectively, through the transferred hyperfine interaction. Note that the $\boldsymbol{q}$-dependence of the hyperfine form factor in Eq. 2.7.1 is due to transferred hyperfine interactions with off-site electrons, as the on-site hyperfine interaction would have $\boldsymbol{r}_i = 0$ and hence no $\boldsymbol{q}$-dependence.

In both NMR and NQR, $1/T_1$ measures the strength of the local fluctuating hyperfine magnetic field along the directions perpendicular to the quantization axis of the nuclear spins. This means $1/T_1$ is related to the dynamic structure factor $S(\boldsymbol{q}, \omega)$

of electron spins at the NMR resonant frequency $\omega_0 = 2\pi\nu_0$, such that

$$\frac{1}{T_1} = \frac{\gamma_N^2 \hbar}{2\mu_B^2} \sum_{\boldsymbol{q}} (|A(\boldsymbol{q})_{\perp,1}|^2 + |A(\boldsymbol{q})_{\perp,2}|^2) S(\boldsymbol{q}, \omega_0), \tag{2.7.5}$$

where $\boldsymbol{q}$ is summed over the first Brillouin zone, and $A(\boldsymbol{q})_{\perp,1}$ and $A(\boldsymbol{q})_{\perp,2}$ are the components of the hyperfine form factor perpendicular to the quantization axis (e.g. 1,2 correspond to $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$ in NMR and $\hat{\boldsymbol{X}}$, $\hat{\boldsymbol{Y}}$ in NQR) [58].

The dynamic structure factor is the space-time Fourier transform of the electron spin-spin correlation function $\langle S(0,0)S(\boldsymbol{r},t)\rangle$[59], and can be expressed as

$$S(\boldsymbol{q}, \omega) = \frac{1}{2\pi} \sum_i \int_{-\infty}^{+\infty} \langle S(0,0)S(\boldsymbol{r},t)\rangle * e^{i(\boldsymbol{q}\cdot\boldsymbol{r}_i - \omega t)} dt, \tag{2.7.6}$$

where the sum is over all electrons i at position $\boldsymbol{r}_i$ relative to the nucleus. In essence, $1/T_1$ probes the $\boldsymbol{q}$-summation of the local low-energy component ($\hbar\omega_0 \sim 0.1$ mK) of the excitation spectrum, whose distribution in real space determines the distribution of $1/T_1$. Meanwhile, reciprocal space probes such as inelastic neutron scattering can measure the sample-averaged $S(\boldsymbol{q}, \omega)$ as a function of energy and momentum.

2.8 NMR Electronics

The experiments in this thesis were performed using commercially available equipment and electronics constructed by Prof. Takashi Imai and previous research group members, who have assembled 2 interchangeable NMR setups using the Tecmag Apollo or Tecmag Redstone as the digital spectrometers. A block diagram of a typical NMR setup is shown in Figure 2.8.

Using the Apollo/Redstone spectrometer, we generate RF pulses which are transmitted through 50 Ω coaxial cables and amplified by a factor of ∼1000 (60dB) by a power amplifier (Kalmus LP1000 1kW). The trigger line of the power amplifier is controlled by an additional "unblanking" transistor-transistor logic (TTL) signal generated by the spectrometer. Before entering the NMR probe, the RF pulses go through a series of paired (back-to-back) diodes, which blocks out any low-voltage noise that may leak from the power amplifier during signal acquisition.

The NMR probe consists of an RLC tank circuit with two variable capacitors, a solenoid (inductance L) containing the sample, and a Q-damping resistor (resistance R) used to control the quality factor Q of the tank circuit (and minimize its temperature dependence). The probe is placed inside a cryostat (Janis Model 11CNDT-SVT) and cooled below room temperature down to $T = 4.2$ K, using liquid N_2 and ^{4}He. Temperatures below 4.2 K (down to 1.7 K) are achieved via evaporative cooling of liquid ^{4}He. For ^{7}Li ($B_{\text{ext}} = 4.5$ T) and ^{19}F ($B_{\text{ext}} = 2.4$ T) NMR experiments, the solenoid and sample are located at the field center of a superconducting electromagnet.

To maximize power transferred to the sample, the capacitance of the tuning (C_T) and matching (C_M) capacitors are manually adjusted for each measured frequency f such that

$$f = \frac{1}{2\pi\sqrt{L(C_M + C_T)}}, \qquad \text{and} \qquad 50\ \Omega = \frac{\omega^2 L^2 (1 + C_M/C_T)^2}{R}, \tag{2.8.1}$$

which matches the frequency and impedance of the RF pulses with the resonant frequency and impedance of the tank circuit. Further details on the NMR tank circuit and effect of Q-damping can be found in Appendix C of Ref. [60].

To detect the faint ($\sim \mu$V) NMR signal generated from nuclear precession, it is amplified by up to 2 RF pre-amplifiers (DOTY LN-2L or LN-2M), and sometimes a 3rd (Mini-circuits ZFL-500LN) amplifier.

These sensitive amplifiers are protected from the high voltage ($\sim$ 300 V) RF pulses by a quarter-wavelength ($\lambda/4$) cable and a set of paired diodes connected to ground. These diodes short any voltage above a specific threshold ($\sim$0.6 V) while allowing past the NMR signal. The $\lambda/4$ cable is inserted so that the RF pulse forms a standing wave with a node located at the grounded diodes, maximizing power transmitted to the probe. For typical 50 Ω coaxial cables ($c \sim 1.8 * 10^8$ m/s), $\lambda/4 \sim 45$ m$/f$, where f is in units of [MHz] [60].

Past the first pre-amplifier are pair of RF mixers, which act as switches to prevent unnecessary saturation of the signal amplifiers and limit the propagation of ringdown within the signal detection circuitry. The switches are triggered by the spectrometer and only turned on during the signal acquisition time window.

Finally, the amplified signal is digitized by the spectrometer. This is traditionally done via a phase-sensitive detection technique known in the NMR literature as quadrature detection [52]. The NMR signal is split into two identical halves, which are mixed with 0°and 90°phase-shifted reference signals created from a continuous RF source. From this, we obtain a pair of near-DC signals (named as the the real and imaginary channels) corresponding to the envelope functions of the sine and cosine components of the original signal.

Another important technique in pulsed NMR, called phase cycling, is used to significantly reduce the cumulative ringdown intensity. Phase cycling relies on the fact that the phase of the ringdown and spin echo are determined by the phase of

the $\pi/2$-pulse and π-pulse. Since the time-evolution of ringdown is dependent on the NMR electronics, its phase is a constant shift from the RF pulse that generated it. On the other hand, we can imagine the phases of the spin echo and RF pulses as corresponding to the directions of $\boldsymbol{M}(t)$ and $\mathbf{B}_1(t)$, respectively, in the rotating reference frame. Based on the phases of the RF pulses, we can predict the phase of the resulting spin echo.

To preform phase cycling, we repeat a cycle of 4 spin echo sequences according to Table 2.2, and add/subtract the measured signal, such that the NMR signal intensity is combined while the ringdown cancels out [61].

$\pi/2$-pulse	π-pulse	Spin echo
$+\hat{\boldsymbol{x}}'$	$+\hat{\boldsymbol{x}}'$	$+\hat{\boldsymbol{y}}'$ (add)
$-\hat{\boldsymbol{x}}'$	$+\hat{\boldsymbol{x}}'$	$-\hat{\boldsymbol{y}}'$ (subtract)
$-\hat{\boldsymbol{x}}'$	$+\hat{\boldsymbol{y}}'$	$+\hat{\boldsymbol{y}}'$ (add)
$+\hat{\boldsymbol{x}}'$	$+\hat{\boldsymbol{y}}'$	$-\hat{\boldsymbol{y}}'$ (subtract)

Table 2.2: Phase cycling table for a standard spin echo sequence.

An additional set of phases (8 total) is needed for the $1/T_1$ inversion recovery sequence, which alternates the phase of the initial π-pulse between $+\hat{\boldsymbol{x}}'$ and $-\hat{\boldsymbol{x}}'$. This is to cancel out the stimulated echos resulting from imperfect pulse lengths [61].

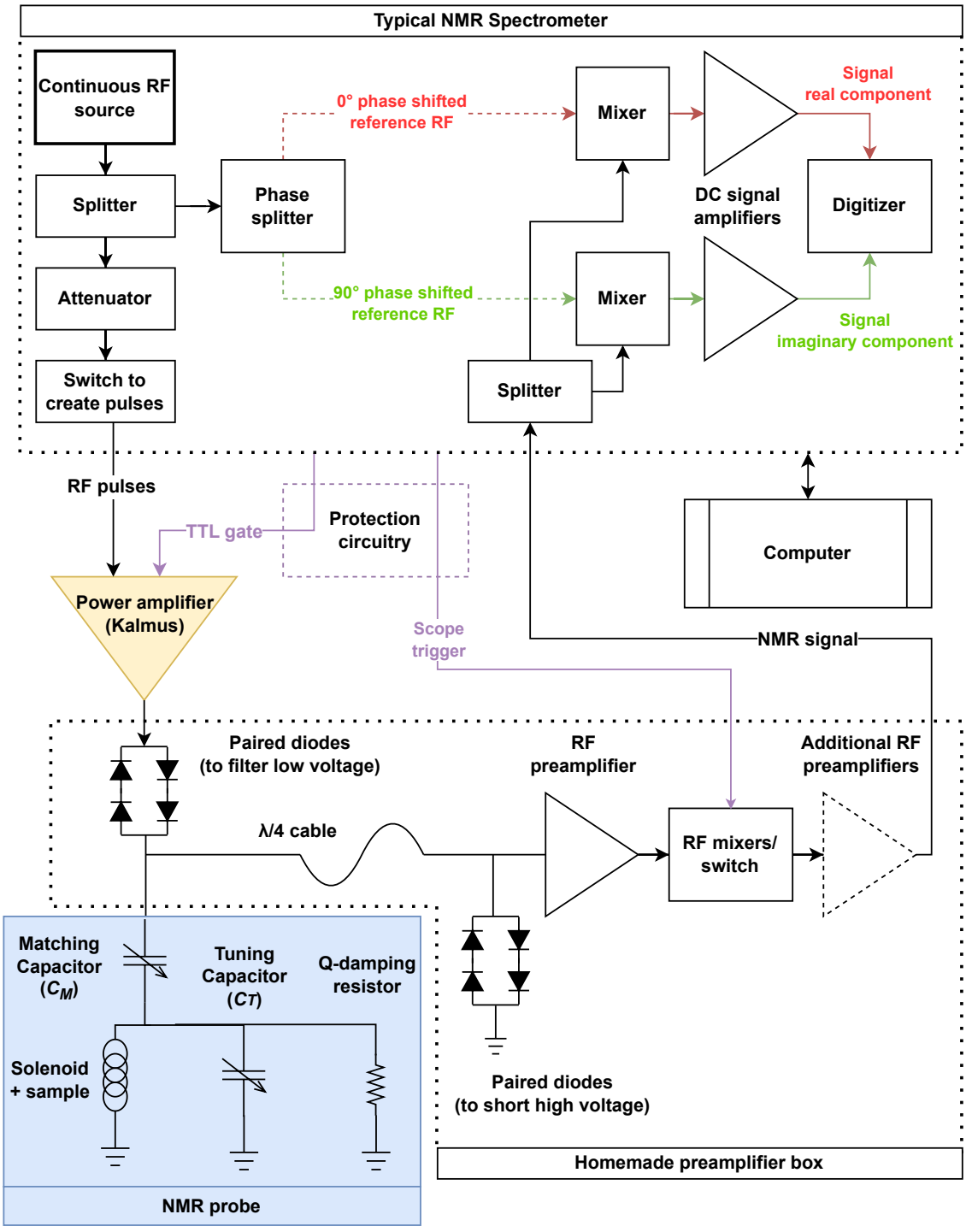


Figure 2.8: Block diagram of a typical NMR spectrometer setup.

Chapter 3

Inverse Laplace Transform

The subset of nuclei probed by a single NMR measurement can be situated in a variety of environments throughout the lattice. The spin-lattice relaxation rate ($1/T_1$) of these nuclei depend on their local electron spin dynamics and/or hyperfine interactions, which can differ greatly in a spatially inhomogeneous material. Assuming these nuclei relax independently of each other, the decaying nuclear magnetization $M(t_D)$ during a $1/T_1$ measurement can be thought of as the sum of many single exponential functions (Eq. 2.5.2), each corresponding to the nuclei exhibiting a specific value of $1/T_1$. With this in mind, we can express $M(t_D)$ as an integral over the nuclear ensemble

$$M(t_D) = \int P(1/T_1)[1 - \lambda \exp(-t_D/T_1)] d\log_{10}(1/T_1), \tag{3.0.1}$$

where $P(1/T_1)$ is the probability density distribution of $1/T_1$. We opt to define $P(1/T_1)$ on a logarithmic scale, as the local $1/T_1$ can range by multiple orders of magnitude within a sample. This means that $P(1/T_1)d\log_{10}(1/T_1)$ corresponds to

the fraction of probed nuclei with relaxation rate $1/T_1$, and adds up to

$$\int P(1/T_1) d\log_{10}(1/T_1) = M_0. \tag{3.0.2}$$

In this chapter, we show how $P(1/T_1)$ can be deduced based on the numerical inverse Laplace transform (ILT) of the measured $M(t_D)$ data, and demonstrate the many advantages of this $1/T_1$ analysis method. ILT is widely employed in NMR of organic materials and porous media, and has significant applications in petrophyics [62–67]. More recently, NMR research on disordered quantum materials used ILT $1/T_1$ analysis to resolve the multiple $1/T_1$ components resulting from their locally inhomogeneous spin dynamics [45–47, 68–71]. In these studies, ILT was able to provide some key insights which were overlooked by traditional $1/T_1$ analysis, as it helped to differentiate the behaviour of nuclear environments affected by different levels of disorder.

3.1 Phenomenological Forms of $M(t_D)$

To understand the motivation for ILT, we start by discussing the more rudimentary methods for analyzing $M(t_D)$ data, mentioned in Sec. 2.5. First, recall that $M(t_D)$ can be fitted with the stretched exponential function defined in Eq. 2.5.3 to obtain the parameters $1/T_{1,\text{str}}$ and β. As an example, we plot the $M(t_D)$ for a stretched exponential with $1/T_{1,\text{str}} = 100$ s^{-1} and $\beta = 0.5$, and corresponding $1/T_1$ distribution (that is, the $P(1/T_1)$ which yields Eq. 2.5.3 when plugged into Eq. 3.0.1) in Fig. 3.1 **(a)** and **(b)**, respectively, and compare them to those of a single exponential with $1/T_1 = 100$ s^{-1}.

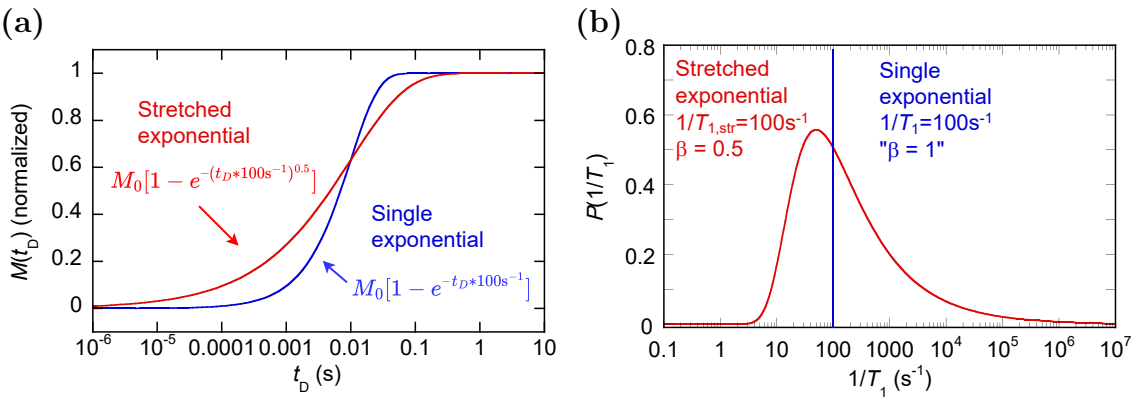


Figure 3.1: **(a)** Relaxing nuclear magnetization $M(t_D)$ in the form of a single exponential with $1/T_1 = 100\text{s}^{-1}$ (blue) vs a stretched exponential with $1/T_{1,\text{str}} = 100\text{s}^{-1}$ and $\beta = 0.5$ (red). **(b)** Corresponding distributions $P(1/T_1)$, plotted with $1/T_1$ on a logarithmic scale. Note that $P(1/T_1)$ of the single exponential is a Dirac δ function.

$1/T_{1,\text{str}}$ is often considered in the NMR literature as a "characteristic" relaxation rate of $M(t_D)$ that ostensibly represents the spatial average of $1/T_1$, while β reflects the extent to which $1/T_1$ is distributed. In actuality, the physical significance of $1/T_{1,\text{str}}$ and β are not well-defined. The $P(1/T_1)$ of stretched exponential functions are shown analytically in Ref. [72] for all rational values of β. Generally speaking, the $P(1/T_1)$ distribution of the stretched exponential appears as a peak whose mean, median, and mode are related, but not necessarily equal to $1/T_{1,\text{str}}$. Meanwhile, β is anti-correlated with the width of $P(1/T_1)$. As β increases towards $\beta = 1$, $P(1/T_1)$ narrows and approaches that of a single exponential (i.e. a Dirac δ function indicated by the vertical blue line in Fig. 3.1 **(b)**).

Certain disordered materials are known to exhibit a $M(t_D)$ following a stretched exponential function [73–75]. Through modeling the inhomogeneous enhancement of $1/T_1$ by random magnetic defects in these materials, the stretched exponential form

of $M(t_D)$ with $\beta = 0.5$ can be derived for the limiting case of low defect concentration and long t_D [75, 76]. Usually however, the stretched exponential is just a phenomenological description of the measured $M(t_D)$ behaviour.

$M(t_D)$ is evidenced to follow a stretched exponential if a log-log plot between $\log(M_0 - M(t_D))$ and t_D yields a straight line, in which case a stretch fit of $M(t_D)$ may be somewhat justified. Even if such justification is absent, one can often fit any arbitrary $M(t_D)$ relaxation data with a stretched exponential reasonably well. As demonstrated by examples in Sec. 3.4 and Ref. [68], the resulting $1/T_{1,\mathrm{str}}$ still acts as a rough estimate of the $P(1/T_1)$ center of gravity (defined in Eq. 3.4.2 below), while β informs us of its logarithmic standard deviation (defined in Eq. 3.4.3 below). These properties make it convenient to use stretch fit analysis when dealing with inhomogeneous $1/T_1$ distributions. In studies on frustrated magnetism over the past few decades, it has become increasingly popular for NMR experimentalists to rely on the stretch fit [41, 77]. However, applying the stretch fit without justification is highly problematic as it naively assumes $P(1/T_1)$ corresponds to that of a stretched exponential, and neglects its actual form. Worse still, it is not uncommon for $1/T_{1,\mathrm{str}}$ to be reported simply as $1/T_1$, which can unintentionally give the false impression that there is no $1/T_1$ distribution and $1/T_1 = 1/T_{1,\mathrm{str}}$ uniformly across the sample.

Beyond the stretched exponential, there exists other forms of $M(t_D)$ that commonly appear in disordered materials. For example, fitting $M(t_D)$ as the sum of two single exponential components can work well in cases where there are two distinct types of nuclear environments [78]. In any case, fitting $M(t_D)$ with a predefined function inherently requires making a significant assumption about the form of $P(1/T_1)$. Whether these assumptions are phenomenological or theoretically justified, there is

always the risk of overlooking an unexpected $1/T_1$ component. Fortunately, we do not have to decide on the specific form of $P(1/T_1)$ because it can instead be extracted from the measured $M(t_D)$ data. The purpose of ILT is then to calculate $P(1/T_1)$ from $M(t_D)$, in a way analogous to how an electromagnetic signal can be decomposed into its oscillatory components through its Fourier transform.

3.2 Numerical Model of $M(t_D)$ Data

Given $M(t_D)$, our goal is to find a non-negative distribution $P(1/T_1) \geq 0$ that satisfies Eq. 3.0.1. The process of numerically inverting $M(t_D)$ to obtain $P(1/T_1)$ has become commonly referred to in NMR community as calculating the ILT of $M(t_D)$. This is because in Eq. 3.0.1, $M(t_D)$ resembles the Laplace transform of $P(1/T_1)$, although technically it is a Fredholm integral equation of the first kind [79]).

Computing the ILT requires redefining Eq. 3.0.1 so that $P(1/T_1)$ and $M(t_D)$ are represented by discrete sets of data. We first define the histogram $P(1/T_{1,j})$, which groups $P(1/T_1)$ into n ($\sim$250) logarithmically-spaced $1/T_1$ bins spanning from $1/T_{1,\min}$ to $1/T_{1,\max}$

$$\{1/T_{1,j}\} = \{1/T_{1,\min} * 10^{(j-1)\Delta_P} \ : \ j = 1 \ldots n\}, \tag{3.2.1}$$

where $\Delta_P = \frac{1}{n-1} \log_{10}(T_{1,\max}^{-1}/T_{1,\min}^{-1})$ is the logarithmic bin separation. We choose the range ($1/T_{1,\min}$ to $1/T_{1,\max}$) $\sim$ (10^{-3} s^{-1} to 10^5 s^{-1}) to encompass all of the $1/T_1$ components detectable by our given range of t_D values, extended by an additional buffer of $\gtrsim 1$ decade. $M(t_D)$ is measured as a set of data points, using a series of m inversion (or saturation) recovery sequences with delay times $t_D = \{t_i \ : \ i = 1 \ldots m\}$

chosen by the experimenter. Typically, m ranges from 16 to 42. This data $M(t_i)$ will also be superimposed by some experimental noise $E(t_i)$. Finally, note that we have so far ignored quadrupole splitting, and Eq. 3.0.1 assumes the case of $I = 1/2$ NMR. To account for multiple normal modes, we replace the single exponential with its more general form: Eq. 2.6.7.

We can now write the discrete form of Eq. 3.0.1 as

$$M(t_i) = \sum_{j=1}^{n} P(1/T_{1,j})\Delta_P \sum_k [1 - \lambda p_k \exp(-q_k t_i/T_{1,j})] + E(t_i), \tag{3.2.2}$$

which we can express in column vector form as

$$\mathbf{M} = \boldsymbol{K}\mathbf{P} + \mathbf{E}, \quad \mathbf{P} \geq 0, \tag{3.2.3}$$

where

$$\begin{aligned} \mathbf{M} &= \{M(t_i) \,:\, i = 1 \ldots m\}, \\ \mathbf{P} &= \{P(1/T_{1,j})\Delta_P \,:\, j = 1 \ldots n\}, \\ \mathbf{E} &= \{E(t_i) \,:\, i = 1 \ldots m\}, \end{aligned} \tag{3.2.4}$$

and

$$\boldsymbol{K} = \{K_{ij}\} = \{\sum_k [1 - \lambda p_k \exp -q_k t_i/T_{1,j}] \,:\, i = 1 \ldots m, j = 1 \ldots n\} \tag{3.2.5}$$

is the matrix of the kernel function [65]. We can quantify $\mathbf{E}$ in terms of the SNR by its standard deviation

$$\sigma_E = ||\mathbf{E}|| = \frac{M_0}{SNR}, \tag{3.2.6}$$

where $||\dots||$ is the vector norm.

To solve for $\mathbf{P}$, we optimize it to fit the equation $\mathbf{M} = \boldsymbol{K}\mathbf{P}$, under the constraint that $\mathbf{P} \geq 0$. We can write this in terms of a cost function as

$$\mathbf{P} = \underset{\mathbf{P}\geq 0}{\arg\min} ||\mathbf{M} - \boldsymbol{K}\mathbf{P}||^2. \tag{3.2.7}$$

In other words, given $\mathbf{M}$ and $\boldsymbol{K}$, our goal is to find a vector $\mathbf{P}$ with non-negative elements $P(1/T_{1,j}) \geq 0$, such that $||\mathbf{M} - \boldsymbol{K}\mathbf{P}||^2$ is minimized. The process of solving for $\mathbf{P}$ belongs to the class of constrained optimization problems, and can be performed using a non-negative least squares algorithm [80].

3.3 Tikhonov Regularization

We ideally want our solution to be unique and stable, like in the case of computing the discrete Fourier transform (and its inverse) using algorithms such as the Fast Fourier Transform. However, the main difficulty inherent to ILT is that it is considered to be an "ill-posed" problem, in which $\mathbf{P}$ varies drastically and discontinuously with the experimental noise $\mathbf{E}$. Eq. 3.2.7 has many over-fitted, near-exact solutions that don't resemble realistic $1/T_1$ distributions (instead appearing like the distribution in Fig. 3.2 **(b)**). To obtain a physically meaningful picture of the $1/T_1$ distribution, we should look for solutions $\mathbf{P}$ which are in some ways "simpler" (i.e. regularization), and deter them from attempting to fit the experimental noise.

Starting off, the direct non-negative least squares fit can be improved by finding the low-rank approximation of the kernel matrix $\boldsymbol{K} \longrightarrow \tilde{\boldsymbol{K}}$ via truncated singular value decomposition (SVD). An outline of this process is summarized in Appendix

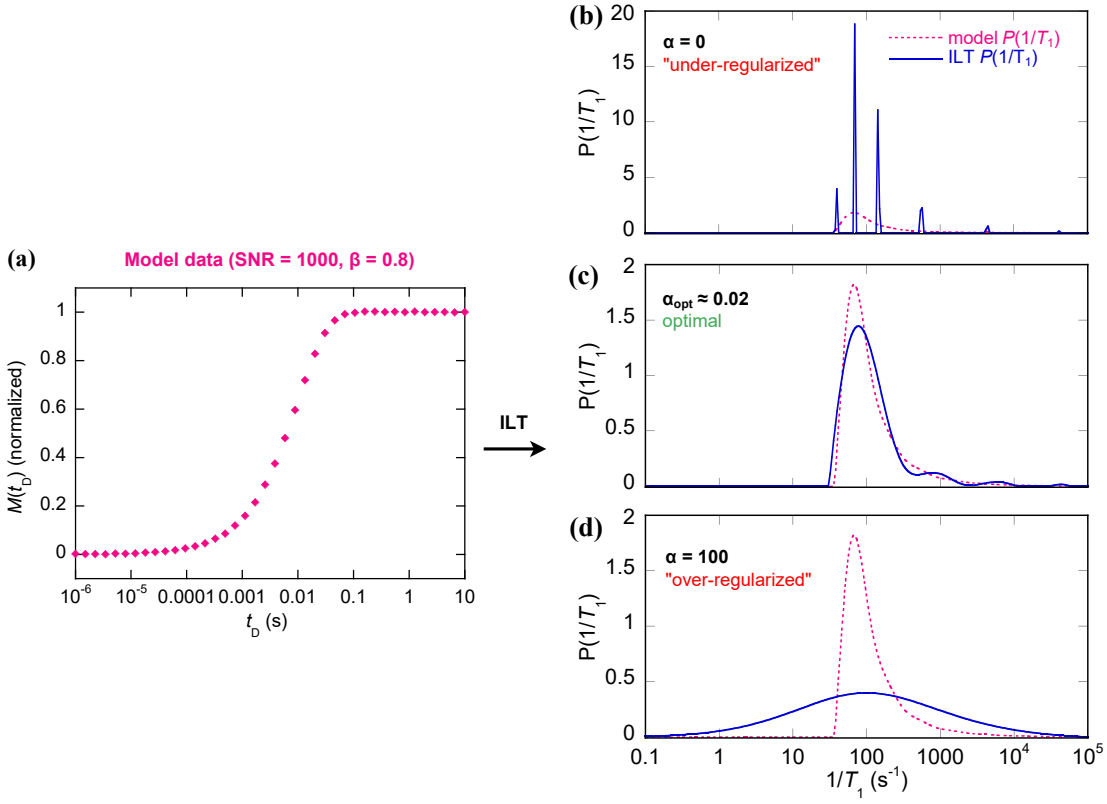


Figure 3.2: **(a)** Synthetic $M(t_D)$ data generated from stretched exponential model function $M(t_D) = 1 - \exp\left(-(t_D * 100\ \mathrm{s}^{-1})^{0.8}\right) + 0.1\%$ noise. **(b-d)** Examples of $P(1/T_1)$ (solid blue curves) computed via Tikhonov Regularization with **(b)** "under-regularized", **(c)** optimal, and **(d)** "over-regularized" values of α, overlaid with the theoretical $1/T_1$ distribution of the model stretched exponential function (dashed pink curves).

A. We do this as a preliminary step in our analysis, as it reduces computing time and provides a basic filter for statistical noise [63].

There exists various approaches to dealing with ill-posed inversion problems, including those based on Tikhonov regularization [81], maximum entropy [82], Monte Carlo [83], and the Mellin transform [84]. We opt to use Tikhonov regularization because of the extensive literature on the topic, as well as its robustness against systematic errors [65]. In Tikhonov regularization, we augment Eq. 3.2.7 with a

"smoothing" parameter $\alpha > 0$, such that

$$\mathbf{P} = \underset{\mathbf{P}\geq 0}{\arg\min} \, ||\mathbf{M} - \boldsymbol{K}\mathbf{P}||^2 + \alpha||\mathbf{P}||^2. \tag{3.3.1}$$

The added term $\alpha||\mathbf{P}||^2$ incentivizes for solutions $\mathbf{P}(\alpha)$ whose vector norms $||\mathbf{P}||$ are smaller, by being more smoothly distributed. The degree of smoothing depends on the choice of α, which should be large enough for $\mathbf{P}(\alpha)$ to be qualitatively unaffected by experimental noise.

In Fig. 3.2 **(b)** and **(d)**, we show examples of $\mathbf{P}(\alpha)$ resulting from poor choices of α. When α is too small, $\mathbf{P}(\alpha)$ develops a "spiky" distribution with multiple unphysical peaks (known as being "under-regularized"). In the case of $\alpha = 0$, Eq. 3.3.1 reduces to Eq. 3.2.7. On the other hand, when α is too large, $\mathbf{P}(\alpha)$ becomes "smeared out" ("over-regularized"), and loses information about the true $1/T_1$ distribution. Fig. 3.2 **(c)** shows the $\mathbf{P}(\alpha)$ calculated using an optimal choice of α, which most accurately recreates the true $1/T_1$ distribution. We outline the process for determining this optimal value α_{opt} in Appendix B.

Eq. 3.3.1 has a unique solution which is best computed via the method described by Butler, Reeds, and Dawson (BRD) [85]. For the purpose of brevity, we outline a simpler, albeit less computationally efficient method of solving Eq. 3.2.7, whose output is identical to that of the BRD method. We extend the vector $\mathbf{M}$ and matrix $\boldsymbol{K}$ via concatenation to produce

$$\mathbf{M}^* = \begin{bmatrix} \boldsymbol{M} \\ \mathbf{0} \end{bmatrix}, \quad \boldsymbol{K}^* = \begin{bmatrix} \boldsymbol{K} \\ \sqrt{\alpha}\boldsymbol{I} \end{bmatrix}, \tag{3.3.2}$$

where $\mathbf{0}$ is a zero vector of length n and I is an $n \times n$ identity matrix. Eq. 3.3.1 can then be rewritten to mimic the form of Eq. 3.2.7, as

$$\mathbf{P} = \underset{\mathbf{P} \geq 0}{\arg\min} ||\mathbf{M}^* - \boldsymbol{K}^* \mathbf{P}||^2, \tag{3.3.3}$$

which can be solved using a widely-available non-negative least squares algorithm [80].

3.4 Example: ILT of Synthetic $M(t_D)$ Data

To demonstrate its efficacy, we perform ILT on synthetic $1/T_1$ relaxation data generated from a model function consisting of two single exponential components

$$M_{model}(t_D) = 1 - 0.6 \exp\left(-t_D * 30 \text{ s}^{-1}\right) - 0.4 \exp\left(-t_D * 1000 \text{ s}^{-1}\right). \tag{3.4.1}$$

We can think of this $M_{model}(t_D)$ as arising from a sample containing two distinct species of nuclei, where 60% of the nuclei are relaxing with $1/T_1 = 30$ s^{-1} while the other 40% are relaxing with $1/T_1 = 1000$ s^{-1}. For simplicity, we assume the case of $I = 1/2$ NMR (i.e. $p_k = q_k = 1$), and normalize to $M_0 = \lambda = 1$. We then simulate 40 measured $M(t_D)$ data points from 10^{-6} s $< t_D < 10^1$ s, and add $\sigma_E/M_0 = 0.1\%$ artificial Gaussian noise. This synthetic data is shown in Fig. 3.3 **(a)**, alongside a basic stretched exponential fit of $M(t_D)$ (Eq. 2.5.3).

Computing the ILT (shown in Fig. 3.3 **(b)**), we see that $P(1/T_1)$ exhibits two peaks centered at $1/T_1 = 30.1$ and 997 s^{-1}, occupying 60% and 40% of the total area (integrated over $\log_{10}(1/T_1)$), respectively. This successfully reproduces the $1/T_1$ and

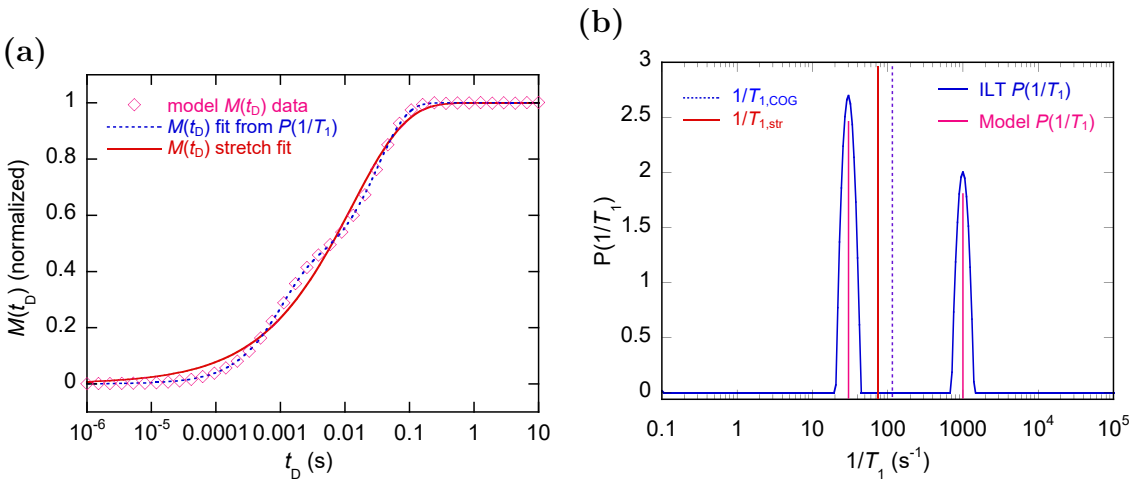


Figure 3.3: **(a)** Synthetic $M(t_D)$ data (pink $\diamond$) fitted with a stretched exponential function (solid red) and the ILT-reconstructed decay curve (dashed blue). **(b)** The $P(1/T_1)$ distribution computed via ILT ($\alpha \approx 0.002$). Also shown are the two $1/T_1$ components of the model function (vertical pink lines). Stretch fit $1/T_{1,\mathrm{str}}$ and $P(1/T_1)$ center of gravity $1/T_{1,COG}$ are indicated by the solid red and dashed blue vertical lines, respectively.

relative intensities of the model function's components. In contrast, the stretch fit fails to preserve this information, since it assumes only a monomodal distribution of $P(1/T_1)$ with a specific shape (see Fig. 3.1 **(b)**). Instead, the stretch fit result $1/T_{1,\mathrm{str}} = 77\ \mathrm{s}^{-1}$ essentially approximates $1/T_{1,COG} = 121\ \mathrm{s}^{-1}$, where $1/T_{1,COG}$ is the log-mean center of gravity of $P(1/T_1)$, defined as

$$\log_{10}(1/T_{1,COG}) = \sum_{j=1}^{n} P(1/T_{1,j})\Delta_P[\log_{10}(1/T_{1,j})]. \tag{3.4.2}$$

The stretch fit also yields $\beta = 0.52$, because a stretched exponential function with lower values of β better resembles the staggered decay of $M(t_D)$ resulting from two highly different $1/T_1$ components. In Supp. Mat. of Ref. [68], it is shown that β is generally anti-correlated with the logarithmic standard deviation of $P(1/T_1)$, defined

as

$$\sigma_{lm}^2 = \sum_{j=1}^{n} P(1/T_{1,j})\Delta_P[\ln(1/T_{1,j}) - \ln(1/T_{1,COG})]^2. \tag{3.4.3}$$

In our example, $\sigma_{lm} \approx 1.38$.

To see how much of the original $M(t_D)$ data is preserved by ILT, we can reverse the ILT process by multiplying $P(1/T_1)$ (i.e. $\mathbf{P}$) with the kernel matrix $\boldsymbol{K}$. The ILT-reconstructed decay curve $\boldsymbol{K}\mathbf{P}$ is plotted as the blue dashed line in Fig. 3.3 **(a)**. Notice that the ILT fit traces the synthetic $M(t_D)$ data points very well, whereas the stretch fit clearly fails to capture the shoulder around $t_D \sim 0.001$. In other words, the ILT fit of $M(t_D)$ yields a much lower residual than the stretch fit. These conclusions are somewhat to be expected, since in the absence of regularization, there are many more degrees of freedom in $P(1/T_1)$ than in the stretched exponential function. The key advantage of ILT over stretch fit analysis (as well as any other method based on fitting $M(t_D)$ with a predefined function) is that it does not make any assumption about the form of $M(t_D)$, other than it being the sum of exponential decay curves within a finite $1/T_1$ range. If the actual $M(t_D)$ data set has one (or two) exponential components, ILT will output one (or two) peaks in the histogram of $P(1/T_1)$. Furthermore, keep in mind that the existence of multiple $1/T_1$ components is not always discernible by eye when looking at $M(t_D)$. If the two $1/T_1$ components in Eq. 3.4.1 were instead only factor 3 apart, both the stretched exponential and ILT will appear to fit $M(t_D)$ fairly well, but only ILT is able to identify the two $1/T_1$ components [68].

We can think of Tikhonov regularization as accounting for non-zero noise by imparting an artificial width to $P(1/T_1)$ that increases in accordance with α [68]. As illustrated in Fig. 3.4, a higher value of α means that Eq. 3.3.1 favours wider, flatter

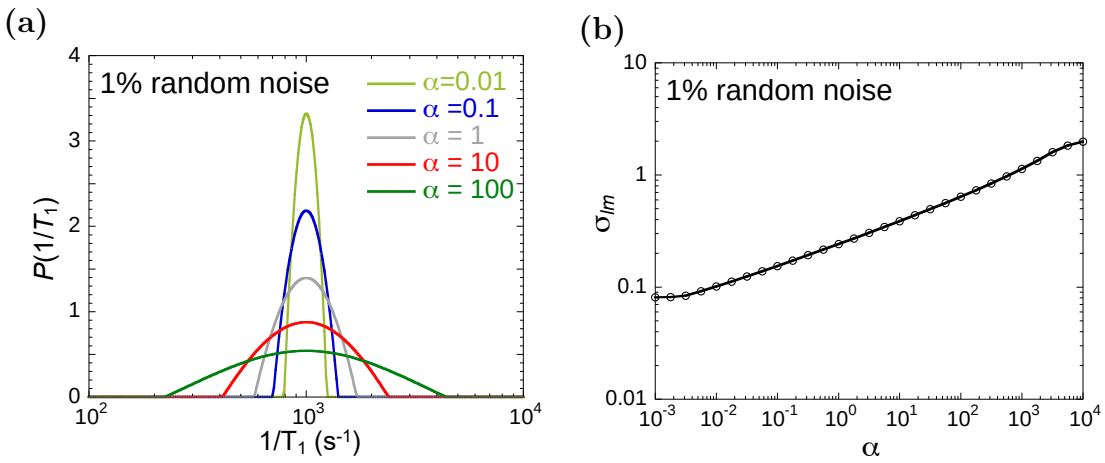


Figure 3.4: **(a)** $P(1/T_1)$ of a single exponential function with $1/T_1 = 1000$ s^{-1} and 1% noise, regularized with different values of α **(b)** Peak width σ_{lm} (in factors of e) of $P(1/T_1)$ as a function of α. Figures adopted from Supp. Mat. of Ref. [46].

solutions more. In essence, the "resolution" of $P(1/T_1)$ is determined by α, which has an optimal value α_{opt} that is anti-correlated with SNR (e.g. Fig. 3.4 **(b)**). Therefore, an inherent trade-off of ILT T_1 analysis is that a high SNR is required to achieve a good resolution in $P(1/T_1)$. Roughly speaking, given ~ 40 $M(t_D)$ data points and an SNR of 100, ILT can distinguish between two single exponential components whose $1/T_1$ are approximately half a decade apart.

Another practical consideration is that the measured $1/T_1$ recovery curve $M(t_D)$ is often not fully inverted in the short t_D limit. Recall that in the ideal case of perfect inversion (or saturation by comb pulses), we expect $M(t_D \to 0) = M_0[1-\lambda] = M_0$ (or 0). In reality, we often find that the measured $M(t_D)$ exhibits $\lambda < 2$ (or $\lambda < 1$). This can be due to things such as an imperfect π-pulse length, or the presence of (almost) immeasurably fast $1/T_1$ components that decay well before the shortest delay time ($T_1 \ll t_{i,\min} \sim 1$ μs). One way to account for incomplete inversions is to also fit for λ as a free parameter (as is the case in stretched fit analysis). However, a more robust

method is to simulate the very fast $1/T_1$ components in $P(1/T_1)$ itself (while λ remains fixed) [68]. We can do this by extending our $\{T_{1,j}\}$ bins during the setup for ILT, so that it includes an extra set of (~ 30) fast $1/T_1$ bins that go up to $1/T_{1,\mathrm{max}} \sim 10^6$ s^{-1}. The ILT algorithm will then use these bins to fit for incomplete inversion, meaning we can ignore the output $P(1/T_1)$ data above a certain cutoff ($1/T_{1,\mathrm{cut}} \sim 10^5$ s^{-1}), and omit them from our analysis.

3.5 Two-Dimensional Correlation Map: C(f,$1/T_1$)

So far in this chapter, we have defined $M(t_D)$ as the relaxing net nuclear magnetization of an arbitrary ensemble of nuclei. However, recall that each $M(t_D)$ measurement is conducted at a specific frequency f somewhere on the NMR spectral lineshape, and probes only the nuclei whose resonant frequencies ν_0 are equal to f. We can obtain $P(1/T_1)$ for this subset of nuclei using ILT, but if the lineshape is broad, this $P(1/T_1)$ only represents a small fraction of all of the nuclei of interest. On the other hand, a standard NMR lineshape measurement can potentially capture the signal arising from all of these nuclei, but lacks any information about their $1/T_1$.

In essence, the NMR lineshape and $P(1/T_1)$ are one-dimensional (1D) histograms which represent the distributions of ν_0 and $1/T_1$, respectively. Combining these, we can imagine a two-dimensional (2D) histogram $C(f, 1/T_1)$, which we define as the probability density distribution of nuclei that have resonant frequency $\nu_0 = f$ and relaxation rate $1/T_1$. Since each slice of $C(f, 1/T_1)$ at a fixed f equals $P(1/T_1)$ measured at that frequency, we can measure the $C(f, 1/T_1)$ of a given sample simply by measuring $M(t_D)$ at multiple frequencies across the NMR lineshape, then plotting the calculated $P(1/T_1)$ as a function of frequency.

The main advantage of analyzing the 2D data of $C(f, 1/T_1)$ is that it allows us to establish the correlations between Knight shift (calculated from ν_0) and $1/T_1$, which are direct measures of the local spin susceptibility and local quantity of low-energy spin excitations, respectively. To illustrate this point, we design a 2D model function $C_{model}(f, 1/T_1)$ to simulate the distributions of ν_0 and $1/T_1$ in a magnetically disordered sample. Shown in Fig. 3.5 **(a)**, this $C_{model}(f, 1/T_1)$ is comprised of three broad, overlapping 2D Gaussian peaks, and is defined as

$$C_{model}(f, 1/T_1) = \sum_{i=1,2,3} \frac{A_i}{\sigma_{f,i}\sigma_{T,i}} \exp\left[-\frac{1}{2}\left(\frac{\log_{10}(T_{1,i}/T_1)}{\sigma_{T,i}}\right)^2 - \frac{1}{2}\left(\frac{f - f_0^i}{\sigma_{f,i}}\right)^2\right]. \tag{3.5.1}$$

These peaks are centered at $1/T_1 = 1/T_{1,i}$ and $f = f_{0,i}$, have relative intensities A_i, and are distributed such that $\sigma_{f,i}$ and $\sigma_{T,i}$ represent their horizontal and logarithmic vertical standard deviations, respectively. The values of these parameters are recorded in Table 3.1. In principle, each peak in $C_{model}(f, 1/T_1)$ is meant to represent a qualitatively distinct domain, whose local spin susceptibility and $1/T_1$ are distributed around some central value.

Peak (i)	A^i	f_0^i (MHz)	$1/T_{1,0}^i$ (s^{-1})	σ_f^i (MHz)	σ_T^i (decades)
A	0.31	103	1500	1	0.5
B	0.61	106	100	1.3	0.7
C	0.08	103	7	0.6	0.3

Table 3.1: Parameters of the model function $C_{model}(f, 1/T_1)$

We can use $C_{model}(f, 1/T_1)$ to generate synthetic $M(t_D)$ data measured at any given frequency. We do this by simulating a set of 21 $1/T_1$ relaxation measurements

at evenly spaced frequencies f ranging from 100 to 110 MHz. For each f, we take a vertical (constant-f) slice of $C_{model}(f, 1/T_1)$ (indicated by the dashed vertical lines in Fig. 3.5 **(a)**), and plug it into Eq. 3.2.2 to obtain a set of synthetic $M(t_D)$ data consisting of 49 data points from 10^{-6} s $< t_D < 10^1$ s mixed with $\sigma_E / \max(M(f, t_D)) = 0.1\%$ noise. These sets of $M(t_D)$ data are then consolidated as a 2D array $M(f, t_D)$, where each column corresponds to the decay curve $M(t_D)$ measured at frequency f (Fig. 3.5 **(b)**).

Performing ILT on the columns of $M(f, t_D)$, we obtain the $P(1/T_1)$ distribution for each measured f (Fig. 3.5 **(c)**). We can combine these to form the calculated 2D correlation map $C(f, 1/T_1)$, shown as a color-contour plot in Fig. 3.6 **(b)**. Notice that this process successfully recreates the original model function $C_{model}(f, 1/T_1)$. We can further test this by fitting our measured $C(f, 1/T_1)$ against Eq. 3.5.1 with A^i, f_0^i, $1/T_{1,0}^i$, σ_f^i, and σ_T^i set as free parameters, and see that it quantitatively reproduces the parameters of the three model components.

A consequence of Eq. 3.0.2 is that integrating $C(f, 1/T_1)$ over the entire $1/T_1$ range gives us the saturated magnetization as a function of frequency $M_0(f)$ (plotted in Fig. 3.6 **(a)**), which is analogous to the conventionally measured NMR lineshape. If we instead integrate $C(f, 1/T_1)$ within a fixed $1/T_1$ interval, we obtain partial lineshapes corresponding to horizontal (constant $1/T_1$) slices of $C(f, 1/T_1)$ (see Fig. 3.6 **(c)** for some examples). These lineshapes can be thought of as resulting from the subset of nuclei with a specific value of $1/T_1$, and are thus we coined them as "ILT T_1-resolved NMR lineshapes".

Regarding our example of 2D ILT using $C_{model}(f, 1/T_1)$, the key takeaway here is that neither the conventional NMR lineshape, nor any individual $P(1/T_1)$ distribution

revealed the true nature of $C_{model}(f, 1/T_1)$. The nearly identical ν_0 of the A and C peaks made them indistinguishable in the lineshape in Fig. 3.6 **(a)**. On the other hand, $P(1/T_1)$ measured at 103 MHz could distinguish the A and C peaks, but their distance from the B peak meant that no individual $P(1/T_1)$ in Fig. 3.5 **(c)** showcased all three peaks. It was only after combining multiple 1D $P(1/T_1)$ measurements to construct $C(f, 1/T_1)$, that the existances of these three peaks were made clear, and we can quantitatively analyze their individual behaviours.

In a real experiment, the time required to measure a 2D data set of $C(f, 1/T_1)$ is generally an order of magnitude longer than conventional 1D NMR, since it involves repeating many $1/T_1$ measurements with sufficient SNR for ILT. However, depending on the material and experimental conditions, such time consuming measurements can be greatly justified, because $C(f, 1/T_1)$ can reveal information beyond the scope of 1D data. In our ^{19}F NMR investigation of Zn-barlowite (chapter 6), this technique was essential in delineating the behaviour of the main component from that of a split-off paramagnetic component.

In the field of NMR petrophysics, this type of 2D data has been previously obtained for narrow NMR lineshapes with the aid of Fast Fourier Transform, in order to show correlations between $1/T_1$ and the chemical shift δ [64]. For NMR research on quantum materials affected by disorder, $C(f, 1/T_1)$ generally reflects the correlations between $1/T_1$ and Knight shift, which both tend to exhibit broad, inhomogeneous distributions. Note that in this section, we focused specifically on $C(f, 1/T_1)$. However, through more advanced applications of ILT, we can visualize the correlations between any two (or more) independent variables which reflect some aspect of a nucleus' local environment, such as $1/T_1$, $1/T_2$, and the self-diffusion coefficient D [62, 63, 86].

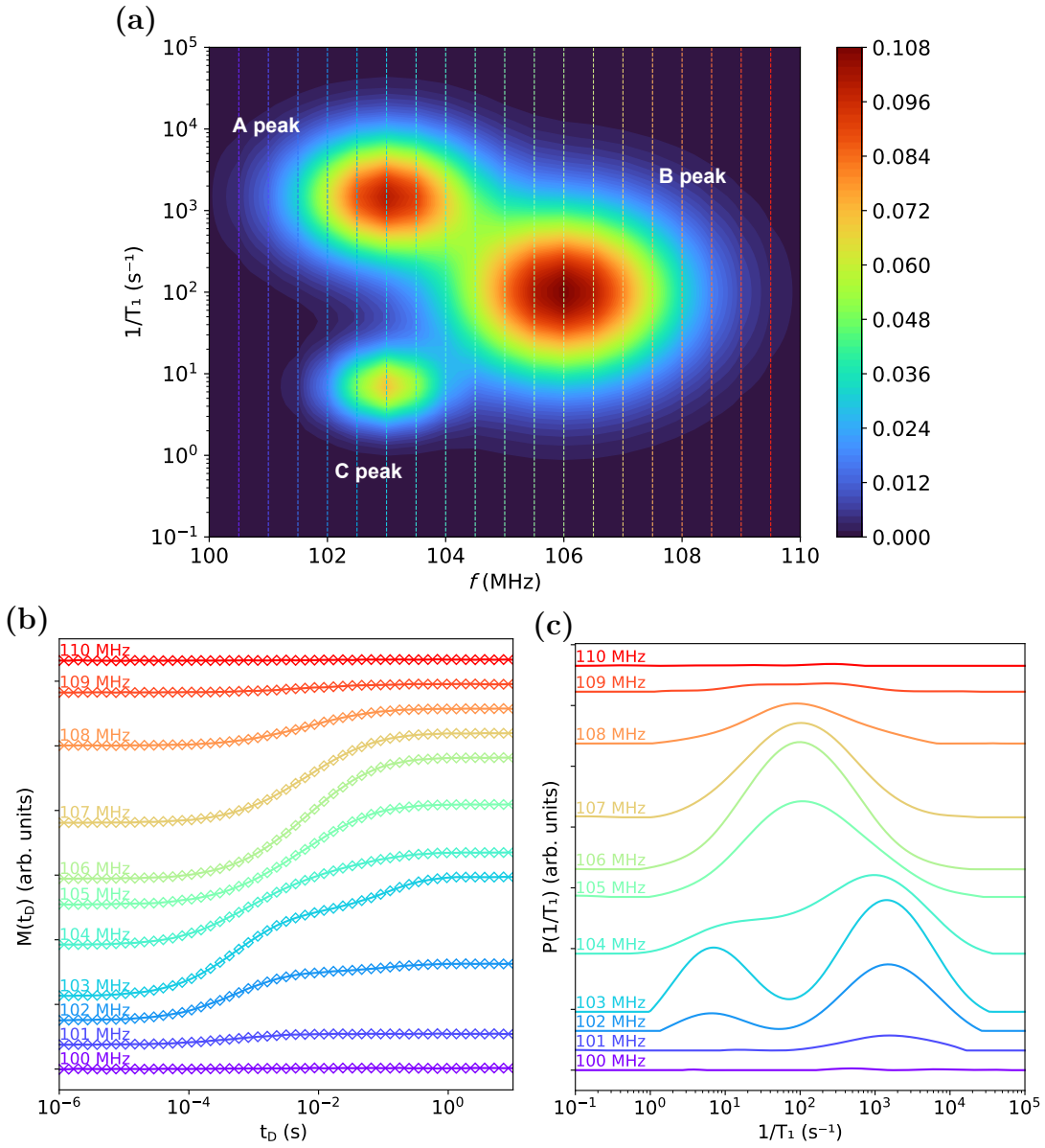


Figure 3.5: **(a)** Model function $C_{model}(f, 1/T_1)$ representing the distributions of local $1/T_1$ and shifted resonance frequency $\nu_0 = f$, the vertical dashed lines correspond to the constant-f slices from which $M(f, t_D)$ is generated. **(b)** Synthetic $M(f, t_D)$ data expected for the model function $C_{model}(f, 1/T_1)$, shown for every 2nd frequency f, and vertically cascaded for visual clarity. **(c)** Constant-f slices of the ILT output $C(f, 1/T_1)$ (i.e. $P(1/T_1)$), computed with fixed $\alpha = 1$.

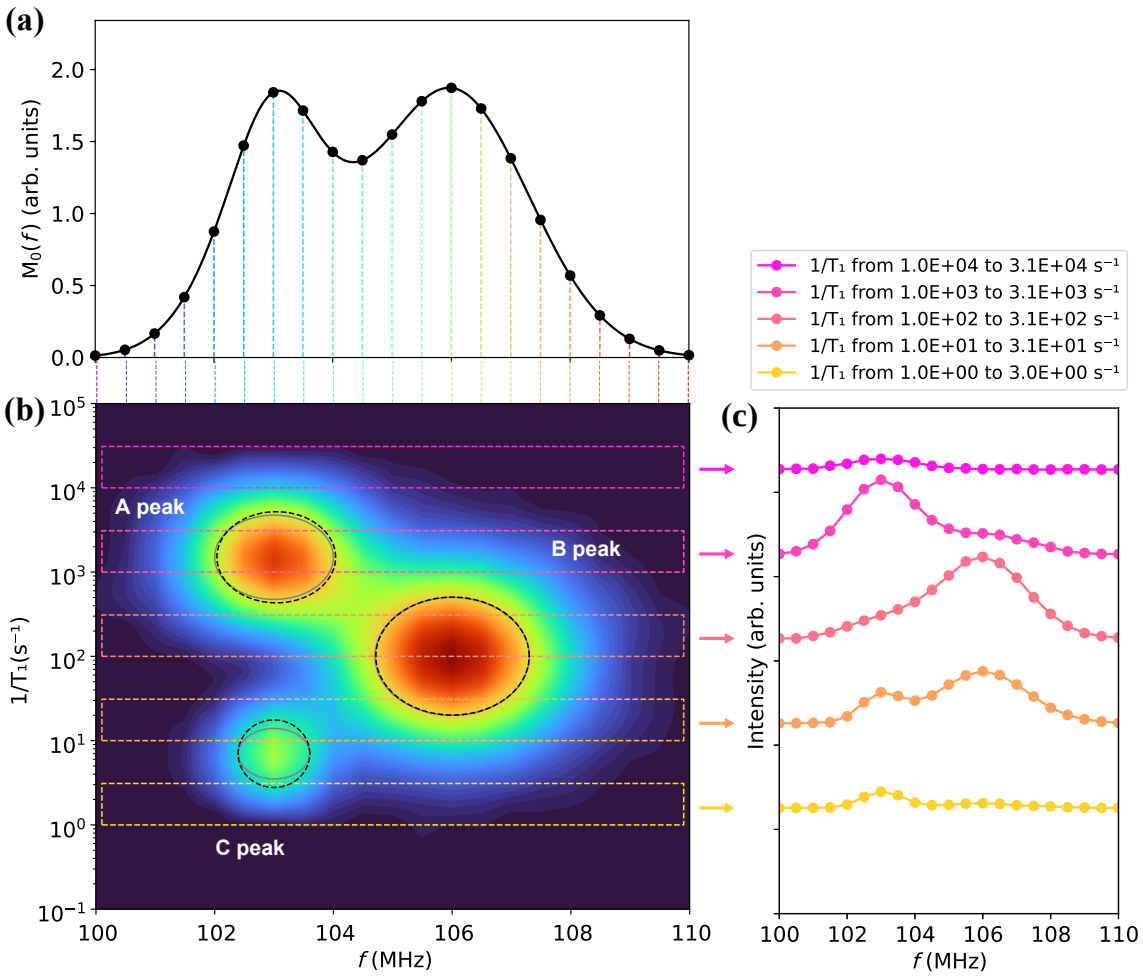


Figure 3.6: **(a)** Conventional 1D NMR lineshape, equal to $M_0(f)$. Each data point is simply the integral of $C(f, 1/T_1)$ along the vertical $1/T_1$ axis over all $1/T_1$ values, at some given frequency f. **(b)** Correlation map $C(f, 1/T_1)$ calculated from ILT. The solid grey (dashed black) ellipses indicate the center and widths of the model (fitted) 2D Gaussian components. The colored dashed boxes indicate the horizontal slices of $C(f, 1/T_1)$ (integrated over a $1/T_1$ range of 0.5 decades) used to generate **(c)** the T_1-resolved NMR lineshapes. For example, the middle pink curve in **(c)** is the horizontal slice of $C(f, 1/T_1)$ for nuclei whose $1/T_1$ ranges from 100 s^{-1} to 300 s^{-1}, and represents the $1/T_1$ resolved lineshape arising mainly from the B peak.

Chapter 4

Kitaev Iridate $Ag_3LiIr_2O_6$

4.1 Preface

The novel Kitaev iridate $Ag_3LiIr_2O_6$ is metastable, meaning its chemical structure is not in its most energetically-favourable state, but rather at a local minimum [39]. Instead of conventional solid-state methods at high-temperatures, $Ag_3LiIr_2O_6$ is synthesized through topochemical exchange. As revealed by X-ray diffraction and transmission electron microscopy, this tends to introduce some amounts of structural and chemical disorder to the material [42, 87]. Muon spin resonance (μSR) measurements also indicate a partial onset of magnetic order whose signatures may be obscured in more disordered samples [87]. To assess $Ag_3LiIr_2O_6$ as a QSL candidate requires differentiating the intrinsic behaviour of its Ir^{4+} electron spins from those situated in regions affected by disorder. These Ir^{4+} spins are coupled to the spins of nearby nuclei, and can be probed using NMR. Our plan is to conduct a ^{7}Li NMR investigation of $Ag_3LiIr_2O_6$, which will greatly benefit from the use of ILT $1/T_1$ analysis. Recall that the nature of low-energy spin excitations is represented in the temperature

dependence of $1/T_1$, and that local $1/T_1$ values are likely influenced by disorder.

For context, the appearance of inhomogeneous $1/T_1$ distribtuions is pervasive amongst Kitaev materials. This is typically dealt with via stretched exponential fitting of $M(t_D)$ [41, 88–90], where a significant decrease in β is often observed at low temperatures. With the introduction of ILT in more recent studies on Kitaev materials, the meaning of the $1/T_1$ values derived from traditional function fitting can be understood in the context of $P(1/T_1)$. In Cu_2IrO_3, the temperature dependence of $P(1/T_1)$ (measured via Cu NQR) identified its intrinsic behaviour as corresponding to a slow $1/T_1$ component, alongside a defect-induced, broad distribution of fast $1/T_1$ components [70]. Similarity, a ^{1}H NMR study on $H_3LiIr_2O_6$ found in $P(1/T_1)$, two $1/T_1$ components whose individual temperature evolutions suggest that the sample approaches a spatially inhomogeneous ground state [71].

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In this article, we present the findings of our ^{7}Li NMR experiments and ILT $1/T_1$ analysis on $Ag_3LiIr_2O_6$ and discuss their implications towards the study of Kitaev materials. My role in this project includes developing the ILT program, conducting NMR measurements, analyzing the data, and writing the manuscript. All of the $Ag_3LiIr_2O_6$ samples and χ_{bulk} data were provided by our collaborators Dr. Faranak Bahrami and Prof. Fazel Tafti at Boston College. The ILT methods we used are based on the work of Prof. Philip M. Singer, whom we have routinely consulted with when developing our algorithm. Prof. Takashi Imai oversaw this project and contributed significantly to the writing of the manuscript.

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NMR investigation on the honeycomb iridate $Ag_3LiIr_2O_6$

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$Ag_3LiIr_2O_6$ is a Kitaev spin-liquid candidate material synthesized from α-Li_2IrO_3 through a topotactic reaction. We investigate the structural and magnetic properties of two samples of $Ag_3LiIr_2O_6$ based on the ^{7}Li nuclear magnetic resonance line shape, Knight shift, and spin-lattice relaxation rate $1/T_1$. The first sample A shows signatures of magnetically ordered spins, and exhibits one sharp ^{7}Li peak with the full width at half maximum increasing significantly below 14 K. $1/T_1^{\text{stretch}}$ of this sample displays a broad local maximum at 40 K, followed by a very sharp peak at $T_N = 9 \pm 1$ K due to a critical slowing down of Ir spin fluctuations, a typical signature of magnetic long-range order. In order to shed light on the position-by-position variation of $1/T_1$ throughout the sample, we use a numerical inverse Laplace transform T_1 analysis based on Tikhonov regularization to deduce the density distribution function $P(1/T_1)$. We demonstrate that ~60% of Ir spins are statically ordered at the NMR measurement timescale but the rest of the sample volume remains paramagnetic even at 4.2 K, presumably because of structural disorder induced primarily by stacking faults. In order to further investigate the influence of structural disorder, we compare these NMR results with those of a second sample B, which has been shown by transmission electron microscopy to have domains with unwanted Ag inclusion at the Li and Ir sites within the Ir honeycomb planes. Sample B displays an additional NMR peak with a relative intensity of ~17%. The small Knight shift and $1/T_1$ of these defect-induced ^{7}Li sites and the enhancement of bulk susceptibility at low temperatures suggest that these defects generate domains of only weakly magnetic Ir spins accompanied by free spins, leading to a lack of clear signatures of long-range order. The apparent lack of long-range order could be easily misinterpreted as evidence for the realization of a spin-liquid ground state in highly disordered Kitaev lattices.

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I. INTRODUCTION

In contrast to conventional magnetism, quantum spin liquids (QSLs) are an exotic state of matter which avoid magnetic order even at absolute zero. They instead have a highly entangled ground state induced by frustrated magnetic interactions [1]. In 2006, Kitaev demonstrated analytically that a honeycomb lattice of effective spin-1/2 moments interacting via bond-dependent Ising interactions $\hat{\mathcal{H}}_K$ has a spin-liquid ground state [2]. This is a promising class of QSLs as it can manifest in certain Mott insulators with strong spin-orbit coupling [3].

Since then, many materials have been shown to exhibit strong Kitaev interactions, providing opportunities for achieving a Kitaev QSL ground state via chemical tuning. Thus, searching for materials with Kitaev honeycomb planes has become the subject of extensive research [4]. Various Kitaev QSL candidates with magnetic transition metal ions have been identified, including Na_2IrO_3 [5–7], α-Li_2IrO_3 [8], Li_2RhO_3 [9], α-$RuCl_3$ [10–15], $H_3LiIr_2O_6$ [16], Cu_2IrO_3 [17–19], and $D_3LiIr_2O_6$ [20].

However, additional interactions such as the isotropic Heisenberg interaction $\hat{\mathcal{H}}_H$ and symmetric off-diagonal exchange term $\hat{\mathcal{H}}_\Gamma$ compete with the frustrated Ising interaction $\hat{\mathcal{H}}_K$ and mask the intrinsic Kitaev behavior [21,22]. For example, materials such as Na_2IrO_3 [23,24] and α-$RuCl_3$ [25,26] undergo long-range antiferromagnetic (AF) order at $T_N \simeq 17$ K or below.

One approach to tune the relative strength of $\hat{\mathcal{H}}_K$, $\hat{\mathcal{H}}_H$, and $\hat{\mathcal{H}}_\Gamma$ is to modify the intralayer orbital overlaps and bond angles by changing the interlayer bonding. This was done in α-Li_2IrO_3 by replacing the interlayer Li^+ ions with Ag^+ ions to produce $Ag_3LiIr_2O_6$ [27], the lattice of which is shown in Fig. 1. Heat capacity and magnetic susceptibility measurements conducted for early samples with significant disorder due both to stacking faults and unwanted Ag inclusion within the Li and Ir sites in the Kitaev layers have shown no evidence for AF long-range order. However, more recent muon spin rotation (μSR) measurements conducted on a cleaner sample without the inclusion of Ag at the in-plane sites exhibit a Bessel function type oscillation below $T_N \sim 8$ K, establishing incommensurate AF ordering for the majority of the sample volume [29].

In this paper, we use ^{7}Li NMR to probe the local structural and magnetic properties of the honeycomb layers in $Ag_3LiIr_2O_6$, and the potential roles played by structural disorder. We also investigate the spatial distribution of Ir spin dynamics by probing the density distribution function $P(1/T_1)$ of the ^{7}Li NMR spin-lattice relaxation rate $1/T_1$ based on an inverse Laplace transform (ILT) T_1 analysis technique [30].

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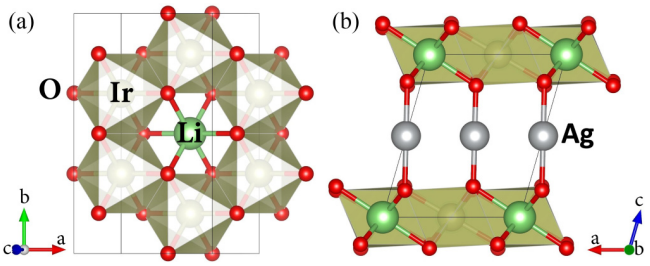


FIG. 1. (a) Crystal structure of Ir^{4+} honeycomb planes in $Ag_3LiIr_2O_6$ [27,28]. Each Ir^{4+} ion is surrounded by six O^{2-} ions (red), forming an octahedron (highlighted in tan). Li^+ (green) ions are situated at the center of the Ir hexagons. (b) Interlayer Ag^+ ions (gray) in between honeycomb planes. The gray lines indicate the unit cell of the lattice.

We will demonstrate that ∼60% of the sample volume undergoes antiferromagnetic long-range order below $T_N = 9 \pm 1$ K, but the rest of the volume remains paramagnetic even in the cleanest sample without the unwanted Ag inclusion. A comparison with a disordered sample indicates that the NMR as well as thermodynamic signatures of the long-range order could be easily washed away in more disordered samples.

II. EXPERIMENTAL METHODS

Powder samples of $Ag_3LiIr_2O_6$ were synthesized for NMR measurements via a topotactic reaction as described in Refs. [27,29]. We present a NMR investigation of two distinct samples, which we refer to as samples A and B. Sample B was synthesized in the early days [27], and has inclusions of unwanted Ag ions occupying the Li and Ir sites within the honeycomb plane, as evidenced by clusters observed in transmission electron microscopy (TEM) measurements [29]. This is the sample that appeared to be a spin-liquid material [27], because the thermodynamic [27], μSR [27], and NMR measurements (this work) do not reveal evidence for magnetic long-range order. Furthermore, sample B exhibits an extra split-off peak in the ^{7}Li NMR line shape (as shown below), indicating that unwanted Ag clusters lead to a different type of Li sites. On the other hand, sample A is a newer sample obtained after the sample synthesis conditions have been refined, and exhibit the signature of magnetic long-range order in the μSR [29] and NMR results (this work). Bulk magnetic susceptibility χ measured using a superconducting quantum interference device (SQUID) of sample A reveals a broad local maximum around 14 K, while χ of sample B increases drastically at low temperatures, as shown in Fig. 2. The overall magnitude of χ of sample B is also lower than χ of sample A at high temperatures.

We also investigated two additional samples with different levels of disorder between those of samples A and B, as indicated by the relative intensity of the split-off NMR peak associated with the unwanted domains with Ag inclusion. Both of these two extra samples exhibit qualitatively similar behavior as sample B, and hence the details of their NMR properties are beyond the scope of this paper. We note that the change of stoichiometry due to a small amount of Ag inclusion is within the margin of error in an energy dispersive

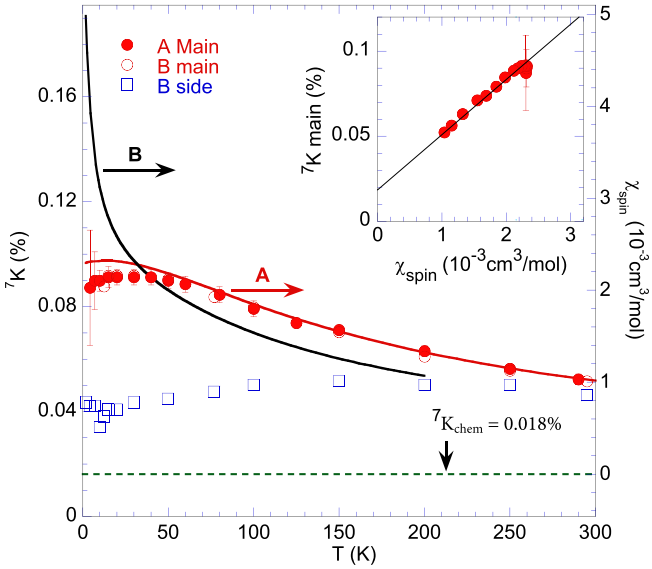


FIG. 2. Knight shift of the main peak for sample A (solid red bullets) along with the main (open red circles) and side (open blue squares) peaks for sample B. The bulk spin susceptibility is plotted as the red line and black line for samples A and B, respectively. The dashed green horizontal line indicates the chemical shift $^7K_{chem} \simeq 0.018\%$ of sample A. The vertical axis of χ_{spin} is scaled in proportion to the Knight shift of sample A, and its origin is vertically shifted to account for the chemical shift of sample A. The inset shows the Knight shift plotted against spin susceptibility χ_{spin} by choosing the temperature as their implicit parameter.

x-ray (EDX) analysis (3%–5%), although the relative intensity of the Li NMR side peak reaches ∼17% for the most disordered sample B. This suggests that each extra Ag site affects multiple Li sites in its vicinity.

We performed ^{7}Li NMR experiments using a standard spin echo pulse sequence in a field of 4.5 T, for temperatures ranging from 2 to 295 K. The $\pi/2$ pulse width ranged from 2 to 4 μs and the pulse separation time τ between the $\pi/2$ and π pulse was 20 μs. We measured the spin-lattice relaxation rate $1/T_1$ based on inversion recovery, by monitoring the nuclear magnetization $M(t)$ at select delay times t. We deduced $1/T_1^{\text{stretch}}$ by fitting $M(t)$ against the conventional stretch exponential. Finally, the density distribution $P(1/T_1)$ of $1/T_1$ is determined from $M(t)$ via the inverse Laplace transform T_1 (ILTT_1) analysis technique outlined in the Appendix.

III. NMR RESULTS AND DISCUSSION

A. ^{7}Li line shape and width

We compare representative ^{7}Li NMR line shapes at 30 K for samples A and B in Figs. 3(a) and 3(b). We also show a cascaded plot of all NMR line shapes observed for sample A at 100 K and below in Fig. 3(c).

There is only one Li site within the honeycomb layer of $Ag_3LiIr_2O_6$ in Fig. 1, and hence we expect to find only one type of ^{7}Li NMR line. In fact, we observed only one sharp ^{7}Li NMR peak for sample A, as shown in Fig. 3(a). However, we can identify a second peak in the total line shape of sample B, as shown by the green curve of Fig. 3(b). NMR is a local

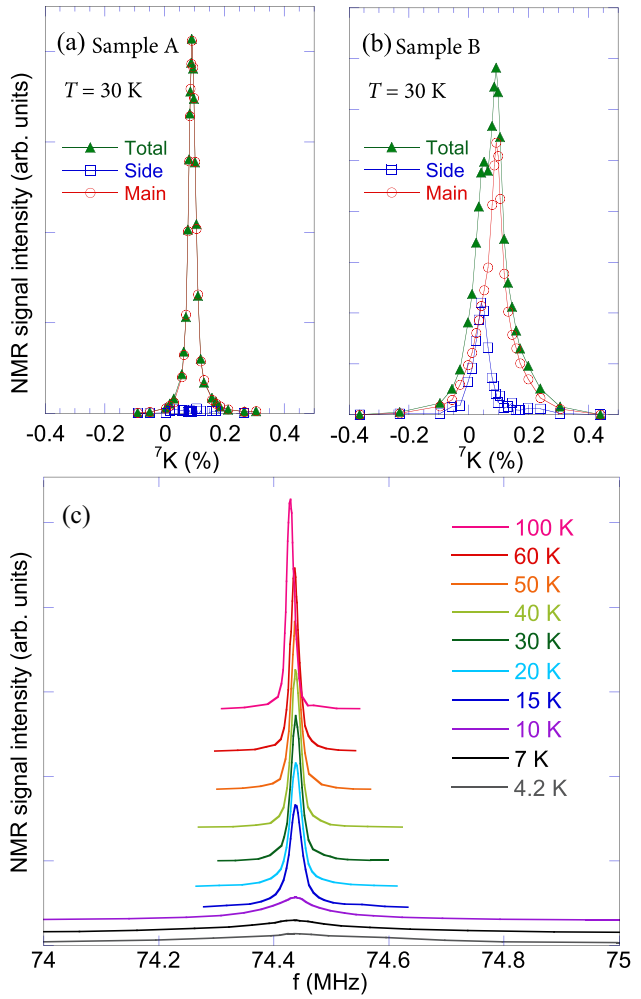


FIG. 3. NMR line shapes of (a) sample A and (b) sample B at 30 K for the fully relaxed signal (solid triangles), along with the line shapes of the fast (open bullets) and slow (open squares) components. (c) Cascaded plot of the NMR line shapes for sample A, observed at temperatures ranging from 4.2 to 100 K.

probe, and hence the presence of the second peak indicates that two different types of ^{7}Li sites exist in the disordered sample B.

We can take advantage of the difference in the NMR spin-lattice relaxation rate $1/T_1$ to isolate this second peak. The aforementioned total NMR signal [green triangles in Figs. 3(a) and 3(b)] is acquired by waiting for the nuclear spins to return to thermal equilibrium between individual spin echo measurements. This takes between ∼200 ms and ∼84 s depending on the temperature. On the other hand, we can selectively capture the fast component of the signal by repeating the spin echo measurements with 31.3 ms intervals as shown by the open red bullets in Figs. 3(a) and 3(b). By subtracting the fast component from the total intensity, we obtain the side peak line shape [blue squares in Figs. 3(a) and 3(b)] with a very slow relaxation rate.

Sample B exhibits a significant fraction (∼17%) of the side peak signal while no side peak is observed in the newest and cleanest sample A. In view of the Ag inclusion at Li and Ir

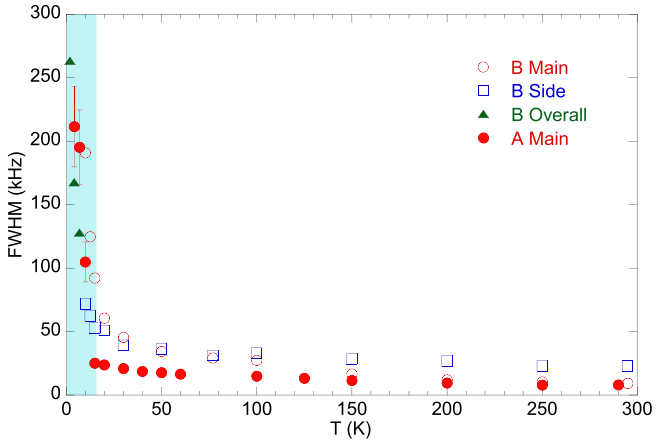


FIG. 4. FWHM of the main peak for both sample A (solid bullets) and sample B (open bullets), along with the FWHM of the side peak (open squares) and overall width (triangles, below 10 K) of sample B. The FWHM begins to broaden rapidly below ∼14 K and this region is highlighted in light blue.

sites as a cluster, we can attribute the side peak to Li sites within or around the areas of the honeycomb layer with these extra Ag sites.

We summarize the full width at half maximum (FWHM) of the line shapes in Fig. 4. We see that as the temperature decreases below ∼14 K, both the main and side peaks broaden rapidly for both samples A and B. The onset of NMR line broadening is generally an indication of the onset of long-range ordering or spin freezing. In fact, spin susceptibility data of sample A exhibit different behavior between field-cooled and zero-field-cooled measurements starting at 14 K, a typical signature of the onset of spin freezing [29]. This is consistent with our observations. Below 10 K, the main and side peaks of sample B merge and are no longer distinguishable via the fast and slow component of the signal; we show the overall linewidth below 10 K with solid triangles.

B. Knight shift

We plot the Knight shift ^{7}K of the main and side peak in Fig. 2 along with the spin contribution to bulk magnetic susceptibility $\chi_{\rm spin}$. We deduce $\chi_{\rm spin}$ from the bulk susceptibility χ using the Van Vleck and diamagnetic contributions such that $\chi = \chi_{\rm spin} + \chi_{\rm dia} + \chi_{\rm vv}$, where $\chi_{\rm dia} \simeq -0.1 \times 10^{-3}$ emu/mol [31] and $\chi_{\rm vv} \simeq 0.14 \times 10^{-3}$ emu/mol [23]. Notice that the ^{7}K of sample A shows a nearly identical temperature dependence as $\chi_{\rm spin}$, and exhibits a broad maximum around 25 K. The slightly less suppression of $\chi_{\rm spin}$ observed below 50 K may be attributed to a small amount of paramagnetic defect spin contributions in the bulk susceptibility data.

The inset in Fig. 2 shows ^{7}K measured for sample A plotted against $\chi_{\rm spin}$. The main peak Knight shift is linear to magnetic susceptibility $\chi_{\rm spin}$ down to ∼100 K, and a linear fit approximates a chemical shift of $^7K_{\rm chem} \simeq 0.018\%$ and the hyperfine coupling constant $A_{\rm hf} = (N_A\mu_B/N_{\rm n.n.}) \times (d\,^7{\rm K})/(d\chi) = 0.31$ kOe/μ_B for sample A, where N_A is Avogadro's number, μ_B is the Bohr magneton, and $N_{\rm n.n.} = 6$ is the number of nearest-neighbor Ir adjacent to each Li site.

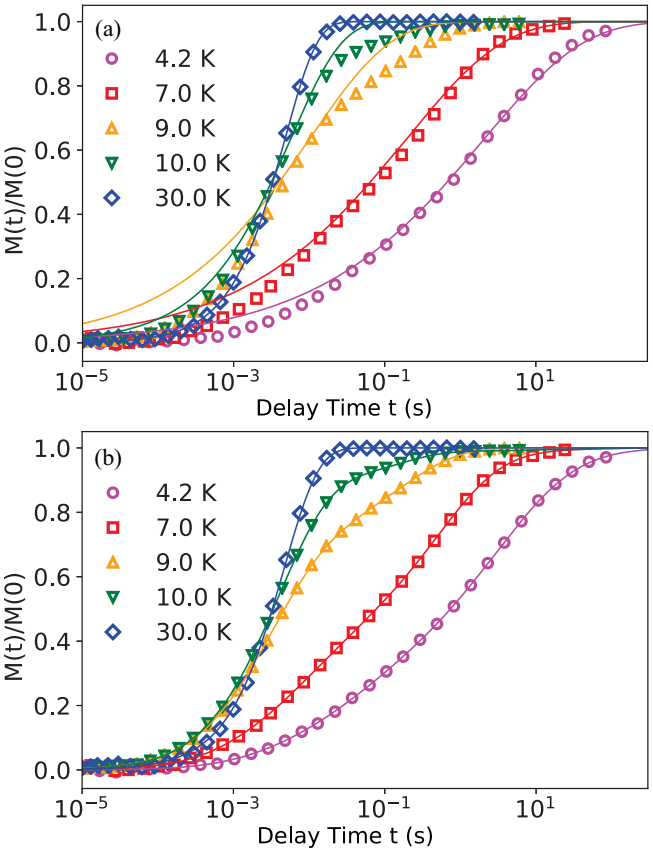


FIG. 5. The normalized recovery curve $M(t)$ observed for sample A at select temperatures. The solid lines in (a) show the respective stretch fit results, and the solid lines in (b) show the ILT fit results. The stretch fit fails to capture all the components when $1/T_1$ becomes highly distributed while the ILT fits the data nicely at all temperatures.

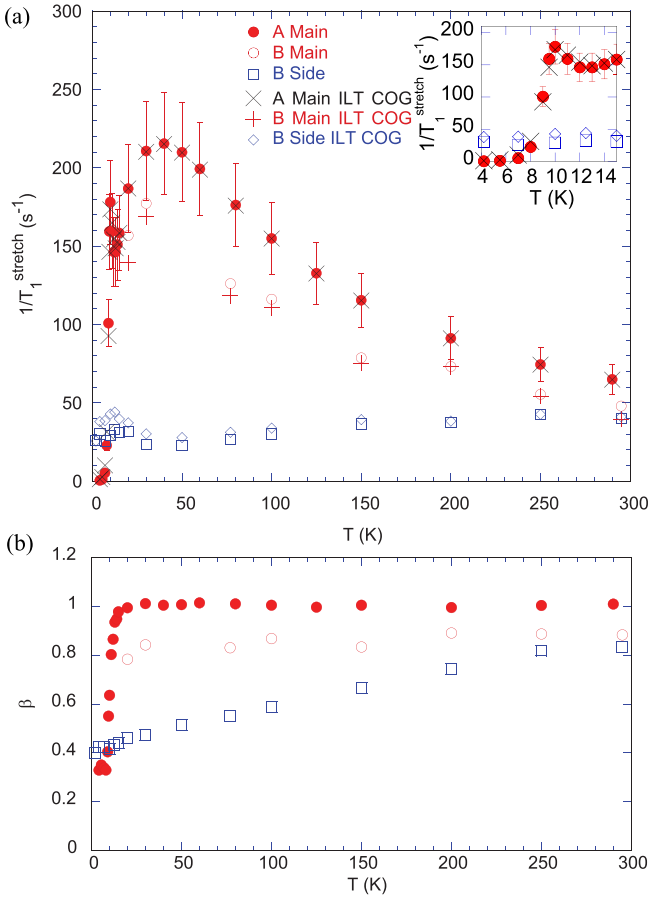


FIG. 6. (a) Spin-lattice relaxation rate $1/T_1^{\text{stretch}}$ estimated from a stretch fit at the main peak of sample A (solid red bullets) and the main (open red circles) and side (open blue squares) peak of sample B. ILT COG of sample A's main peak (black ×'s) and sample B's main (red +'s) and side (open blue diamonds) peaks seen in Fig. 7 are also plotted. Note that below 10 K in sample B, the rapidly broadening main peak is overshadowed by the side peak, and thus $1/T_1$ is measured at the peak of the overall NMR line shape. The inset shows the low-temperature data zoomed in to better display the sharp $1/T_1$ peak at 9 ± 1 K of sample A's main peak. (b) The stretch fit parameter β for samples A and B. For sample A's main peak, we found $\beta = 1$ within experimental uncertainties down to 20 K, implying that there is no distribution in $1/T_1$.

The ^{7}K of sample B's main peak shows nearly identical behavior to that of sample A. This indicates that the local spin susceptibility of the cleaner parts of sample B far from the Ag clusters exhibits nearly identical behavior as a clean sample A. On the other hand, ^{7}K at the side peak of sample B decreases at lower temperatures and levels off, suggesting that spin susceptibility gets locally suppressed near the Ag clusters. Moreover, the absence of a ^{7}K component that increases dramatically below ∼10 K indicates that a greater concentration of unpaired spins is present in sample B and responsible for the steep upturn of bulk χ at low temperatures. These findings are consistent with theoretical speculations that disorder in the Kitaev planes may cause the appearance of local spin singlets, accompanied by orphaned paramagnetic spins [32,33].

C. Spin dynamics and $1/T_1$ distribution

In order to investigate the Ir spin dynamics, we measured the spin-lattice relaxation rate $1/T_1$ at the main peaks of samples A and B, and the side peak of sample B. Representative $1/T_1$ recovery curves $M(t)$ at sample A's main peak are plotted in Fig. 5.

To obtain an estimate of $1/T_1$, $M(t)$ is fitted against the phenomenological stretched exponential function

$$M(t) = M_0 - A \exp\left[- \left(t/T_1^{\text{stretch}}\right)^{\beta} \right], \qquad (1)$$

where M_0, A, T_1^{stretch}, and β are the free parameters. The stretch fit exponent β accounts for the distribution of $1/T_1$ with $\beta = 1$ corresponding to no distribution.

We summarize the fit results of $1/T_1$ and β in Fig. 6 along with the center of gravity (COG) of the distribution of $1/T_1$ estimated from ILT, as discussed later in this section. We see that $1/T_1^{\text{stretch}}$ measured at the main peaks of samples A and B increases with decreasing temperature and reaches a broad maximum around 40 K. For sample A, $1/T_1^{\text{stretch}}$ then exhibits a second, sharp peak at $T_N = 9 \pm 1$ K and becomes vanishingly small below it. Similar sharp peaks of $1/T_1$ are generally

observed in materials undergoing magnetic long-range order, where $1/T_1$ diverges toward the transition temperature due to a critical slowing down of spin fluctuations. Our finding is consistent with recent μSR measurements, where a static hyperfine field arising from incommensurate AF order emerges below ∼8 K [29]. As discussed above in Sec. III A, these signatures of long-range order are preceded by the aforementioned typical signatures of spin freezing in the bulk $\chi_{\rm spin}$ and NMR line broadening below 14 K, presumably because the residual disorder effects caused by stacking faults are suppressing T_N even in a cleaner sample A.

The broad peak in the $1/T_1$ of sample A around 40 K is accompanied by the aforementioned broad peak in ^{7}K. This finding is similar to the case in Cu_2IrO_3, where $1/T_1$ and the Knight shift at ^{63}Cu sites also exhibit broad peaks around 40 K but $1/T_1$ does not diverge [18]. Since the suppression of χ usually signals the short-range order of spins in low-dimensional systems, it may be perplexing to find that $1/T_1$ is not starting to diverge below ∼40 K. An interesting scenario is that the spin excitation spectrum develops a gap below ∼40 K at a fraction of the Ir-Ir Kitaev interaction $|J_K| \sim 10^2$ K [21], as expected for Majorana fermions and fluxes [34–37]. However, the Li atoms are situated at the high-symmetry position of the center of six Ir sites. Therefore fluctuating hyperfine magnetic fields arising from incommensurate antiferromagnetic spin fluctuations may nearly cancel out at the ^{7}Li sites, suppressing $1/T_1$ below 40 K even without a spin excitation gap. An analogous situation was encountered in the ^{89}Y NMR $1/T_1$ data in $YBa_2Cu_3O_6$ [38], and we cannot entirely rule out such a purely geometrical scenario for the observed suppression of $1/T_1$ below ∼40 K.

The $1/T_1^{\rm stretch}$ and β measured at the main peak of sample B show roughly the same behavior as sample A with both being slightly smaller. However, $1/T_1^{\rm stretch}$ measured at the side peak of sample B shows qualitatively different behavior as it is much lower and does not peak near 40 K. Instead, $1/T_1^{\rm stretch}$ of the side peak gradually decreases. This is consistent with the gradual decrease of ^{7}K of the side peak, and seems consistent with the nearly nonmagnetic regions that emerge near domains with Ag inclusion. The emergence of these nonmagnetic regions also explains why the magnitude of the overall bulk susceptibility data in Fig. 2 is somewhat suppressed for sample B, except in the low-temperature region dominated by isolated spins.

$1/T_1^{\rm stretch}$ is generally only a crude approximation of the center of gravity (COG) of the distributed values of $1/T_1$, as we recently demonstrated for various materials [18,30,39]. Accordingly, a more generalized analysis technique is needed to understand the behavior of a highly distributed $1/T_1$ with multiple components.

To get the precise distribution of $1/T_1$, we can apply the ILTT$_1$ analysis technique to our $M(t)$ data and deduce the density distribution function of $P(1/T_1)$ [30,39]. We define $P(1/T_1)$ for a discrete range of relaxation rates $1/T_{1,i}$ as

$$M(t) = \sum_i [1 - 2\exp(-t/T_{1,i})]P(1/T_{1,i}) + E(t), \quad (2)$$

where $P(1/T_{1,i})$ is the non-negative ILT spectrum weight and $E(t)$ is Gaussian noise. The incomplete inversion of $M(t)$ is taken into account as explained in the Supplemental Material of Ref. [30]. We numerically invert $M(t)$ to obtain $P(1/T_1)$ based on ILT via the method outlined in the Appendix; see Ref. [30] and its Supplemental Material for a brief review and the complete details of the ILT procedures.

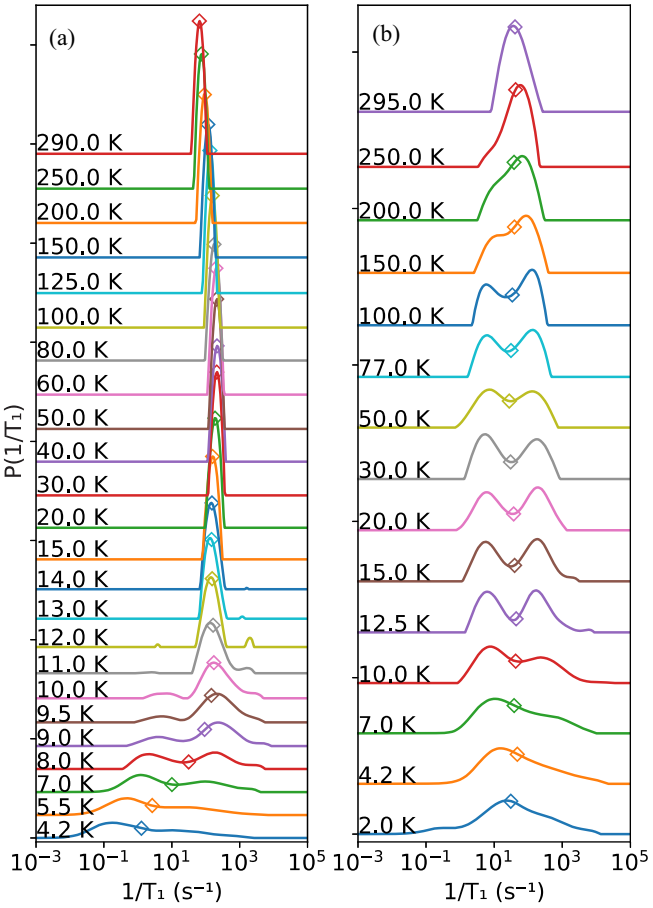


FIG. 7. Cascaded ILT density distributions $P(1/T_1)$ for (a) sample A's main peak and (b) sample B's side peak $1/T_1$ relaxation measurements. The distributions are normalized to have equal areas. The temperatures corresponding to each curve are labeled on the left side of the plot. The open diamonds indicate the calculated COG of the ILT distribution. The $P(1/T_1)$ results deduced for $M(t)$ measured at the main peak of sample B (not shown) are similar to the side peak results in (b), except that the relative intensity of the slower peak of $P(1/T_1)$ is much smaller.

The ILT approach has major advantages over the stretch fit analysis as it is model independent and thus naturally distinguishes separate components of $1/T_1$ in its distribution. The total integral of the $P(1/T_1)$ curves is normalized to 1. Therefore, the area underneath the $P(1/T_1)$ curves between two values of $1/T_1 = a$ and $1/T_1 = b$ represents the fraction of ^{7}Li nuclear spins that relaxes with $1/T_1$ values between a and b. We summarize the results for the main peak of sample A in Fig. 7(a); see Fig. 8 for additional details in the low-temperature region and Fig. 9(a) for the color contour plot.

Looking closely at Fig. 8, we see that for sample A, some components of $1/T_1$ as marked by black downward arrows indeed peak at 9 ± 1 K with $1/T_1 \approx 200$ s^{-1}, in agreement with the sharp peak observed for $1/T_1^{\rm stretch}$ in Fig. 6(a). In addition, notice that an increasing fraction of ^{7}Li sites with $1/T_1$ two

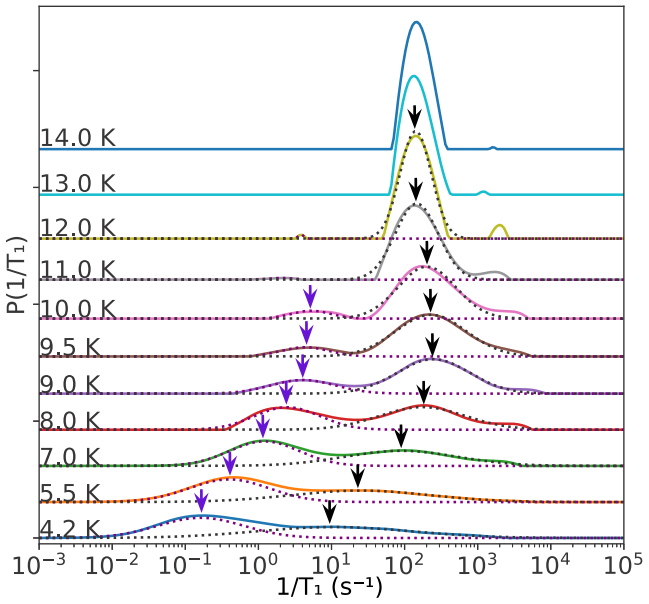


FIG. 8. Zoomed-in view of sample A's main peak ILT below 15 K. The black and purple dotted lines below 13 K indicate double Gaussian fits used to deduce the relative fraction of the extra-slow component. The black and purple arrows indicate the peak of the fast and slow components, respectively.

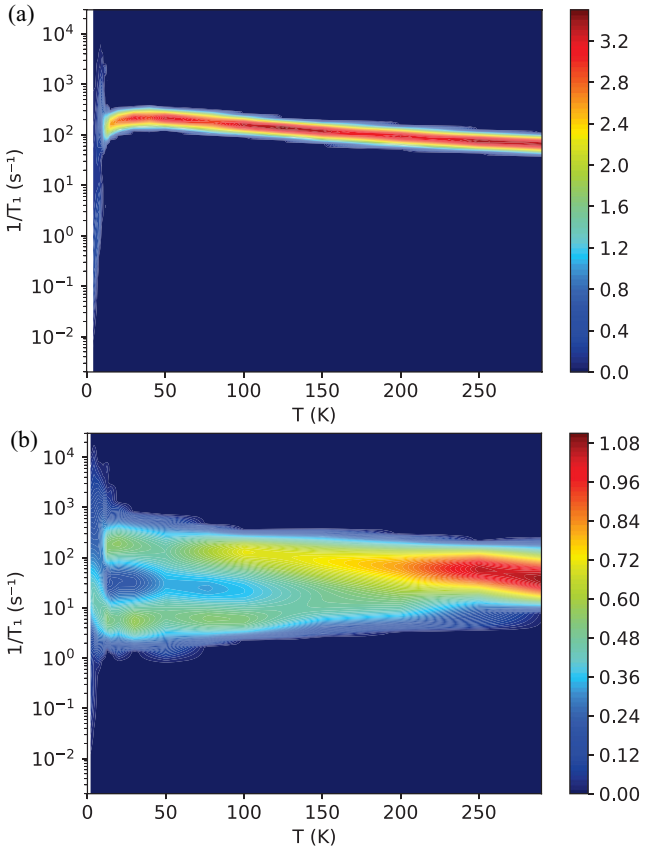


FIG. 9. Contour plot of the density distribution function $P(1/T_1)$ deduced from $M(t)$ by ILT for (a) the main peak of sample A and (b) side peak of sample B. Note that the faster component in (b) shows nearly identical behavior as the result in (a). This means that the fast components observed for the side peak of sample B correspond to the intrinsic behavior of the material arising from the superposed signals from the main peak.

orders of magnitude lower emerges at ∼10 K. These Li sites are surrounded by Ir spins whose spin fluctuation time scale has slowed below the NMR frequency (∼74.4 MHz), owing to static order that is already underway at 10 K. These findings are not revealed by the conventional stretch fit. The ILTT$_1$ analysis is better suited to probe these multiple components of $1/T_1$ with qualitatively different behavior.

At 4.2 K in sample A, a large fraction of the $P(1/T_1)$ (∼60%) appears to be part of the extra-slow component centered around $1/T_1 \sim 20$ s^{-1} or less, although a substantial fraction (∼40%) of ^{7}Li nuclear spins still has $1/T_1$ greater than 1 s^{-1}. We apply a double Gaussian deconvolution of $P(1/T_1)$ below 13 K (dotted curves in Fig. 8) to estimate the relative fraction f_s of this slow component, and plot the results in Fig. 10. From this, we estimate the fraction of static Ir moments f_s at 4.2 K as approximately 60%. This explains the significant decrease in $1/T_1^{\text{stretch}}$ below 10 K.

For comparison, we present the ILT curves observed at the side peak of sample B in Fig. 7(b), and its color contour plot in Fig. 9(b). Notice that the $P(1/T_1)$ curve splits into two distinct components with $1/T_1$ differing by up to two orders of magnitude. This is simply because the side peak of the NMR line shape is superposed by the tail of the main peak as shown in Fig. 3(b), and the latter shows nearly identical behavior as sample A, as readily seen in the similarity between the result in Fig. 9(a) and the upper branch of Fig. 9(b). We also confirmed that the ILT results of $P(1/T_1)$ calculated for the $M(t)$ data measured for sample B at the main peak of the line shape in Fig. 3(b) resemble the results in Fig. 7(b), except the relative intensity of the fast component ($1/T_1 > 10$ s^{-1}) is much higher; this assures us that the fast component peak seen in Fig. 7(b) indeed arises from the superposed signals of the main peak in the line shape. The peak of the slow component in Fig. 7(b) is located at ∼50 s^{-1} near room temperature, and quickly drops to the baseline value of ∼10 s^{-1} by ∼100 K. Combined with the relative intensity ∼17% in the NMR line

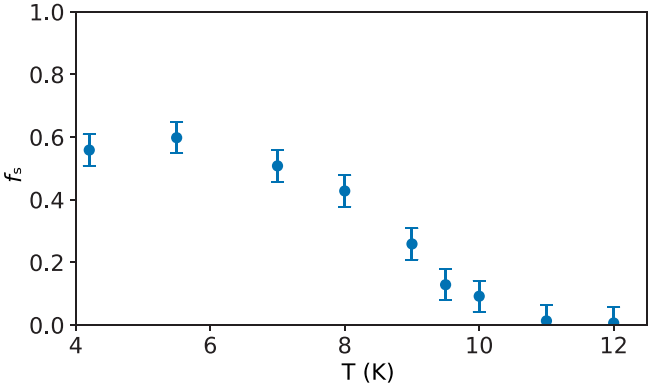


FIG. 10. Fraction of extra-slow component f_s in $P(1/T_1)$ below 13 K for sample A, estimated by double Gaussian deconvolution of $P(1/T_1)$ in Fig. 8.

shape in Fig. 3(b) and the small values of the Knight shift in Fig. 2 observed for the side peak, this finding establishes that ∼17% of Li sites are located in a region with suppressed magnetism. Theoretically, the disorder effects in a Kitaev lattice are predicted to induce randomly emerging singlets accompanied by isolated free spins, generally referred to as orphaned spins in the literature [32,33]. If we attribute the side peak to such singlets, we can estimate their excitation gap to be of the order of ∼100 K in this material. The significant volume of nearly nonmagnetic domains accompanied by orphaned spins explains why the clear signatures of magnetic long-range order can be easily masked by disorder.

IV. CONCLUSIONS

We have probed the intrinsic spin susceptibility and spin dynamics of a clean sample A of $Ag_3LiIr_2O_6$ using ^{7}Li NMR. The sharp peak in $1/T_1^{\text{stretch}}$ along with the behavior of the very fast component in the ILT is consistent with ∼60% of the sample exhibiting long-range ordering starting at $T_N = 9 \pm 1$ K. However, the ILT analysis of the highly distributed $1/T_1$ revealed that some parts are already entering the ordered phase below ∼10 K while some other fraction is still fluctuating even at 4.2 K for a nominally defect-free sample A. This is presumably due to the residual disorder effects arising from stacking faults revealed in TEM measurements.

To elucidate the influence of structural disorder, we also compare the NMR results for a highly disordered sample B with clusters of unwanted Ag occupying the Li and Ir sites in addition to stacking faults. We demonstrated that these domains with Ag inclusion give rise to a new side peak in the NMR line shape with a suppressed Knight shift and $1/T_1$, indicating that disorder locally suppresses magnetism.

The highly disordered sample of $Ag_3LiIr_2O_6$ is similar to other Kitaev materials such as Cu_2IrO_3 and $H_3LiIr_2O_6$ in that there is no definitive evidence of long-range order, such as diverging $1/T_1$ [16,18]. The absence of such a clear-cut signature of long-range order led to earlier proposals for the quantum spin-liquid ground state in these materials, but our findings for the disordered sample B here and elsewhere [29] indicate that disorder could easily mask the signature of long-range order.

ACKNOWLEDGMENTS

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APPENDIX: INVERSE LAPLACE TRANSFORM

We can numerically deduce the density distribution function $P(1/T_1)$ in Eq. (2) using ILTT$_1$ analysis technique without relying on phenomenological models by computationally inverting $M(t)$. This was done previously in Cu_2IrO_3, $La_{1.875}Ba_{0.125}CuO_6$, and $La_{1.885}Sr_{0.115}CuO_4$ and had revealed information beyond what was shown by the stretch fit and two-component fit [18,30,39,40].

Here, we provide an outline of the ILTT$_1$ analysis. For a discrete range of time steps $\{t_i\}$, we can reduce Eq. (2) to its vector form,

$$\mathbf{M} = K\mathbf{P} + \mathbf{E}, \quad \mathbf{P} = \{P(1/T_{1,i}) \geqslant 0\}, \tag{A1}$$

where K is the kernel matrix and $\mathbf{E}$ is a vector representing Gaussian noise. For a sufficient number of $1/T_{1,i}$, this is an ill-posed problem with nonunique solutions that are sensitive to noise [41,42]. We thus use Tikhonov regularization to introduce a smoothing parameter α, such that we can find the unique solution $\mathbf{P}$ under the constraint $\mathbf{P} \geqslant \mathbf{0}$ which minimizes the cost function,

$$\mathbf{P} = \underset{\mathbf{P}\geqslant 0}{\arg\min} |\mathbf{M} - K\mathbf{P}|^2 + \alpha|\mathbf{P}|^2, \tag{A2}$$

where $|\cdots|$ is the vector norm. To prevent overfitting (the ILT distribution becomes greatly affected by the noise) and underfitting (important information in $M(t)$ becomes lost), α is chosen such that the ILT fit deviates from $M(t)$ in proportion to the experimental noise. We refer readers to Sec. II of Ref. [30] and its Appendix for more details.

[1] L. Balents, Nature (London) **464**, 199 (2010).
[2] A. Kitaev, Ann. Phys. **321**, 2 (2006).
[3] G. Jackeli and G. Khaliullin, Phys. Rev. Lett. **102**, 017205 (2009).
[4] H. Takagi, T. Takayama, G. Jackeli, G. Khaliullin, and S. E. Nagler, Nat. Rev. Phys. **1**, 264 (2019).
[5] Y. Singh and P. Gegenwart, Phys. Rev. B **82**, 064412 (2010).
[6] G. Simutis, N. Barbero, K. Rolfs, P. Leroy-Calatayud, K. Mehlawat, R. Khasanov, H. Luetkens, E. Pomjakushina, Y. Singh, H.-R. Ott, J. Mesot, A. Amato, and T. Shiroka, Phys. Rev. B **98**, 104421 (2018).
[7] R. Sarkar, Z. Mei, A. Ruiz, G. Lopez, H.-H. Klauss, J. G. Analytis, I. Kimchi, and N. J. Curro, Phys. Rev. B **101**, 081101(R) (2020).
[8] Y. Singh, S. Manni, J. Reuther, T. Berlijn, R. Thomale, W. Ku, S. Trebst, and P. Gegenwart, Phys. Rev. Lett. **108**, 127203 (2012).
[9] P. Khuntia, S. Manni, F. R. Foronda, T. Lancaster, S. J. Blundell, P. Gegenwart, and M. Baenitz, Phys. Rev. B **96**, 094432 (2017).
[10] K. W. Plumb, J. P. Clancy, L. J. Sandilands, V. V. Shankar, Y. F. Hu, K. S. Burch, H.-Y. Kee, and Y.-J. Kim, Phys. Rev. B **90**, 041112(R) (2014).
[11] L. J. Sandilands, Y. Tian, K. W. Plumb, Y.-J. Kim, and K. S. Burch, Phys. Rev. Lett. **114**, 147201 (2015).
[12] S.-H. Baek, S.-H. Do, K.-Y. Choi, Y. S. Kwon, A. U. B. Wolter, S. Nishimoto, J. van den Brink, and B. Büchner, Phys. Rev. Lett. **119**, 037201 (2017).

[13] J. Zheng, K. Ran, T. Li, J. Wang, P. Wang, B. Liu, Z.-X. Liu, B. Normand, J. Wen, and W. Yu, Phys. Rev. Lett. **119**, 227208 (2017).
[14] N. Janša, A. Zorko, M. Gomilšek, M. Pregelj, K. W. Krämer, D. Biner, A. Biffin, C. Rüegg, and M. Klanjšek, Nat. Phys. **14**, 786 (2018).
[15] Y. Nagai, T. Jinno, J. Yoshitake, J. Nasu, Y. Motome, M. Itoh, and Y. Shimizu, Phys. Rev. B **101**, 020414(R) (2020).
[16] K. Kitagawa, T. Takayama, Y. Matsumoto, A. Kato, R. Takano, Y. Kishimoto, S. Bette, R. Dinnebier, G. Jackeli, and H. Takagi, Nature (London) **554**, 341 (2018).
[17] M. Abramchuk, C. Ozsoy-Keskinbora, J. W. Krizan, K. R. Metz, D. C. Bell, and F. Tafti, J. Am. Chem. Soc. **139**, 15371 (2017).
[18] S. K. Takahashi, J. Wang, A. Arsenault, T. Imai, M. Abramchuk, F. Tafti, and P. M. Singer, Phys. Rev. X **9**, 031047 (2019).
[19] E. M. Kenney, C. U. Segre, W. Lafargue-Dit-Hauret, O. I. Lebedev, M. Abramchuk, A. Berlie, S. P. Cottrell, G. Simutis, F. Bahrami, N. E. Mordvinova, G. Fabbris, J. L. McChesney, D. Haskel, X. Rocquefelte, M. J. Graf, and F. Tafti, Phys. Rev. B **100**, 094418 (2019).
[20] K. Geirhos, P. Lunkenheimer, M. Blankenhorn, R. Claus, Y. Matsumoto, K. Kitagawa, T. Takayama, H. Takagi, I. Kézsmárki, and A. Loidl, Phys. Rev. B **101**, 184410 (2020).
[21] J. c. v. Chaloupka, G. Jackeli, and G. Khaliullin, Phys. Rev. Lett. **105**, 027204 (2010).
[22] J. G. Rau, E. K.-H. Lee, and H.-Y. Kee, Annu. Rev. Condens. Matter Phys. **7**, 195 (2016).
[23] J. Chaloupka, G. Jackeli, and G. Khaliullin, Phys. Rev. Lett. **110**, 097204 (2013).
[24] X. Liu, T. Berlijn, W.-G. Yin, W. Ku, A. Tsvelik, Y.-J. Kim, H. Gretarsson, Y. Singh, P. Gegenwart, and J. P. Hill, Phys. Rev. B **83**, 220403(R) (2011).
[25] H. B. Cao, A. Banerjee, J.-Q. Yan, C. A. Bridges, M. D. Lumsden, D. G. Mandrus, D. A. Tennant, B. C. Chakoumakos, and S. E. Nagler, Phys. Rev. B **93**, 134423 (2016).
[26] J. A. Sears, M. Songvilay, K. W. Plumb, J. P. Clancy, Y. Qiu, Y. Zhao, D. Parshall, and Y.-J. Kim, Phys. Rev. B **91**, 144420 (2015).
[27] F. Bahrami, W. Lafargue-Dit-Hauret, O. I. Lebedev, R. Movshovich, H.-Y. Yang, D. Broido, X. Rocquefelte, and F. Tafti, Phys. Rev. Lett. **123**, 237203 (2019).
[28] S. Bette, T. Takayama, V. Duppel, A. Poulain, H. Takagi, and R. Dinnebier, Dalton Trans. **48**, 9250 (2019).
[29] F. Bahrami, E. M. Kenney, C. Wang, A. Berlie, O. I. Lebedev, M. J. Graf, and F. Tafti, Phys. Rev. B **103**, 094427 (2021).
[30] P. M. Singer, A. Arsenault, T. Imai, and M. Fujita, Phys. Rev. B **101**, 174508 (2020).
[31] T. Mitsui and E. Nakamura, in *Numerical Data and Functional Relationships in Science and Technology,* edited by K. H. Hellwege, Landolt-Börnstein, New Series, Group III, Vol. 16 (Springer, Berlin, 1982), p. 402.
[32] I. Kimchi, J. P. Sheckelton, T. M. McQueen, and P. A. Lee, Nat. Commun. **9**, 4367 (2018).
[33] J. Knolle, R. Moessner, and N. B. Perkins, Phys. Rev. Lett. **122**, 047202 (2019).
[34] J. Knolle, D. L. Kovrizhin, J. T. Chalker, and R. Moessner, Phys. Rev. B **92**, 115127 (2015).
[35] J. Yoshitake, J. Nasu, and Y. Motome, Phys. Rev. Lett. **117**, 157203 (2016).
[36] J. Yoshitake, J. Nasu, Y. Kato, and Y. Motome, Phys. Rev. B **96**, 024438 (2017).
[37] J. Yoshitake, J. Nasu, and Y. Motome, Phys. Rev. B **96**, 064433 (2017).
[38] T. Ohno, H. Alloul, and P. Mendels, J. Phys. Soc. Jpn. **59**, 1139 (1990).
[39] A. Arsenault, T. Imai, P. M. Singer, K. M. Suzuki, and M. Fujita, Phys. Rev. B **101**, 184505 (2020).
[40] P. M. Singer, D. Asthagiri, W. G. Chapman, and G. J. Hirasaki, J. Chem. Phys. **148**, 204504 (2018).
[41] J. Mitchell, T. Chandrasekera, and L. Gladden, Prog. Nucl. Magn. Reson. Spectrosc. **62**, 34 (2012).
[42] D. Asthagiri, W. G. Chapman, G. J. Hirasaki, and P. M. Singer, J. Phys. Chem. B **124**, 10802 (2020).

Chapter 5

Inhomogeneous Ground States of $ZnCu_3(OD)_6FBr$ and $ZnCu_3(OD)_6Cl_2$ (^{63}Cu NQR)

5.1 Spin Gap Controversy

Though it is well-established that the ground states of herbertsmithite and Zn-barlowite do not undergo magnetic LRO or (complete) spin-freezing, their exact nature has yet to be fully understood. A key question is whether their spin excitation spectrum is intrinsically gapped [91] or gapless [92], for which there exists conflicting experimental evidence in favour of either case. Adding the fuel to this experimental controversy was the disagreement amongst theorists about whether the ground state of a pristine KHA is gapped or gapless (see Sec. 1.3).

At the heart of this issue is the disorder present in these materials. In the highest-quality samples known to date, site-specific X-ray scattering experiments found that $S = 1/2$ Cu^{2+} defects occupy 15% and 5% of the non-magnetic interlayer Zn^{2+} sites in herbertsmithite [93] and Zn barlowite [94], respectively. Their chemical compositions

are thus $Zn_{0.85}Cu_{3.15}(OD)_6Cl_2$ for herbertsmithite and $Zn_{0.95}Cu_{3.05}(OD)_6FBr$ for Zn-barlowite. These experiments also show no indication of Zn^{2+} defects occupying the kagome Cu^{2+} sites, setting an upper bound of less than $\sim 1\%$ Zn^{2+} occupancy.

Each interlayer Zn^{2+} site neighbors three Cu^{2+} sites in each of the adjacent kagome planes, as shown in Fig. 5.1. This means that the magnetic moments of the interlayer Cu^{2+} defects can each interact with six nearest-neighbour Cu^{2+} spins in the adjacent kagome planes, potentially disrupting the formation of a QSL ground state. This spatial inhomogeneity motivates the use of local probes such as NMR in studying these materials.

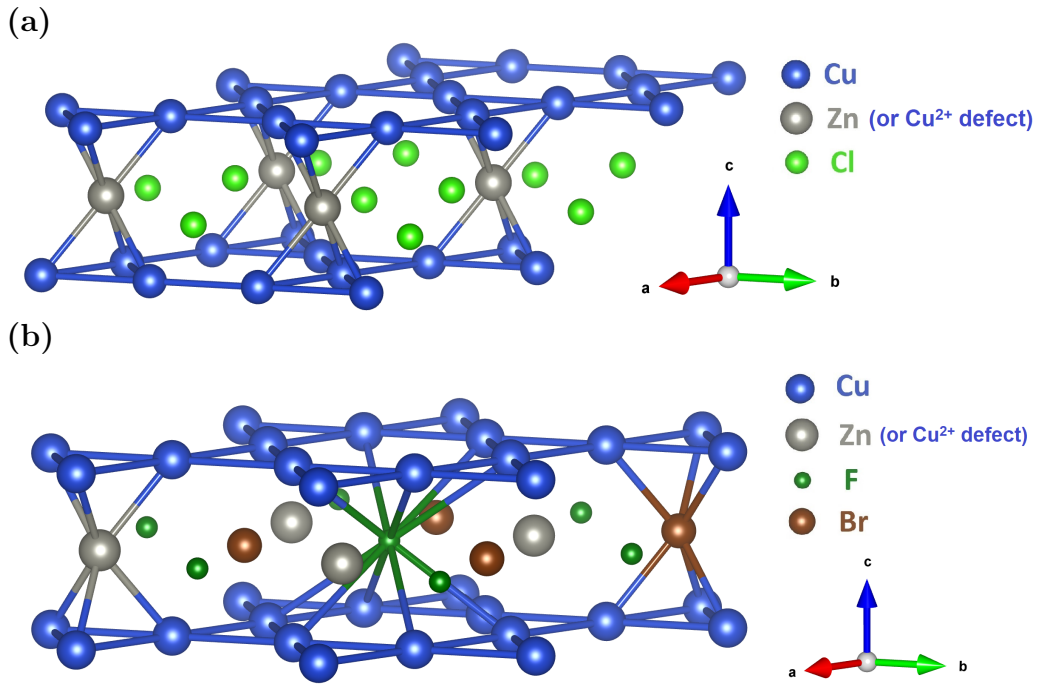


Figure 5.1: Cu^{2+} nearest-neighbors of interlayer atoms in **(a)** herbertsmithite and **(b)** Zn-barlowite. Each Zn^{2+} (or Cu^{2+} defect) neighbors 6 Cu^{2+} sites in the adjacent kagome planes. Also in Zn-barlowite, each Br^- and F^- neighbors 6 and 12 Cu^{2+} sites, respectively.

5.2 Spin Dynamics Probed via $1/T_1$

There are several ways of elucidating the KHA ground state using NMR. A primary method, which we will discuss in Chpt. 6, is to probe the intrinsic spin susceptibility of the kagome planes χ_{kagome} by measuring the Knight shift of nuclei coupled to χ_{kagome}. The other key method is through the nuclear spin-lattice relaxation rate $1/T_1$, which reveals the local behaviour of low-energy spin fluctuations (see Sec. 2.7).

If all of the Cu^{2+} spins were paired into magnetically-inert spin singlets and a spin gap Δ was homogeneously formed in the system, $1/T_1$ becomes dramatically suppressed, such as in the cases of $SrCu_2O_3$ (2-leg spin-ladder), $CuGeO_3$ (spin-Peierls chain), and $SrCu_2(BO_3)_3$ (Shastry-Sutherland lattice) [95–98]. Near $T = 0$, $1/T_1$ follows an activation law, such that $1/T_1 \propto \exp(-\Delta/k_BT)$.

On the other hand, for Heisenberg antiferromagnets in the high-temperature paramagnetic regime, $1/T_1$ generally behaves as $1/T_1 \propto |A_{hf}|^2/J_H$ [99], unless suppressed by a spin gap or enhanced by the onset of magnetic order. In the case of Cu^{2+} nuclei at the sites of the paramagnetically fluctuating spins, the value of $1/T_1$ is generally measured to be $\sim 10^3$ s^{-1} to $\sim 10^4$ s^{-1} [95, 97, 100].

Soon after the discovery of herbertsmithite as a QSL candidate, initial ^{63}Cu, ^{35}Cl, and ^{17}O NMR measurements (on powder samples) appeared to exhibit power law behaviours in $1/T_1$ at low temperatures, such that $1/T_1 \propto T^{\eta}$, where $0 < \eta \leq 1$ [77, 101], which are consistent with a gapless spin liquid picture based on a Dirac fermions model [26]. Subsequent ^{17}O NMR measurements on single crystal samples of herbertsmithite [92] also claimed to have found $1/T_1$ following this power law. However, the disorder in herbertsmithite and Zn-barlowite causes them to develop locally inhomogeneous spin dynamics, manifesting as a distribution in $1/T_1$ which becomes

especially broad at low temperatures. To account for this distribution, all these past $1/T_1$ studies employed a stretched exponential fit of the recovery curve $M(t_D)$, which can overlook crucial information about the distribution of $1/T_1$ (demonstrated in Sec. 3.4).

As will be discussed in Chpt. 6, magnetic disorder can be accounted for by careful analysis of the distribution of the Knight shift (i.e. NMR lineshape). Through this, previous ^{17}O and ^{2}D NMR studies on herbertsmithite single crystals were able to differentiate the intrinsic and defect-induced behaviours [91, 102]. It stands to reason that the same care should be applied to the analysis of $1/T_1$. This is a key motivation for our development of the ILT $1/T_1$ analysis program, enabling us to revisit these materials and deduce the distribution of $1/T_1$.

Since the the physics of the kagome QSL is defined by the behaviour of the electron spins located at the Cu^{2+} sites within the kagome planes, our most direct NMR probe is via the ^{63}Cu and ^{65}Cu nuclei, which have the strongest hyperfine coupling to these spins. Since the large nuclear quadrupole moments of Q of ^{63,65}Cu ($I = 3/2$) splits their powder NMR lineshapes at high fields, a more effective probe of T_1^{Cu} is via ^{63,65}Cu NQR (i.e. $B_{\mathrm{ext}} = 0$).

5.3 Conventional NQR Results

Deuterated powder samples of herbertsmithite ($ZnCu_3(OD)_6Cl_2$) and Zn-barlowite ($ZnCu_3(OD)_6FBr$) were synthesized by Prof. Young S. Lee's group at Stanford University, based on the procedures outlined in Ref. [16] and [21]. We conducted NQR measurements on both samples using spin echo and $1/T_1$ inversion recovery pulse sequences, using typical pulse widths of $t_{180} = 2t_{90} \approx 5\mu$s and pulse separation times

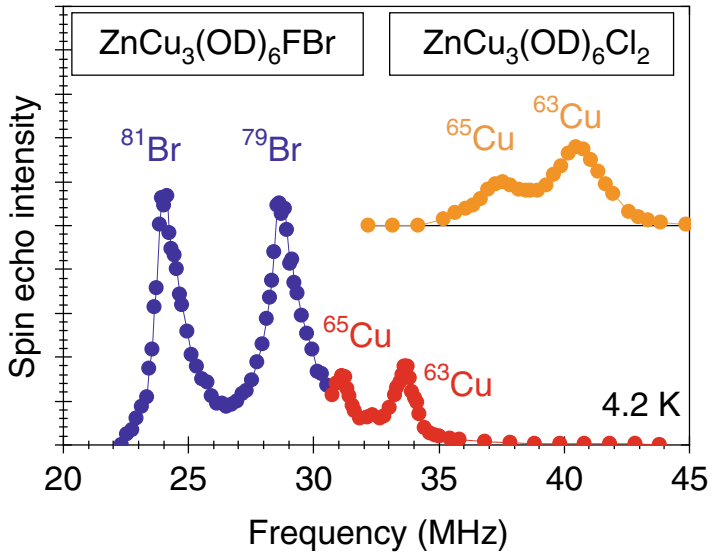


Figure 5.2: NQR lineshapes of $ZnCu_3(OD)_6FBr$ (bottom left) and $ZnCu_3(OD)_6Cl_2$ (top right), measured at $T = 4.2$ K. Figure adopted from Ref. [46]

ranging from $\tau = 6$ μs to $\tau = 30$ μs.

We show the NQR lineshapes from 20 to 45 MHz (measured with $\tau = 6$ μs) in Fig. 5.2. Both lineshapes exhibit a pair of peaks corresponding to ^{63}Cu and ^{65}Cu, which have similar values of Q (listed in Table 2.1). The ^{65}Cu peak for $ZnCu_3(OD)_6FBr$ also overlaps with those of ^{79}Br and ^{81}Br, which are both also $I = 3/2$ and have a smaller but comparable Q.

Our findings related to ^{79,81}Br NQR also have significant implications on the disorder and lattice dynamics of $ZnCu_3(OD)_6FBr$. Specifically, they evidence the gradual development of local lattice distortions which start around 160 K and become frozen below 50 K [103]. A similar lattice freezing is known to occur in $ZnCu_3(OD)_6Cl_2$ [101]. An in-depth discussion of ^{79,81}Br NQR is beyond the scope of this thesis, but can be found in Ref. [46, 103].

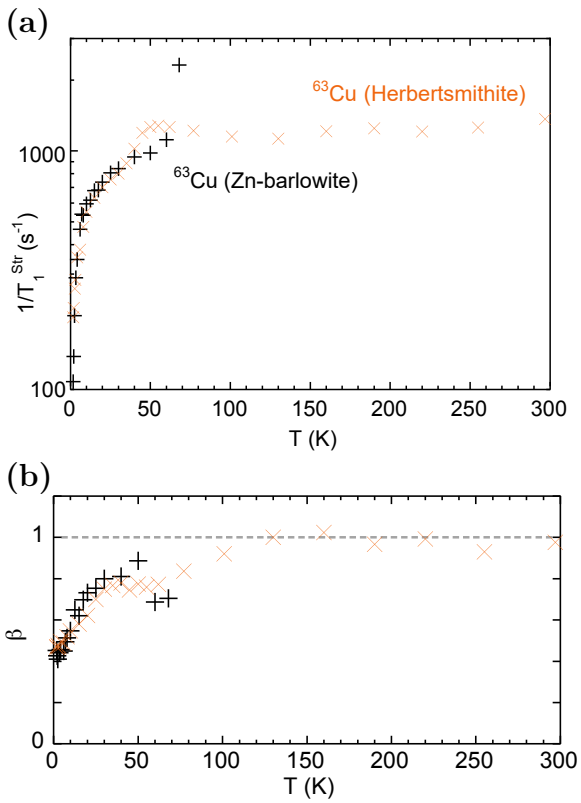


Figure 5.3: Temperature dependence of ^{63}Cu **(a)** $1/T_{1,str}^{Cu}$ and **(b)** β of $ZnCu_3(OD)_6FBr$ (black +) and $ZnCu_3(OD)_6Cl_2$ (orange $\times$) measured at the ^{63}Cu NQR peak. Figure adapted from Supplemental Materials of Ref. [46]

We measured the $1/T_1$ recovery curve $M(t_D)$ (using $\tau = 15$ μs) at the ^{63}Cu NQR peak for both samples (since it has a higher intensity than the ^{65}Cu peak), using the inversion recovery method outline in Sec. 2.5. We obtain $1/T_{1,str}^{Cu}$ via a stretched exponential fit of $M(t_D)$ to Eq. 2.6.6, and plot its temperature dependence in Fig. 5.3 **(a)**.

Note the freezing of lattice distortions causes $1/T_{1,str}^{Cu}$ to be enhanced near 100 K in $ZnCu_3(OD)_6FBr$ and near 50 K in $ZnCu_3(OD)_6Cl_2$, where the fluctuation rate of the slowing EFG crosses over the NMR/NQR frequency as the fluctuations of the

lattice freezes.

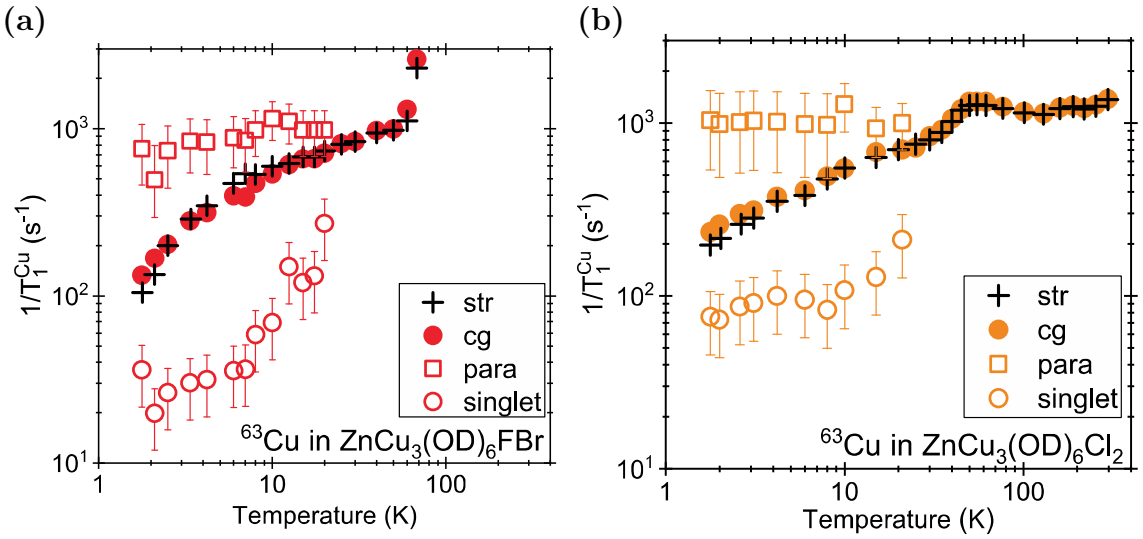


Figure 5.4: Temperature dependence of $1/T_{1,str}^{Cu}$ (black +) measured at the ^{63}Cu peak of **(a)** Zn-barlowite and **(b)** Herbertsmithite. Also plotted are $1/T_{1,para}^{Cu}$ (red □) and $1/T_{1,singlet}^{Cu}$ (red ○) corresponding to the singlet and paramagnetic components, respectively, as well as the $P(1/T_1^{Cu})$ center of gravity $1/T_{1,COG}^{Cu}$ (red •). Figure adopted from Ref. [46]

On a log-log plot of $1/T_1^{Cu}$ vs T (Fig. 5.4), we see that below $\sim$ 50 K, $1/T_{1,str}^{Cu}$ of $ZnCu_3(OD)_6Cl_2$ (represented by + symbols) is consistent with the power-law behaviour previously observed in Ref. [101]. On the other hand, $1/T_{1,str}^{Cu}$ of $ZnCu_3(OD)_6FBr$ does not appear to follow an activation curve, nor the power-law expected for gapless Dirac fermions. In any case, the decreases in $1/T_{1,str}^{Cu}$ (from $\sim 10^3$ s^{-1} to $\sim 10^2$ s^{-1}) exhibited by both materials at low temperatures, are much milder compared to what is expected for homogeneously gapped Cu^{2+} spin singlets.

Loosely speaking, these are the conclusions which can be drawn based solely on $1/T_{1,str}^{Cu}$. However, we know that $1/T_1^{Cu}$ develops a broad distribution starting below $\sim$ 40 K, as is evidenced by the decrease in the stretch fit parameter β in Fig. 5.3 **(b)**, which becomes as low as $\sim$ 0.4 at 1.8 K. Our next step should therefore be to

analyze the full distribution of $1/T_1^{Cu}$ using ILT.

5.4 ILT Analysis of ^{63}Cu $1/T_1$

We compute the ILT of the $M(t_D)$ via the Tikhonov regularization method described in Chpt. 3, where we opt to use a fixed smoothing parameter value of $\alpha = 3$. The temperature evolution of the resulting probability density distributions $P(1/T_1^{Cu})$ are combined and plotted in Fig. 5.5.

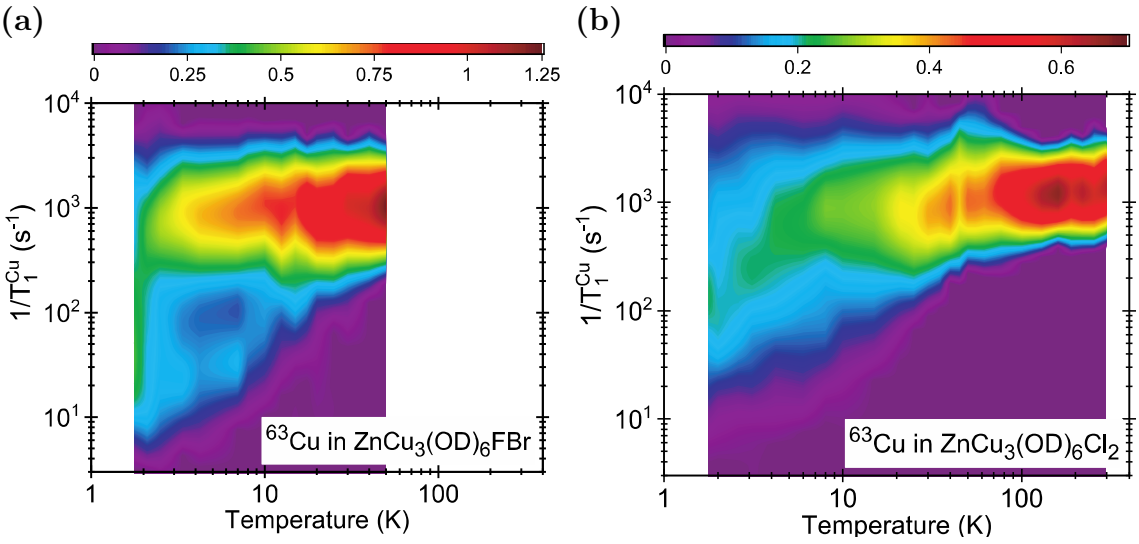


Figure 5.5: Color contour maps showing the temperature evolution of $P(1/T_1^{Cu})$ of **(a)** $ZnCu_3(OD)_6FBr$ and **(b)** $ZnCu_3(OD)_6Cl_2$, where each vertical slice corresponds to $P(1/T_1^{Cu})$ measured at the ^{63}Cu NQR peak for a given temperature. The integrated area of $P(1/T_1^{Cu})$ at each temperature is normalized to 1. Figure adopted from Ref. [46]

Above $T = 40$ K, $P(1/T_1^{Cu})$ of both samples consists of a single peak whose $1/T_1$ remains relatively constant (~ 1000 s^{-1}) across all temperatures (excluding the region where lattice freezing takes place). This temperature-independent $1/T_1$ value is typical for paramagnetic Cu^{2+} sites.

Starting below $\sim$ 30 K in both samples, a new component in $P(1/T_1^{\mathrm{Cu}})$ with progressively slower values of $1/T_1$ splits off (this corresponds to the diagonal blue ridges in Fig. 5.5 **(a)** and **(b)**). Note that the paramagnetic component in $P(1/T_1^{\mathrm{Cu}})$ persists at low temperatures and coexist alongside the split-off component, meaning that the decrease in $1/T_{1,str}^{\mathrm{Cu}}$ can be thought of as due to the increasing spectral weight and decreasing $1/T_1$ of the split-off component.

We can deduce the individual $1/T_1$ of the 2 components by fitting $P(1/T_1^{\mathrm{Cu}})$ with a double Gaussian function

$$P(1/T_1^{\mathrm{Cu}}) = \sum_{i=para,singlet} \frac{A_i}{\sigma_{T,i}} \exp\left[-\frac{1}{2}\left(\frac{\log_{10}\left(T_{1,i}^{\mathrm{Cu}}/T_1^{\mathrm{Cu}}\right)}{\sigma_{T,i}}\right)^2\right], \tag{5.4.1}$$

where the paramagnetic and split-off components are modeled as a Gaussian distributions centered at $1/T_{1,para}^{\mathrm{Cu}}$ and $1/T_{1,singlet}^{\mathrm{Cu}}$, respectively. We plot $1/T_{1,para}^{\mathrm{Cu}}$ and $1/T_{1,singlet}^{\mathrm{Cu}}$ in Fig. 5.4, alongside $1/T_{1,str}^{\mathrm{Cu}}$ and the $P(1/T_1^{\mathrm{Cu}})$ center of gravity $1/T_{1,COG}^{\mathrm{Cu}}$.

We show the double Gaussian deconvolution of $P(1/T_1^{\mathrm{Cu}})$ at select temperatures for both samples in Fig. 5.6. Below $\sim$ 20 K, $1/T_{1,singlet}^{\mathrm{Cu}}$ deviates significantly from $1/T_{1,para}^{\mathrm{Cu}}$, becoming as slow as $\sim$ 20 s^{-1} for $ZnCu_3(OD)_6FBr$ and 80 s^{-1} for $ZnCu_3(OD)_6Cl_2$ at the base temperature (1.8 K). Such a slow value of $1/T_{1,singlet}^{\mathrm{Cu}}$ are comparable to the $1/T_1$ exhibited by Cu^{2+} in the aforementioned spin-dimer materials with gapped spin-singlet ground states. The gradual formation of local spin singlets has been known to occur in disordered antiferromagnets [104]. We therefore deduce that the emergent slow component in $P(1/T_1^{\mathrm{Cu}})$ originate from Cu^{2+} sites which have been paired into spin-singlets.

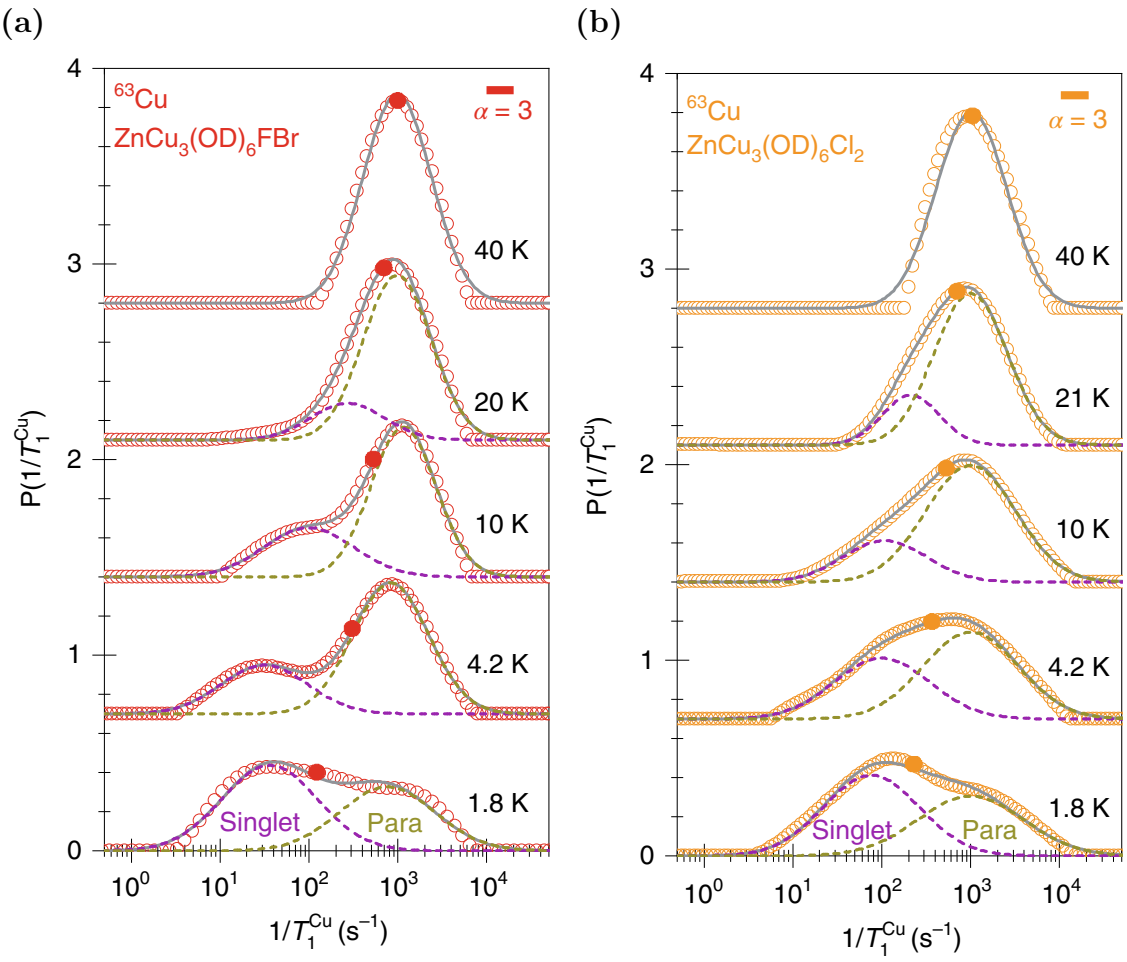


Figure 5.6: Normalized $P(1/T_1^{Cu})$ distributions of **(a)** $ZnCu_3(OD)_6FBr$ (red ∘) and **(b)** $ZnCu_3(OD)_6Cl_2$ (orange ∘) at select temperatures (origins are vertically cascaded for visual clarity). The double Gaussian fit of $P(1/T_1^{Cu})$ (solid grey curve) equals the sum of the singlet (dashed purple curve) and paramagnetic (dashed gold curve) components. The filled circles indicate $1/T_{1,COG}^{Cu}$. Figure adopted from Ref. [46]

The floor value of $1/T_{1,singlet}^{Cu}$ being larger in $ZnCu_3(OD)_6Cl_2$ than in $ZnCu_3(OD)_6FBr$ suggests it may be influenced by the factor of ∼3 greater Cu^{2+} defect spin concentration. This is because the intrinsic magnetic fluctuations of the kagome planes could potentially be overshadowed by a background contribution to $1/T_1^{Cu}$ originating from

the interlayer Cu^{2+} defect spins, which exists everywhere in the sample. This difference in the lower limit of $1/T_{1,singlet}^{Cu}$ also explains why the singlet component is less distinguishable in $ZnCu_3(OD)_6Cl_2$.

5.5 Spin Singlets with Inhomogenous Gaps

From the relative spectral weight of the singlet and paramagnetic components in the double Gaussian deconvolution of $P(1/T_1^{Cu})$ ($A_{singlet}$ and A_{para}), we can estimate the fraction of Cu^{2+} sites in the kagome planes that are involved in spins-singlet formation as $F_{singlet} = A_{singlet}/(A_{singlet} + A_{para})$, and plot its temperature dependence in Fig. 5.7. Remarkably, the behaviour of $F_{singlet}$ is nearly identical in $ZnCu_3(OD)_6FBr$ and $ZnCu_3(OD)_6Cl_2$.

In both materials, kagome Cu^{2+} spins begin to form singlets around ~ 30 K and accumulate with decreasing temperature, reaching up to $F_{singlet} \sim 50\%$ at 1.8 K. This growth in $F_{singlet}$ implies that some parts of these samples indeed approach a ground state with a spin-excitation gap, but the energy of the gap Δ varies throughout the sample. We estimate Δ to be less than ~ 10 K for most of the singlets, but varies significantly as to be as large $\Delta \sim 30$ K for some singlets.

Bear in mind that $F_{singlet}$ estimated for the data acquired with a fixed $\tau = 6$ μs should be considered as an upper bound for the true fraction of Cu^{2+} spin singlets. This is because the $1/T_2$ relaxation rate is faster for paramagnetic Cu^{2+} than for singlets. This means that when using finite values of τ, the paramagnetic component in $P(1/T_1^{Cu})$ is suppressed (as demonstrated in Fig. 5.8). Since $F_{singlet}$ is measured for $\tau = 6$ μs, it should be smaller when extrapolated to $\tau = 0$ μs. Furthermore, note that $F_{singlet}$ was measured at the peak of the NQR lineshape, and $P(1/T_1^{Cu})$ can potentially

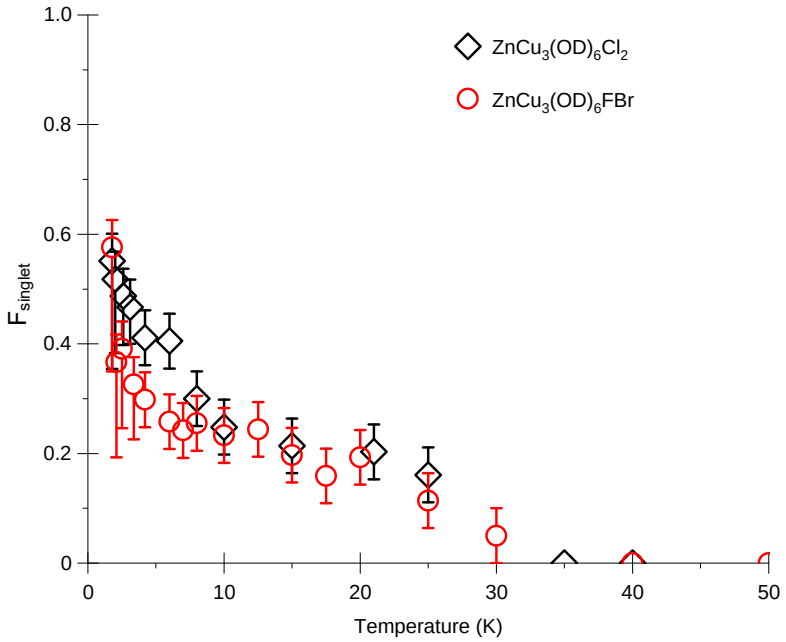


Figure 5.7: Temperature dependence of (the upper bound of) the spin-singlet fraction $F_{singlet}$ in $ZnCu_3(OD)_6FBr$ (red $\circ$) and $ZnCu_3(OD)_6Cl_2$ (black $\diamond$). Figure adapted from Ref. [46]

vary as a function of ν_Q (this occurs in Chpt. 6 for ^{19}F). Specifically, it is possible that the spectral width $\Delta\nu_Q$ can be different for the singlet and paramagnetic components, which scales their total intensity when integrated over frequency. Regardless of the exact value of $F_{singlet}$, it is clear that below 30 K, some fraction of Cu sites are involved in spin singlet formation, while the rest remain paramagnetic with a roughly constant value of $1/T^{\mathrm{Cu}}_{1,para} \approx 10^3 \mathrm{s}^{-1}$.

Notice that once the spin-singlet component emerges, $1/T^{\mathrm{Cu}}_{1,str}$ reflects neither $1/T^{\mathrm{Cu}}_{1,para}$ nor $1/T^{\mathrm{Cu}}_{1,singlet}$, but instead resembles their weighted average (i.e. $1/T^{\mathrm{Cu}}_{1,COG}$). This means that the behaviour of $1/T^{\mathrm{Cu}}_{1,str}$ can fail to represent the intrinsic spin dynamics, analogous to how χ_{bulk} can fail to represent the intrinsic magnetic susceptibility, and can be outright deceptive if taken at face value. Recall that in

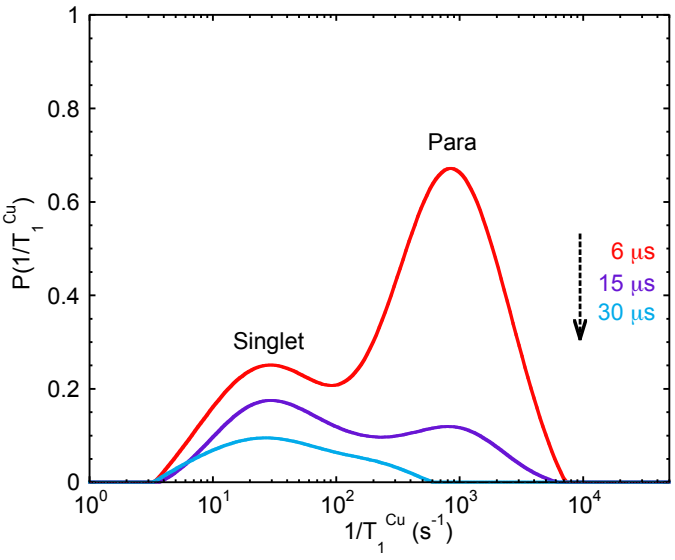


Figure 5.8: $P(1/T_1^{\mathrm{Cu}})$ of $ZnCu_3(OD)_6FBr$ measured at 4.2 K using various different pulse separation $\tau = 6$ μs, 15 μs, and 30 μs, Figures adopted from Supplemental Materials of Ref. [46]

$ZnCu_3(OD)_6Cl_2$, even though $1/T_{1,str}^{\mathrm{Cu}}$ showed an apparent power law behaviour expected for gapless spin liquids, neither $1/T_{1,para}^{\mathrm{Cu}}$ nor $1/T_{1,singlet}^{\mathrm{Cu}}$ did so. This multi-component behaviour in $P(1/T_1^{\mathrm{Cu}})$ therefor rules out the possibility of a uniformly gapless ground state.

It appears very unlikely that these samples of $ZnCu_3(OD)_6FBr$ and $ZnCu_3(OD)_6Cl_2$ approach a homogeneous ground state in which 100% of Cu^{2+} are paired into spin-singlets, as even the upper bound of the singlet fraction $F_{singlet}$ does not exceed 60% down to 1.8 K (≈ 0.01 J). Thus, theoretical models which attempt to describe the KHA ground state in should take into account the presence of disorder, such as in [22, 29–31].

Chapter 6

Emergence of Spin-Polarized Domains in $ZnCu_3(OD)_6FBr$ (^{19}F NMR)

6.1 Interlayer Defect Spins

The spin susceptibility of a pristine kagome lattice χ_{kagome} is predicted to decrease at low temperatures, as is typical in correlated antiferromagnets. Its behaviour near $T = 0$ depends on whether or not there is a spin gap, where χ_{kagome} is expected to decrease exponentially in gapped pictures [25], and exhibit T-linear behaviour in gapless pictures [26].

However, χ_{kagome} is not fully characterized by the bulk magnetic susceptibility χ_{bulk}, especially in the low-temperature region of key interest. χ_{bulk} of $ZnCu_3(OD)_6Cl_2$ and $ZnCu_3(OD)_6FBr$ both exhibit an upturn below $\sim$ 50 K. This appears to be a Curie-Weiss component arising from uncoupled (or weakly-coupled) paramagnetic

moments, such as the interlayer Cu^{2+} defect spins. In the random-singlet picture, orphaned spins within the kagome planes are shown to also contribute to the upturn in χ_{bulk} [22, 31, 104].

Early ^{35}Cl and ^{17}O NMR measurements on $ZnCu_3(OD)_6Cl_2$ powder samples exhibit magnetically broadened lineshapes which indicate a highly distributed local spin susceptibility [77, 101]. At low temperatures ($T <\sim 50$ K), some components appear to approach zero shift, while the overall lineshape width diverges. Later ^{17}O NMR and ^{2}D NMR experiments on $ZnCu_3(OD)_6Cl_2$ single crystal samples were able to better differentiate the Knight shifts corresponding to different magnetic domains [91, 92, 102]. These studies show that the interlayer Cu^{2+} defects induce a negative hyperfine magnetic field B_{hyp} in its vicinity, within the nearby kagome planes. Thus, sites neighboring the defects have NMR lineshapes which become (on average) negatively shifted and significantly broadened at low temperatures, in accordance with χ_{bulk}.

In contrast, sites that are farther away from the defects exhibit Knight shifts which decrease towards zero, and better represent the intrinsic χ_{kagome}. The ^{17}O NMR studies (which employ different methods in their experiments and analyses) drew opposing conclusions about whether the intrinsic component of the Knight shift demonstrates gapped [91] or gapless [92] behaviour. Nonetheless, both studies are consistent with our ^{63}Cu NQR results, which show that $ZnCu_3(OD)_6Cl_2$ is neither uniformly gapped nor gapless, and instead some fraction of kagome Cu^{2+} spins are inhomogeneously gapped. The gapped behaviour displayed in the central ^{17}O peak in Ref. [91] can be attributed to ^{17}O which are situated between 2 kagome Cu^{2+} sites involved in spin-singlet formation. Meanwhile, the apparent $1/T_{1,str}^{O}$ power law seen in Ref. [92] is likely an average of multiple $1/T_1$ components.

In any case, it is well-established that the interlayer Cu^{2+} defects greatly influence the kagome planes, but the spatial extent of this influence, as well its effect on the ground state, still warrants further investigation for both $ZnCu_3(OD)_6FBr$ and $ZnCu_3(OD)_6Cl_2$. A key question is whether or not these defects are ultimately responsible for the non-singlet fraction which remain at low temperatures.

6.2 ^{19}F NMR in $Zn_{0.95}Cu_{3.05}(OD)_6FBr$

For Zn-barlowite, the presence of interlayer ^{19}F sites presents a new opportunity for to study KHAs via NMR [18, 47]. A major advantage of probing ^{19}F is that its nucleus has spin $I = 1/2$, making it a good probe of local spin susceptibility via the NMR Knight shift. This is because ^{19}F NMR is not subject to the quadrupole shift experienced by nuclei with $I > 1/2$ (i.e. ^{63,65}Cu, ^{79,81}Br, ^{17}O, and ^{2}D), which can be orders of magnitude larger than the Knight shift corresponding to χ_{kagome} at low temperatures. Furthermore, the lack of an orientation-dependent quadrupole shift allows us to conduct ^{19}F NMR using the more abundant powder samples.

However, there is a drawback: ^{19}F couples (via hyperfine interactions) with as many as 12 nearest-neighbor Cu^{2+} spins within the kagome planes (Fig. 5.1), and hence the ^{19}F Knight shift can probe only the summation of their contributions, such that

$$^{19}K = \sum_{i=1}^{12} \frac{A_{hf}}{N_A \mu_B} \chi_s^i + {}^{19}K_{chem}, \tag{6.2.1}$$

where A_{hf} is the hyperfine coupling constant between the ^{19}F nuclei and each of the 12 kagome Cu^{2+} spins (whose local spin susceptibility are individually defined as χ_s^i). The temperature-independent chemical shift $^{19}K_{chem}$ is known to be fairly small

($\sim 0.01\%$) [18]. This means that for ^{19}F to exhibit the magnetically inert behaviour of a gapped system, all 12 neighboring Cu^{2+} spins must do so as well (i.e. be paired into singlets).

We performed ^{19}F NMR measurements on our powder sample of Zn-barlowite in an external magnetic field of $B_{\text{ext}} = 2.4$ T, with typical $t_{180} = 2t_{90} \approx 6\mu$s and $\tau = 10$ μs. As mentioned in Sec. 5.1, the exact composition of Zn-barlowite is $Zn_{0.95}Cu_{3.05}(OD)_6FBr$, which we abbreviate as "$Zn_{0.95}$" from here on.

We obtain the ^{19}F NMR lineshapes of $Zn_{0.95}$ at temperatures ranging from 2 K to 250 K and show them in Fig. 6.1 **(a)**. Starting at 250 K the lineshape consists of a single narrow peak which appears to broaden asymmetrically (extending farther towards the left side) with decreasing temperature.

We define the Knight shift of the peak frequency $^{19}f_{peak}$ as

$$^{19}K_{peak} = \frac{^{19}f_{peak} - ^{19}f_0}{^{19}f_0} \tag{6.2.2}$$

where $^{19}f_0 = ^{19}\gamma_n B_{\text{ext}} = 96.02$ MHz is the zero-shift frequency, and shows its temperature dependence in Fig. 6.2 along with the bulk susceptibility χ_{bulk} of $Zn_{0.95}$ (measured at 2.4 T).

We see that $^{19}K_{peak}$ is proportional to χ_{bulk} down to $\sim$50 K, below which, $^{19}K_{peak}$ begins to deviate towards zero while χ_{bulk} increases rapidly, likely due to the contribution of the interlayer Cu^{2+} defects.

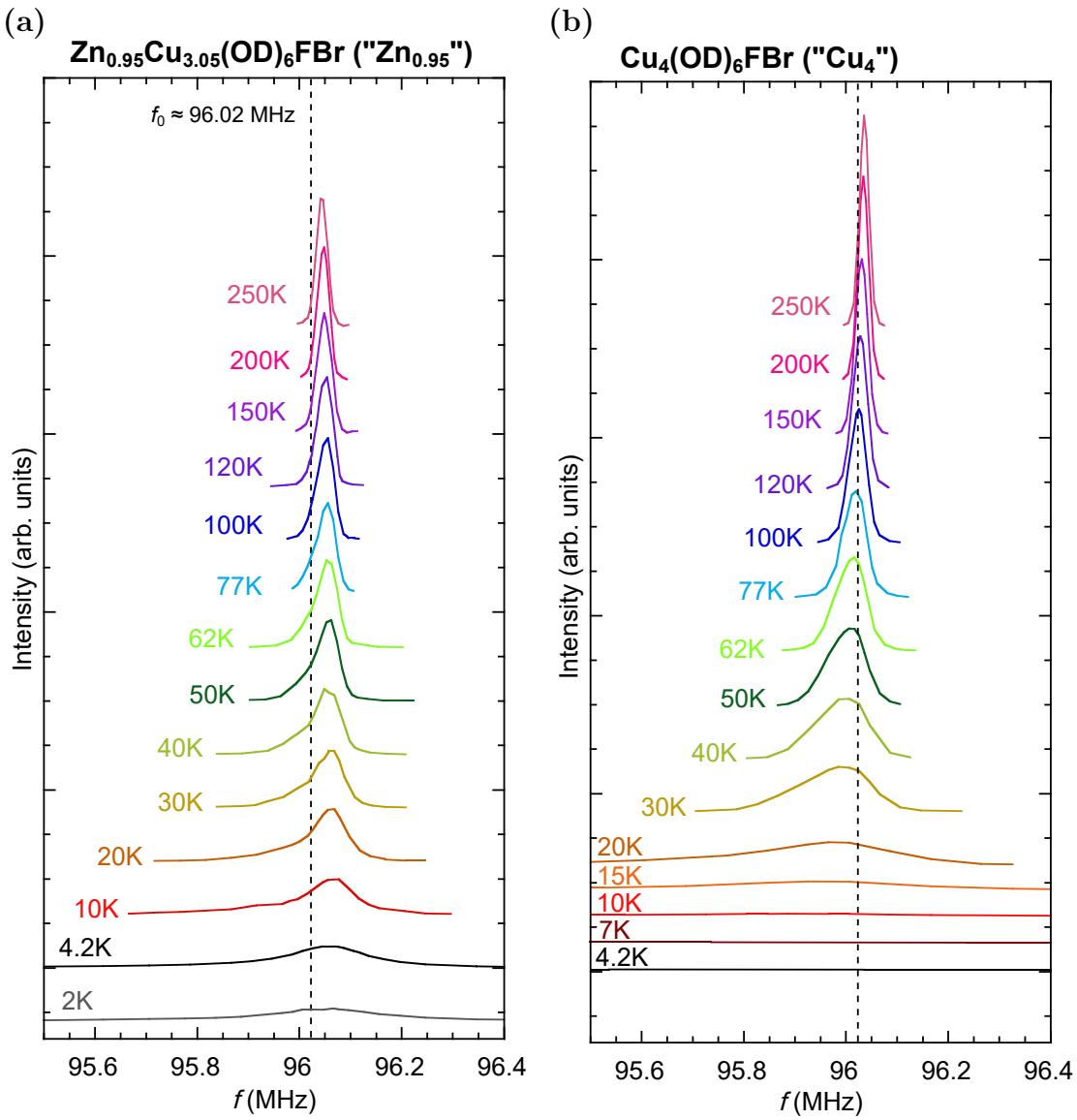


Figure 6.1: ^{19}F NMR lineshapes of **(a)** $Zn_{0.95}Cu_{3.05}(OD)_6FBr$ ($Zn_{0.95}$) and **(b)** $Cu_4(OD)_6FBr$ (Cu_4) measured in B_{ext} = 2.4 T at various temperatures. The lineshapes are vertically cascaded and their intensities are normalized by the Boltzmann factor. The vertical dashed lines indicate the zero-shift frequency $^{19}f_0$.

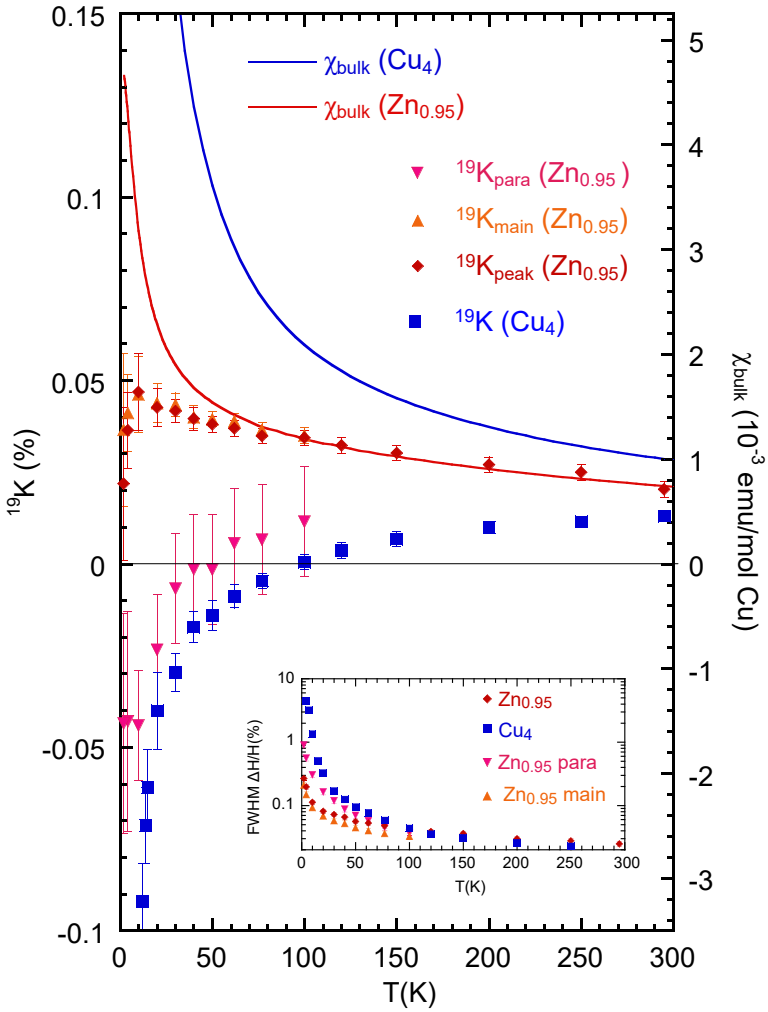


Figure 6.2: Knight Shift ^{19}K of the ^{19}F lineshape of $Zn_{0.95}$ (red ♦) and Cu_4 (blue ■). The solid lines show the bulk susceptibility χ_{bulk} (measured at 2.4 T) of $Zn_{0.95}$ (red) and Cu_4 (blue). The Knight shift of the fast (pink ▼) and slow (orange ▲) components of $Zn_{0.95}$ separated via 2D double Gaussian deconvolution. The inset shows the FWHM of the $Zn_{0.95}$, Cu_4, and ILT T_1-resolved peaks.

Our ^{19}F lineshapes and $^{19}K_{peak}$ are consistent with those reported by by Feng *et. al.* [18]. In their study, the low-temperature $^{19}K_{peak}$ data was directly fitted against the activation curve $^{19}K \propto \exp(-\Delta/k_BT)$ expected for a homogeneously

gapped system. However, it's important to keep in mind that the FWHM of the ^{19}F lineshape (shown in the inset of Fig. 6.2) grows rapidly below ∼50 K, becoming several times larger than $^{19}K_{peak}$ near the base temperature. This means that instead of going to zero Knight shift like $^{19}K_{peak}$, the majority of ^{19}F sites are shifted by large, distributed hyperfine fields. The broadness of the lineshape makes it dangerous to assume that χ_{kagome} is directly represented by $^{19}K_{peak}$. Thus, the decrease in $^{19}K_{peak}$ does not necessarily evidence a gapped system, or even rule out the gapless spin liquid picture.

6.3 $1/T_1$ and ILT

We measured $M(t_D)$ at $^{19}f_{peak}$ of $Zn_{0.95}$ based on the saturation recovery method by applying comb pulses separated by 400 μs, and obtained $1/T^{\mathrm{F}}_{1,str}$ and β via the stretched exponential fit. We summarize $1/T^{\mathrm{F}}_{1,str}$ and β by plotting them in Fig. 6.3 **(a)**, **(b)**), as well as the $1/T_1$ density distribution $P(1/T^{\mathrm{F}}_1)$ using ILT (shown in Fig. 6.4 **(a)**).

Similar to the behaviour of $1/T^{\mathrm{Cu}}_{1,str}$, $1/T^{\mathrm{F}}_{1,str}$ decreases with temperature, while the deviation in β indicates the development of a broad distribution in $1/T_1$ below ∼50 K. This decrease in $1/T^{\mathrm{F}}_{1,str}$ is not consistent with a power law nor an activation curve expected for a homogeneously gapless or gapped ground state, respectively. Recall that for the ^{19}F sites, the fluctuations from all 12 n.n. Cu^{2+} spins contribute to $1/T^F_1$ via their transferred hyperfine fields. This means that for $1/T^F_1$ to show gapped behaviour, all 12 n.n. Cu^{2+} must be paired into spin singlets, as their fluctuations would be overshadowed by those of paramagnetic Cu^{2+}. Since the ^{63}Cu NQR results in Chpt. 5 established that the fraction of singlets is less than 50%, a negligible

fraction of ^{19}F is expected to show gapped $1/T_1^F$, assuming the singlets are randomly distributed.

Turning to $P(1/T_1^{\rm F})$ in Fig. 6.4, we see that above $\sim$ 100 K, $P(1/T_1^{\rm F})$ is narrow (recall that the width of $P(1/T_1^{\rm F})$ is dependent on α) and symmetrically centered on $1/T_{1,str}^{\rm F}$. With decreasing temperature, $P(1/T_1^{\rm F})$ gradually skews towards the left, until a distinctly faster component appears to emerge below 20 K. These $P(1/T_1^{\rm F})$ results match those of our earlier ^{19}F NMR study on $Zn_{0.95}$ at $B_{\rm ext} = 0.72\ T$ [46].

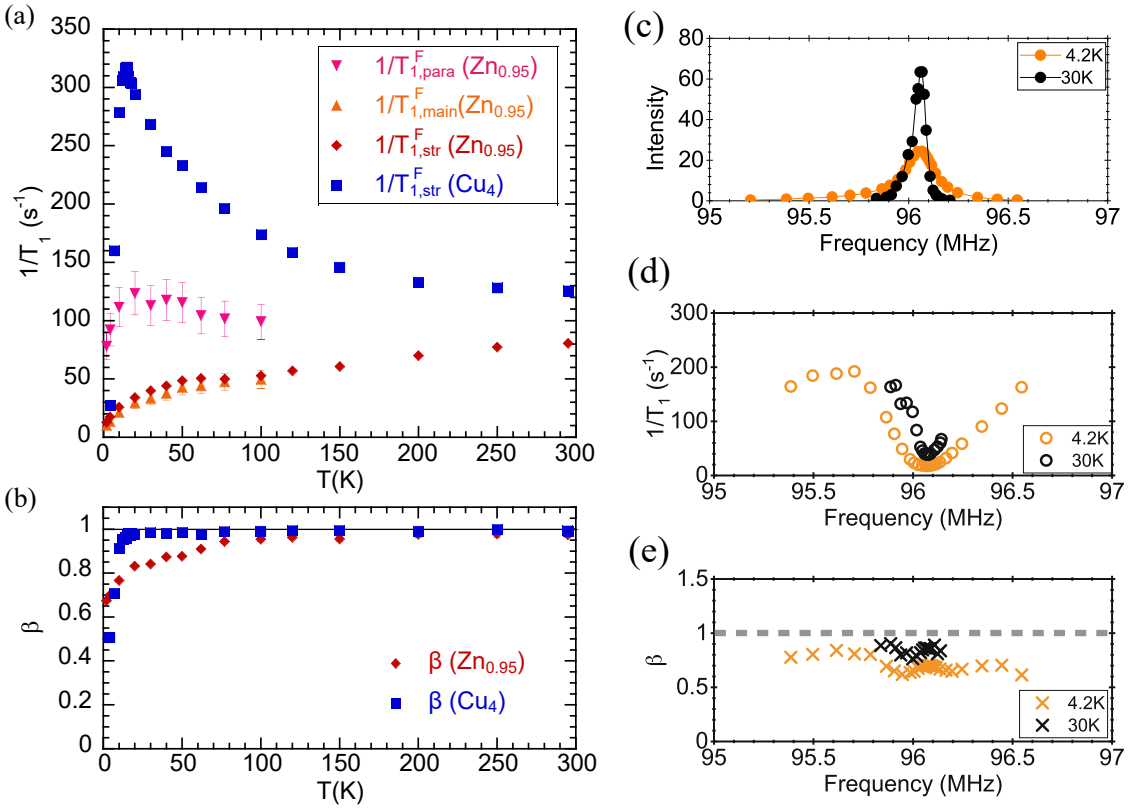


Figure 6.3: **(a)** $1/T_{1,str}^{\rm F}$ and **(b)** the stretch parameter β measured at $^{19}f_{peak}$ for $Zn_{0.95}$ (red ♦) and Cu_4 (blue ■), plotted alongside $1/T_{1,para}^{\rm F}$ (pink ▼) and $1/T_{1,main}^{\rm F}$ (orange ▲) obtained from deconvolution of $C(f, 1/T_1^{\rm F})$. **(d)** $1/T_{1,str}^{\rm F}$ of $Zn_{0.95}$ and $1/T_{1,str}^{\rm F}$ of Cu_4 and **(e)** β measured at various frequencies across **(c)** the $Zn_{0.95}$ lineshape at 30 K (black) and 4.2 K (orange). Figures **(c,d,e)** adopted from Ref. [47].

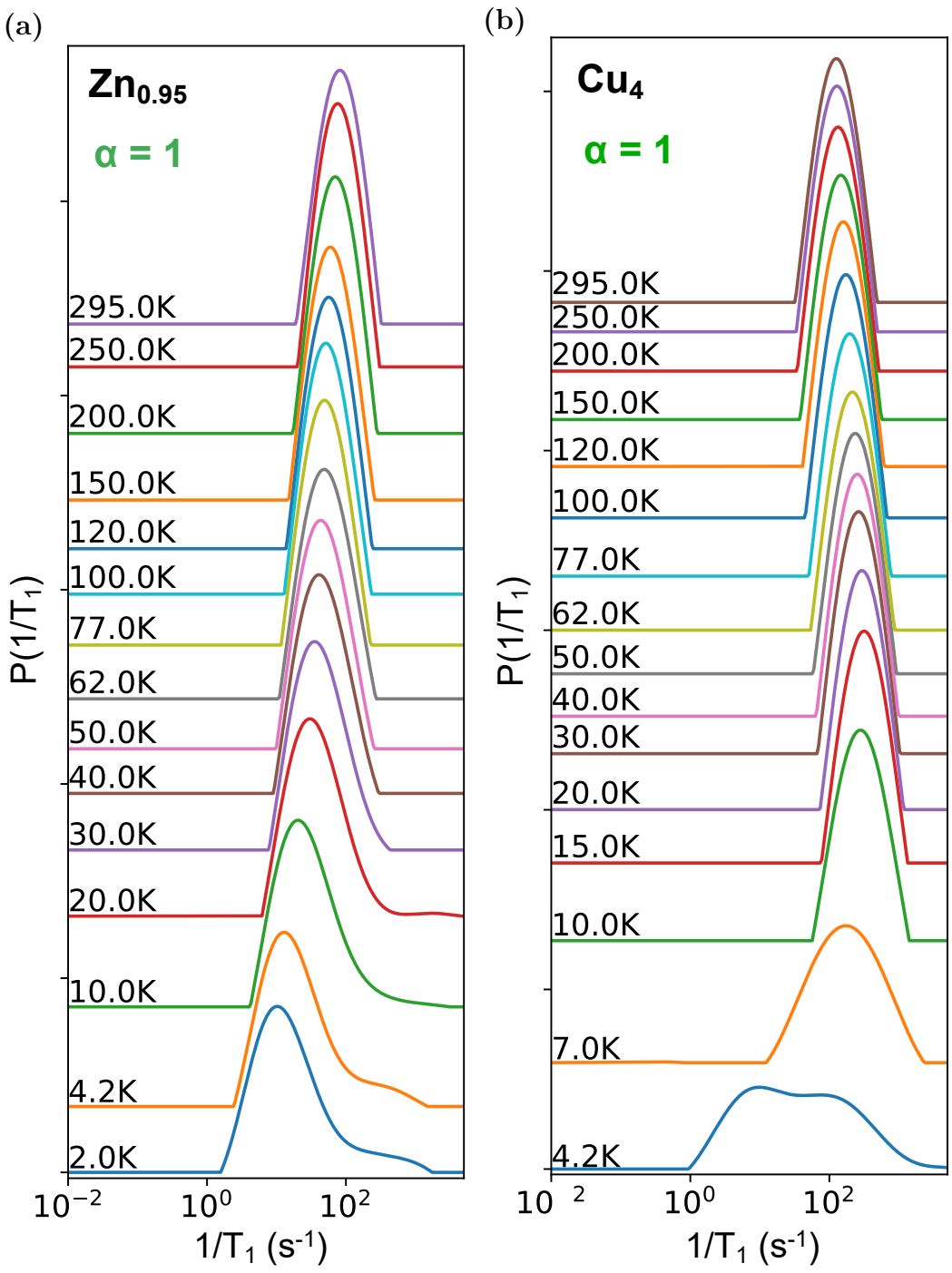


Figure 6.4: $P(1/T_1^{\mathrm{F}})$ of **(a)** $Zn_{0.95}$ and **(b)** Cu_4 measured at ${}^{19}f_{peak}$. The origin is vertically shifted for different temperatures. A fixed smoothing parameter $\alpha = 1$ was used for the ILT analysis. Note that in the context of Fig. 6.5, these distributions represent the vertical slices $C({}^{19}f_{peak}, 1/T_1)$.

We can make a comparison between these $P(1/T_1^{\mathrm{F}})$ results and those obtained from ^{7}Li NMR in $Ag_3LiIr_2O_6$(Chpt. 4). Both $P(1/T_1)$ exhibit 2 distinct components whose $1/T_1$ differ by up to 2 orders of magnitude. Recall that in the case of $Ag_3LiIr_2O_6$, the lineshape of the slower $1/T_1$ component was centered around a lower Knight shift, which can be separated by shortening the recovery time t_{end}. Unfortunately, this method is not as effective for $Zn_{0.95}$ due to the broader $P(1/T_1^{\mathrm{F}})$ distributions, but one can hypothesize a similar situation, where the partial ^{19}F lineshape for a given range of $1/T_1$ can vary.

Considering the broadness of the lineshape at low temperatures, the information provided by $1/T_{1,str}^{\mathrm{F}}$ and $P(1/T_1^{\mathrm{F}})$ at $^{19}f_{peak}$ represents only a small fraction of ^{19}F sites. We can repeat $1/T_1$ measurements at various points across the lineshape in order to obtain $1/T_{1,str}^{\mathrm{F}}$ as a function of frequency (shown for 30 K and 4.2 K in Fig. 6.3 **(c-e)**). Notice that $1/T_{1,str}^{\mathrm{F}}$ becomes higher towards both the left and right ends of the lineshape, which implies a correlation between the shifting of local hyperfine fields and enhancement of local $1/T_1^{\mathrm{F}}$.

6.4 2D ILT Correlation maps

It is clear $Zn_{0.95}$ is highly distributed in both ^{19}K and $1/T_1^{\mathrm{F}}$. Recall that a generalized method of resolving the NMR lineshape as a function of $1/T_1$ is to construct the 2D correlation map $C(f, 1/T_1^{\mathrm{F}})$ (see Sec. 3.5), in which each vertical slice of $C(f, 1/T_1^{\mathrm{F}})$ represents the $P(1/T_1^{\mathrm{F}})$ measured at that frequency. We measured $P(1/T_1^{\mathrm{F}})$ at multiple frequencies across the ^{19}F lineshape to obtain $C(f, 1/T_1^{\mathrm{F}})$ for temperatures from 2 K to 250 K, as shown in Fig. 6.5 **(a)**.

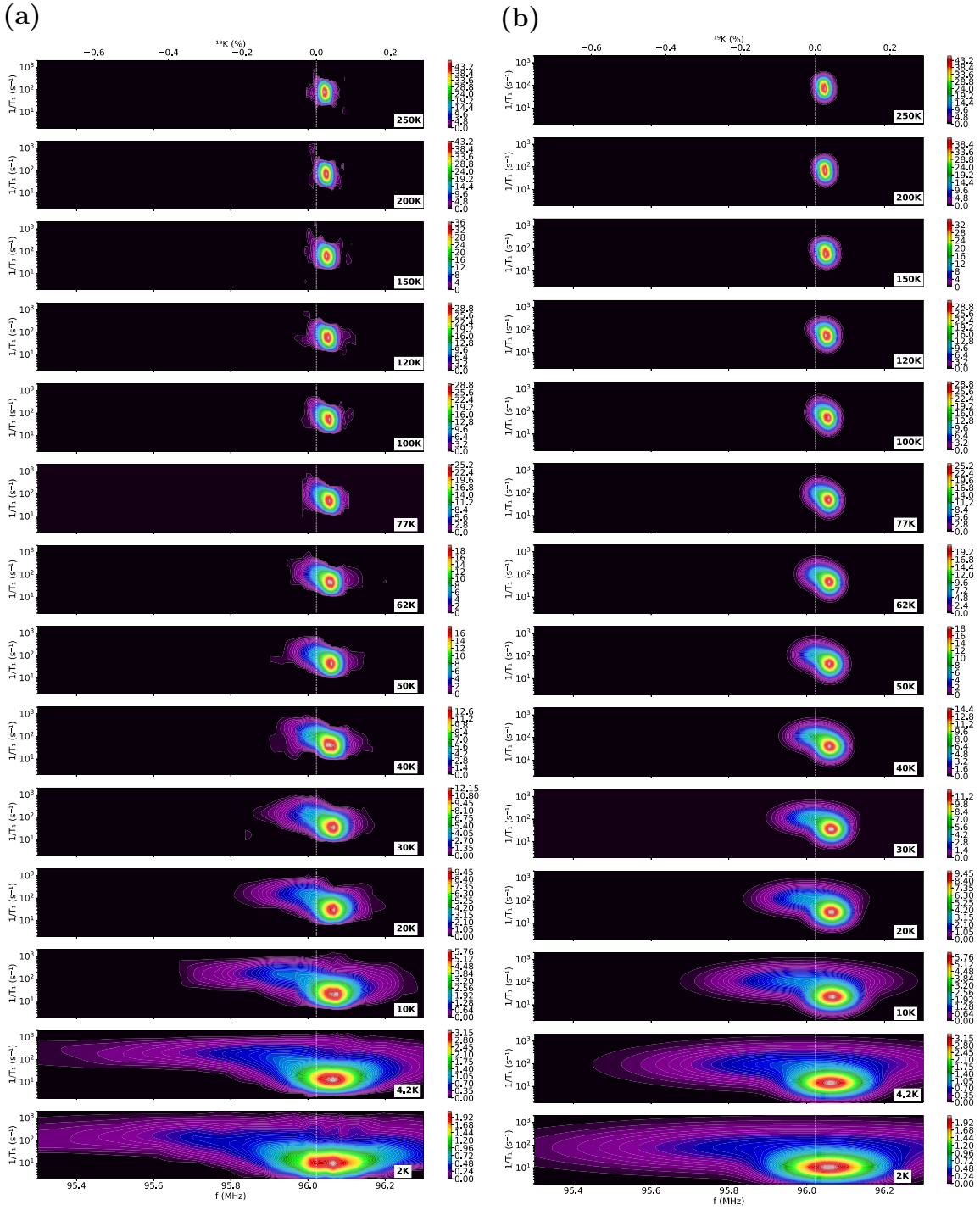


Figure 6.5: **(a)** $C(f, 1/T_1^{\mathrm{F}})$ of $Zn_{0.95}$ measured from 2 K to 250 K. **(b)** Corresponding 2D double Gaussian fits $C_{DGmodel}(f, 1/T_1^{\mathrm{F}})$. Figures adopted from supplemental materials of Ref. [47].

At high temperatures, $C(f, 1/T_1^{\mathrm{F}})$ is a relatively sharp and symmetrical peak, its center of gravity (f_{COG} $1/T_{1,COG}^{\mathrm{F}}$) accurately reproduces their 1D counterparts ($^{19}f_{peak}$, $1/T_{1,str}^{\mathrm{F}}$). Below ∼120 K, a distinct new region in $C(f, 1/T_1^{\mathrm{F}})$ (with faster $1/T_1^{\mathrm{F}}$) emerges from the top-left side of the original ("main") peak. This faster component (which we denote as the paramagnetic component for reasons that will become clear later in this chapter) broadens with decreasing temperature, eventually extending past both the left and right side of the main peak and accounting for the majority of the lineshape width at 4.2 K.

The emergence of the paramagnetic component in $C(f, 1/T_1^{\mathrm{F}})$ is consistent with our 1D results, accounting for the split-off peak in Fig. 6.4 **(a)**, the asymmetric tails in Fig. 6.1 **(a)**, and the enhanced $1/T_{1,str}^{\mathrm{F}}$ in Fig. 6.3 **(c-e)**. However, the key improvement over 1D is that the two components are far more distinguishable in $C(f, 1/T_1^{\mathrm{F}})$. This allows us to perform a stable deconvolution of $C(f, 1/T_1^{\mathrm{F}})$ via the process described in Sec. 3.5, by modeling the 2 components each as a 2D Gaussian function in the form of Eq. 3.5.1, where the sum is instead over $(i = 1, 2)$, representing the main and paramagnetic components, respectively. We denote this double Gaussian model function as $C_{DGmodel}(f, 1/T_1^{\mathrm{F}})$.

We show the resulting 2D double Gaussian fits $C_{DGmodel}(f, 1/T_1^{\mathrm{F}})$ in Fig. 6.5 **(b)**, as well as an example of the deconvolution process for 30 K in Fig. 6.6. Overall, this 2 component model fits our $C(f, 1/T_1^{\mathrm{F}})$ data fairly well and reproduces its key features. A difference between the data and fit appears at lower temperatures ($\lesssim$ 10 K), mainly due to the asymmetry of the fast component along the frequency axis (i.e. the long tail extending towards negative shift). However, this discrepancy does not significantly affect the results of our deconvolution analysis.

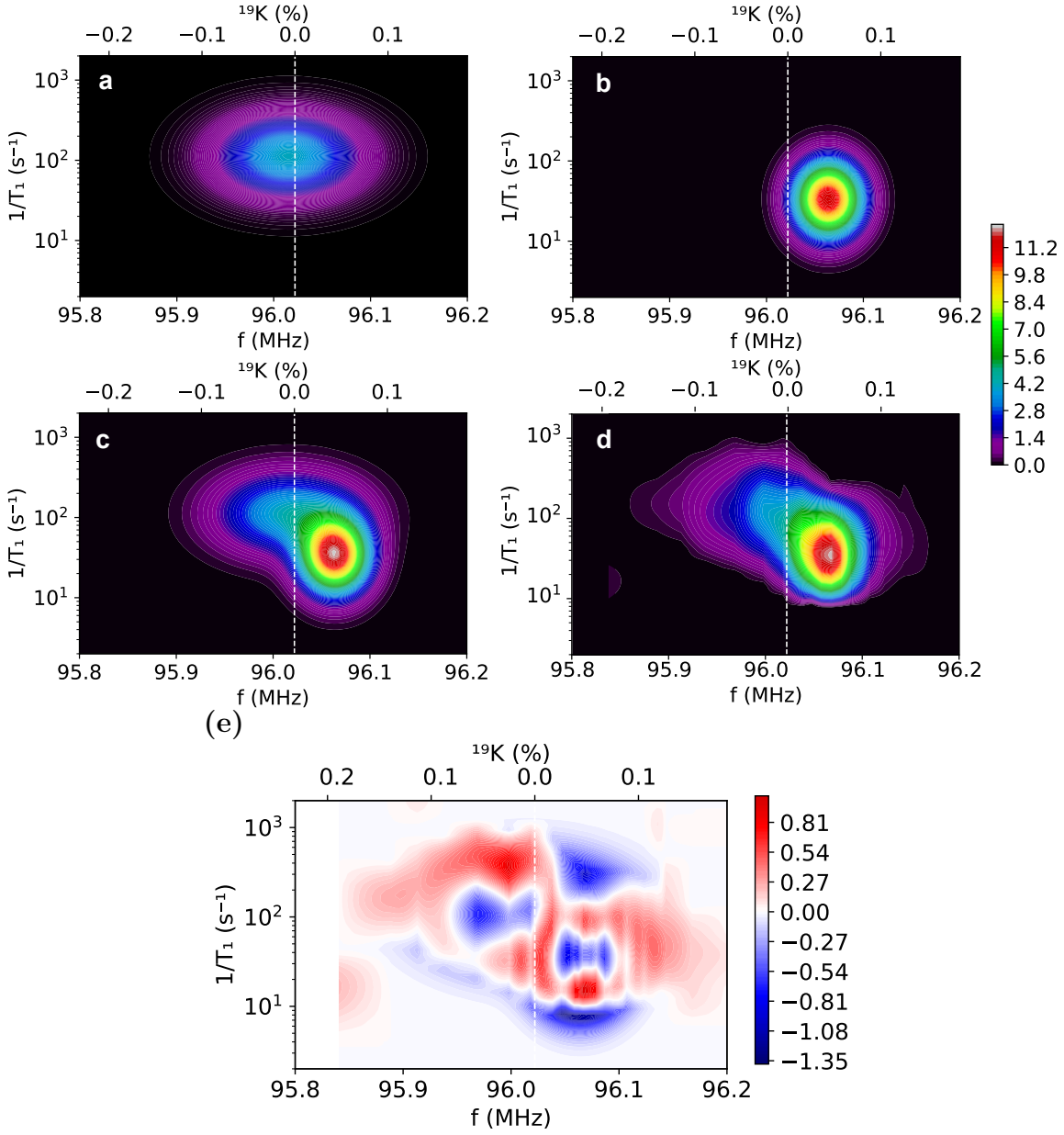


Figure 6.6: Example of 2D double Gaussian deconvolution of $C(f, 1/T_1^{\mathrm{F}})$ for $Zn_{0.95}$ at 30 K. **(a)** The separate "*para*" and **(b)** "*main*" components of **(c)** the double Gaussian function fit of **(d)** the measured $C(f, 1/T_1^{\mathrm{F}})$. **(e)** Color contour map of the fit residual $(C(f, 1/T_1^{\mathrm{F}}) - C_{DGmodel}(f, 1/T_1^{\mathrm{F}}))$. Figures adopted from supplemental materials of Ref. [47].

From our optimal values of the fit parameters defined in Eq. 3.5.1 (i.e. A_i, $f_{0,i}$, $\sigma_{f,i}$, and $1/T_{1,i}$), we can individually obtain the Knight shifts ($^{19}K_{main}$, $^{19}K_{para} = (f_{0,i} - ^{19} f_0)/^{19}f_0$), peak widths ($\Delta H/H = 2.355\sigma_{f,i}$), and central $1/T_1$ values ($1/T_{1,main}^{\mathrm{F}}$, $1/T_{1,para}^{\mathrm{F}} = 1/T_{1,i}$) for the main and paramagnetic components, and plot them in Fig. 6.2, and 6.3, alongside our 1D results. We can estimate the relative fraction of the paramagnetic component as $F_{para} = A_2/(A_1 + A_2)$, and show its temperature dependence in Fig. 6.7 **(a)** (plotted as $1 - F_{para}$).

Firstly, note the near-identical behaviours of $^{19}K_{peak}$ and $^{19}K_{main}$, as well as $1/T_{1,str}^{\mathrm{F}}$ and $1/T_{1,main}^{\mathrm{F}}$. This means our 1D results were mostly representative of the main component, which makes sense, as the main component is sharper and dominates the NMR signal measured at $^{19}f_{peak}$. In contrast, $^{19}K_{para}$ drops off towards negative shift with decreasing temperature, while $1/T_{1,para}^{\mathrm{F}}$ appears roughly temperature-independent (~ 100 s^{-1}) within the estimated uncertainty.

F_{para} grows with decreasing temperature, occupying roughly $\sim 60\%$ of the sample at 2 K. Correcting for the difference in $1/T_2$ between the main and paramagnetic component (like in case of $F_{singlet}$) does not appear to affect our value of F_{para} beyond the range of uncertainty.

Notice in Fig. 6.2 that the decrease in $^{19}K_{para}$ from positive to negative appears to correlate with the increase in χ_{bulk} at low temperatures. This behaviour is strongly reminiscent of the peaks originating from the nearest-neighbors of the 15% interlayer Cu^{2+} defects observed in the ^{2}D and ^{17}O NMR lineshapes of $Zn_{0.85}Cu_{3.15}(OD)_6Cl_2$ [91, 102]. This suggests that the paramagnetic component in $Zn_{0.95}$ similarly comes from ^{19}F sites within some proximity of its 5% Cu^{2+} defects, while the main component represents the fraction ^{19}F sites that are situated away from the defects.

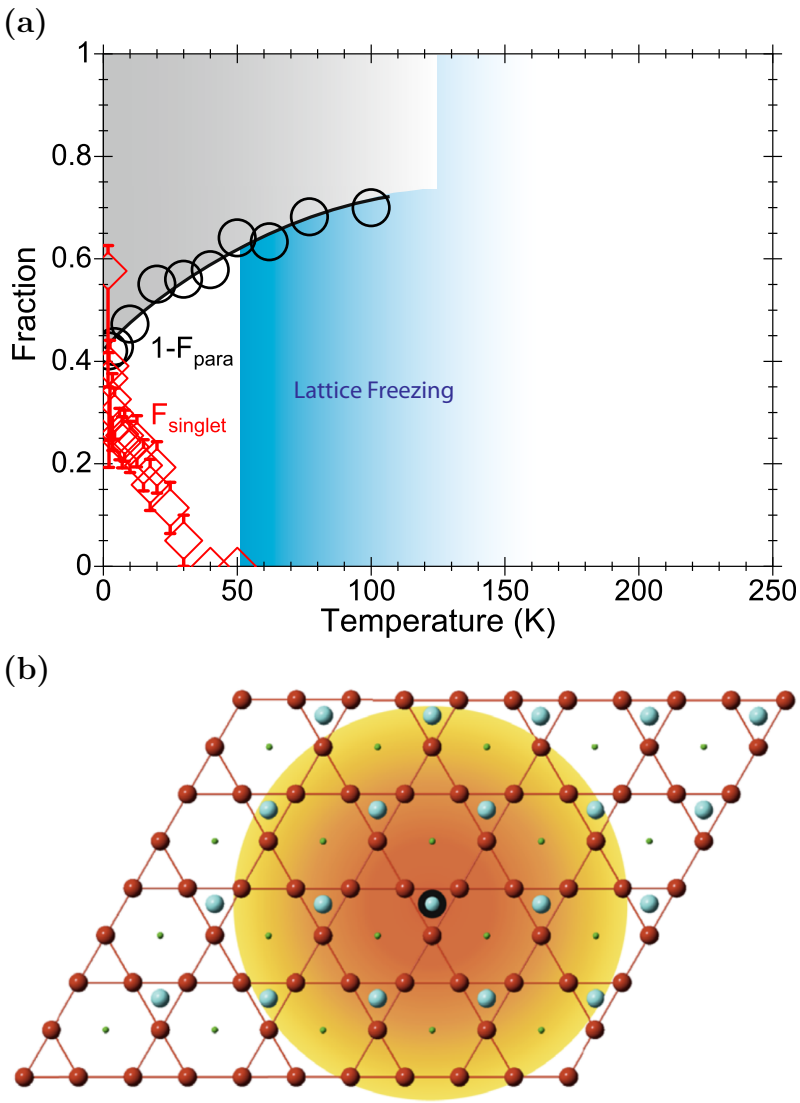


Figure 6.7: **(a)** $1 - F_{para}$ (black $\circ$), where F_{para} is the relative fraction of spin-polarized domains in $Zn_{0.95}$-barlowite. This is compared to $F_{singlet}$ (red $\diamond$) measured at the ^{63}Cu NQR peak in $Zn_{0.95}$. **(b)** Conceptual picture showing the approximate extent spin-polarized domains (orange circle) covering up to $\sim 60\%$ of ^{19}F sites at 2 K, which centered around the 5% interlayer Cu^{2+} defects (black dot). Figures adopted from Ref. [47].

6.5 Comparison to $Cu_4(OD)_6FBr$

To test the hypothesis that the paramagnetic component is associated with interlayer Cu^{2+} in $Zn_{0.95}$, we can investigate its "undoped parent compound" $Cu_4(OD)_6FBr$ (denoted as "Cu_4"), in which all of the interlayer sites are occupied by Cu^{2+} instead of Zn^{2+} (see Sec. 1.3).

We conduct an analogous set of ^{19}F NMR measurements on a deuterated polycrystalline sample of Cu_4 in $B_{\rm ext} = 2.4$ T, and compare our results to $Zn_{0.95}$. We present the ^{19}F lineshape in Fig. 6.1 **(b)**, $^{19}K_{peak}$, χ_{bulk}, and FWHM in Fig. 6.2, peak $1/T_{1,str}^{\rm F}$ and β in Fig. 6.3 **(a)**, and peak $P(1/T_1^{\rm F})$ in Fig. 6.4 **(b)**. Our ^{19}F measurements on Cu_4 are consistent with those of an earlier study by Ranjith *et.al.*, conducted on a protonated sample of Cu_4 [11].

Most notable about the behaviour of Cu_4 is its clear resemblance to that of the paramagnetic component of $Zn_{0.95}$, both exhibiting a lineshape which rapidly broadens with decreasing temperature, and shifts towards negative ^{19}K in tandem with the upturn in χ_{bulk}.

Recalling that Cu_4 undergoes a Néel transition at $T_N \approx 15$ K, we see that $1/T_{1,str}^{\rm F}$ of Cu_4 is greatly enhanced as the temperature approaches T_N, owing to the gradual onset of short-range magnetic order [21, 105]. Even so, the high-temperature asymptotic value of $1/T_{1,str}^{\rm F}$ for Cu_4 is approximately equal to $1/T_{1,para}^{\rm F}$ observed for $Zn_{0.95}$.

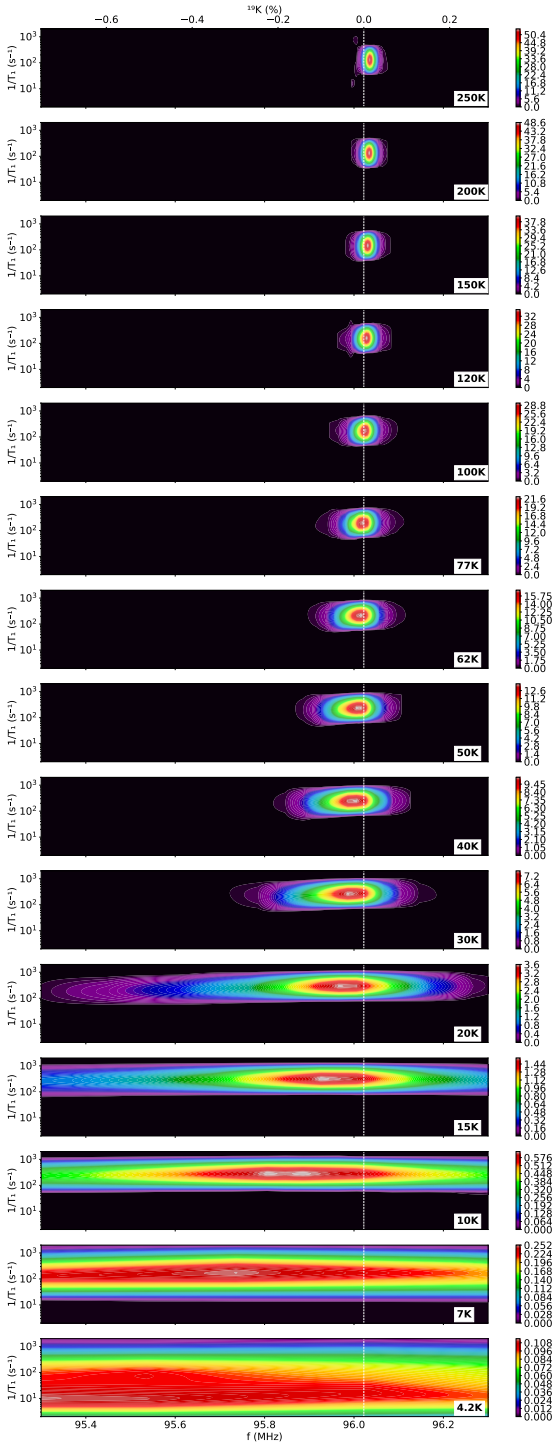


Figure 6.8: **(a)** $C(f, 1/T_1^{\mathrm{F}})$ of Cu_4 measured from 4.2 K and 250 K. Figure adopted from supplemental materials of Ref. [47].

We also obtain $C(f, 1/T_1^{\mathrm{F}})$ for Cu_4 and show its temperature evolution in Fig. 6.8. In the paramagnetic state above T_N, Cu_4 shows only a single component in $C(f, 1/T_1^{\mathrm{F}})$, affirming that the conventional measurements of $^{19}K_{peak}$ and $1/T_{1,str}^{\mathrm{F}}$ are decent proxies of the overall behavior of Cu_4. This is consistent with our expectations, as the number of nearby interlayer Cu^{2+} sites is no longer variable for the (crystallographically unique) F sites in Cu_4. Below T_N, $1/T_{1,str}^{\mathrm{F}}$ slows down while $C(f, 1/T_1^{\mathrm{F}})$ becomes highly distributed in both ^{19}K and $1/T_1$, as expected for an antiferromagnetically ordered ground state.

6.6 Spin-Polarized Domains

Our findings support the idea that the paramagnetic component of $Zn_{0.95}$ originate from regions of the sample which locally behave like Cu_4. Below ∼120 K, these regions start to differentiate themselves and gradually increase in size. This happens to occur within the temperature range where lattice distortions develop in $Zn_{0.95}$ [103]. Similarly, the lattice of Cu_4 also becomes distorted at low temperatures [21], although unlike $Zn_{0.95}$, the distortions in Cu_4 reduce the overall crystal symmetry. In $ZnCu_3(OD)_6Cl_2$, local distortions were shown to form near the interlayer Cu^{2+} defects [91]. Together, these facts suggest that domains with distinct spin polarization emerge in $Zn_{0.95}$ near the Cu^{2+} defects as a result of their influence. In Fig. 6.7 **(b)** we illustrate an approximate radius around the 5% interlayer Cu^{2+} defects in which spin-polarized domains must form, such that they cover $\sim 60\%$ of ^{19}F sites at 2 K.

The configuration of spins in these domains still remains uncertain. The dramatic broadening of the $Zn_{0.95}$ and Cu_4 lineshapes indicate that the large B_{hyp} experienced by ^{19}F can be both positive or negative. Based on the early ^{17}O NMR results, a

picture was suggested in which $5\% - 10\%$ of the non-magnetic interlayer Zn^{2+} sites in $ZnCu_3(OD)_6Cl_2$ are switched with kagome Cu^{2+} sites (i.e. antisite exchange) [77, 92]. Regions of staggered magnetic susceptibility in the kagome planes, centered around Zn^{2+} antisite defects, had been proposed to explain the broadening of the ^{17}O NMR lineshapes. This picture is contradicted by our ^{19}F results, since Cu_4 does not contain Zn^{2+}, yet its lineshape broadens similarly to $Zn_{0.95}$. Furthermore, recall that antisite defects are not observed by site-specific X-ray scattering [93]

Another key comparison can be made between the paramagnetic components observed by ^{19}F NMR and ^{63}Cu NQR. Both $1/T_{1,para}^{F}$ and $1/T_{1,para}^{Cu}$ hover around large, nearly temperature-independent values. This suggests that both components correspond to regions of the sample in which $1/T_1^F$ is locally enhanced by paramagnetic fluctuations. We plot the upper bound of the Cu singlet fraction $F_{singlet}$ alongside $1 - F_{para}$ in Fig. 6.7 **(a)**. Since we expect to find the intrinsic kagome behaviour (in which spin-singlets are formed) to be outside of the defect-induced spin-polarized domains, it makes sense that $F_{singlet}$ does not exceed $1 - F_{para}$.

Chapter 7

Conclusion

In this chapter, we summarize our key results on the QSL candidates $Ag_3LiIr_2O_6$, $ZnCu_3(OD)_6Cl_2$, and $ZnCu_3(OD)_6FBr$. In common among these materials are the presence of chemical and/or structural disorder which non-uniformly alter the local magnetism and spin dynamics. Although their magnetic ground state can be experimentally deduced by measuring the NMR spin-lattice relaxation rate $1/T_1$, we have routinely found that the popular method of estimating $1/T_{1,str}$ via a stretched exponential fit is grossly insufficient in dealing with the broad and multimodal distributions of $1/T_1$ that often emerge at low temperatures. To overcome this, we developed a numerical ILT $1/T_1$ analysis program to compute the $1/T_1$ probability density distribution $P(1/T_1)$, and were able to successfully individuate the qualitatively distinct magnetic environments present in these materials [45–47].

We used ILT to reveal the effects of disorder in our ^{7}Li NMR study on $Ag_3LiIr_2O_6$, where we compared samples with varying amounts of stacking faults and unwanted Ag inclusion in the Kitaev layers. The most disordered sample B lacks clear signatures of LRO in its heat capacity and χ_{bulk}, as they are washed out by the contribution

of nearly-nonmagnetic regions with Ag defects [42]. We distinguished the suppressed ^7K and $1/T_{1,str}^{\mathrm{Li}}$ exhibited by these regions, since they correspond to an additional component in both the ^{7}Li lineshape and $P(1/T_1^{\mathrm{Li}})$. On the other hand, in the nominally defect-free sample A, the peak in the temperature dependence of $1/T_{1,str}^{\mathrm{Li}}$ at $T_N = 9$ K$\pm$1 K evidences the onset of antiferromagnetic LRO, and is consistent with the appearance of μSR oscillations below $T_N \approx 8$ K observed in Ref. [87]. However, analysis of $P(1/T_1^{\mathrm{Li}})$ revealed that parts of the sample are already undergoing static order at 10 K while others ($\sim 40\%$) remain fluctuating even at 4.2 K. Overall, these results provide a key example in which the lack of magnetic order should not be interpreted as evidence for a QSL ground state. Furthermore, they emphasize the importance of ILT, since the multimodal $1/T_1$ distribution of $Ag_3LiIr_2O_6$ is emblematic of other second-generation Kitaev iridates Cu_2IrO_3 and $H_3LiIr_2O_6$ [41, 70, 71].

We probed the KHAs $ZnCu_3(OD)_6FBr$ and $ZnCu_3(OD)_6Cl_2$ using ^{63}Cu NQR, and discovered the gradual formation of Cu spin-singlets with spatially varying excitation gaps occurring below ~ 30 K in both materials. The fraction of the kagome planes occupied by singlets grow with decreasing temperature, eventually reaching up to $\lesssim 60\%$ at 1.8 K, while the rest continue to fluctuate paramagnetically. Using ILT, we identify these two as qualitatively distinct components in $P(1/T_1^{\mathrm{Cu}})$. The exponentially suppressed $1/T_{1,singlet}^{\mathrm{Cu}}$ and roughly temperature-independent $1/T_{1,para}^{\mathrm{Cu}}$ correspond to the local $1/T_1^{\mathrm{Cu}}$ of singlet and paramagnetic Cu sites, respectively. The conclusion of a spatially inhomogeneous and partially gapped ground state provides a strong explanation for the $1/T_1$ observed in past NMR studies on $ZnCu_3(OD)_6Cl_2$ [77, 91, 92, 101], in which the long history of conflicting interpretations is partly due to the lack of information about $P(1/T_1)$. With the development of ILT, we are no

longer limited to assuming a monomodal $1/T_1$ distribution, and can properly account for an inhomogeneous ground state.

In our ^{19}F NMR investigation of $ZnCu_3(OD)_6FBr$, we expanded the applications of ILT and developed a novel tool which shows the correlations between Knight shift and $1/T_1$. In the $1/T_1$-resolved NMR lineshape $C(f, 1/T_1^F)$ of $ZnCu_3(OD)_6FBr$, we identified a broad and fast component whose resemblance to the overall behaviour of ^{19}F in $Cu_4(OD)_6FBr$ indicate the existence of spin-polarized paramagnetic domains emerging near the 5% interlayer Cu^{2+} defects. We show that at 2 K, these paramagnetic domains envelop up to $\sim 60\%$ of the ^{19}F sites in $ZnCu_3(OD)_6FBr$. In addition to better understanding the influence of interlayer Cu^{2+}, our results exemplify the advantages of acquiring multidimensional NMR data. Although the feasibility of such experiments is often limited by the SNR, the insights gained from analyzing $C(f, 1/T_1)$ as a whole can potentially outclass those gained from conventional 1D NMR data, and could be crucial in elucidating the behaviour of a disordered material.

The findings reported in this thesis, along with those of Ref. [68–71], demonstrate the groundbreaking utility of both 1D and 2D ILT, and argues for the mass adoption of ILT $1/T_1$ analysis in this field of condensed matter NMR. In additional to disordered QSL systems, it will be very interesting to see what ILT can reveal about other disordered materials such as randomly diluted magnets [74] and heavy fermion systems [75]. At the very least, the potential inhomogeneity of $1/T_1$ and the limitations of stretch fit analysis should always be considered in future NMR studies.

Appendix A

Singular Value Decomposition

The kernel function matrix $\boldsymbol{K}$ (defined in Eq. 3.2.5) can be factorized to obtain its SVD

$$\boldsymbol{K} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^T, \tag{A.0.1}$$

where $\boldsymbol{U}$ and $\boldsymbol{V}$ are unitary, orthogonal matrices ($\boldsymbol{U}^{-1} = \boldsymbol{U}^T$ and $\boldsymbol{V}^{-1} = \boldsymbol{V}^T$), and $\boldsymbol{S}$ is an $m \times n$ rectangular diagonal matrix whose diagonal entries $S_{i,i}$ are arranged in descending order.

We want to compress $\boldsymbol{K}$ as to only include significant singular values above a chosen threshold corresponding to the noise floor σ_E [65]. To do this, we find the largest value of $\tilde{m}$ such that

$$S_{\tilde{m},\tilde{m}} > \frac{\sigma_E}{M_0}, \tag{A.0.2}$$

and truncate $\boldsymbol{S}$, $\boldsymbol{U}$, and $\boldsymbol{V}$ to obtain $\tilde{\boldsymbol{S}}$ (size $\tilde{m} \times \tilde{m}$), $\tilde{\boldsymbol{U}}$ (size $m \times \tilde{m}$), and $\tilde{\boldsymbol{V}}$ (size $\tilde{m} \times n$), respectively.

To perform Tikhonov regularization, we project $\boldsymbol{K}$ and $\mathbf{M}$ onto the truncated

singular basis

$$\begin{aligned} \boldsymbol{K} &\longrightarrow \tilde{\boldsymbol{K}} = \tilde{\boldsymbol{S}}\tilde{\boldsymbol{V}}^T, \\ \mathbf{M} &\longrightarrow \tilde{\mathbf{M}} = \tilde{\boldsymbol{U}}^T\mathbf{M}, \end{aligned} \tag{A.0.3}$$

and compute $\mathbf{P}$ using the SVD-truncated form of Eq. 3.3.1

$$\mathbf{P} = \underset{\mathbf{P}\geq 0}{\arg\min} \, ||\tilde{\mathbf{M}} - \tilde{\boldsymbol{K}}\mathbf{P}||^2 + \alpha||\mathbf{P}||^2. \tag{A.0.4}$$

To reconcile with Eq. 3.3.3, we perform the concatenation after the SVD transformation.

When dealing with the $\mathbf{M}$ data acquired by our experimental setup, SVD is a non-essential step used to improve computation efficiency, and as an initial layer of noise reduction before Tikhonov regularization. For much larger data sets, SVD truncation and other data reduction methods are needed in order to achieve reasonable computing times [63].

Appendix B

Optimal Value of α

The Tikhonov parameter α in Eq. 3.3.1 should be chosen carefully as to prevent over-fitting or under-fitting of $P(1/T_1)$ (Demonstrated in Fig. 3.2). We determine the optimal value $\alpha = \alpha_{opt}$, based on 2 distinct criteria which we denote as the BRD and "heel" conditions [68].

The residual $\chi(\alpha)$ between our data and ILT fit result is defined as

$$\chi(\alpha) = ||\mathbf{M} - \boldsymbol{K}\mathbf{P}(\alpha)||. \tag{B.0.1}$$

The BRD condition aims to match $\chi(\alpha)$ to its expected value, given the level experimental noise σ_E [85], such that

$$\chi(\alpha_{opt}^{BRD}) = \sigma_E\sqrt{m}. \tag{B.0.2}$$

We can estimate σ_E by measuring $M(t_D)$ in the absence of a spin echo signal and

calculating its standard deviation. This can be done by either turning off the transmitted RF pulses or integrating the imaginary channel of the NMR signal (measured via quadrature detection).

When using the compressed kernel matrix obtained via SVD, the fit residual and BRD condition are redefined as

$$\tilde{\chi}(\alpha) = ||\tilde{\mathbf{M}} - \tilde{\boldsymbol{K}}\mathbf{P}(\alpha)||, \qquad \text{and} \qquad \tilde{\chi}(\alpha_{opt}^{BRD}) = \sigma_E\sqrt{\tilde{m}}. \tag{B.0.3}$$

The BRD condition works well when $M(t_D)$ is dominated by random noise, but breaks down for higher levels of SNR (and/or m) or when the kernel function is not exact (such as in the case of quadrupole-split $1/T_1$ normal modes). We employ the "heel" condition to account for systematic errors in $P(1/T_1)$, which defines α_{opt}^{heel} such that

$$\left.\frac{d\log\chi(\alpha)}{d\log\alpha}\right|_{\chi(\alpha_{opt}^{heel})} = \text{TOL}. \tag{B.0.4}$$

The tolerance, chosen as TOL $= 10^{-1}$, represent the "heel" of the S-shaped χ vs α curve (B.1 **(a)**). α_{opt}^{heel} acts as a lower bound on α that is independent of the experimental conditions (i.e. σ_E) [106]. We therefore choose α_{opt} such that

$$\alpha_{opt} = \max(\alpha_{opt}^{BRD}, \alpha_{opt}^{heel}). \tag{B.0.5}$$

To find the value of α_{opt} which satisfies Eq. B.0.5, we calculate $P(1/T_1)$ with α varied from 10^{-4} to 10^5 in ascending quarter-decade steps and terminate the search once both the BRD and "heel" conditions are met. α_{opt} is then refined via linear interpolation.

In addition to the BRD and heel conditions, more complex methods such as L-curve optimization or generalized cross-validation can also be used to optimize α [65]. For our purposes, the exact value of α_{opt} is relatively unimportant, as the key ILT-obtained results discussed in this thesis are independent of α to within a decade of α_{opt}.

We elect to use a fixed value of α when performing ILT on 2D sets of related $1/T_1$ measurements, such as $C(f, 1/T_1)$, or the temperature-dependence of $P(1/T_1)$. The fixed α is chosen to be close to the average α_{opt}, which can differ between the individual $P(1/T_1)$ due to varying levels of relative noise σ_E/M_0 (see supplemental materials of Ref. [68] and [46]). Since the artificial width introduced to $P(1/T_1)$ by regularization (i.e. σ_{lm} in Fig. 3.4) depends on α, fixing α significantly improves the visual coherence of the 2D data and color contour maps.

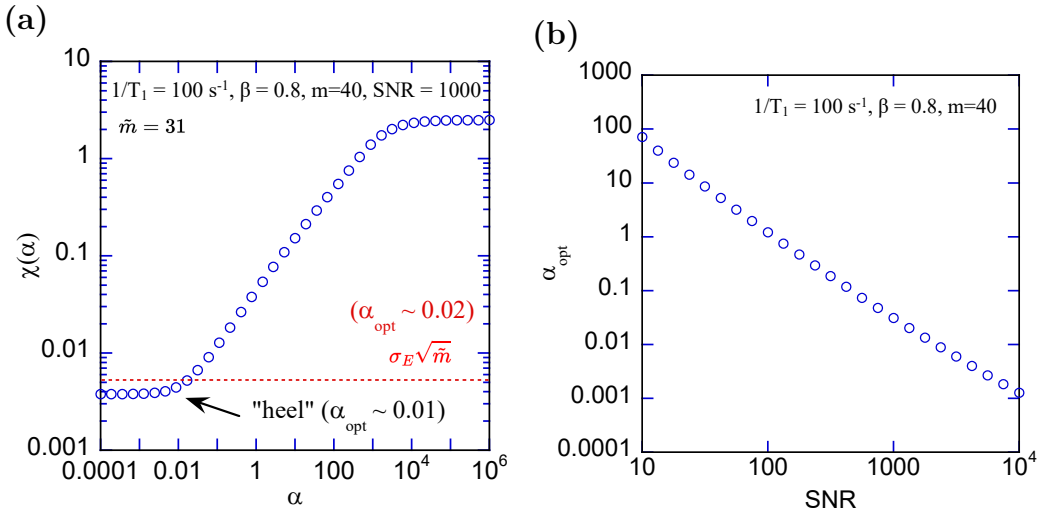


Figure B.1: ILT fit of model $M(t_D)$ data from Fig. 3.2. **(a)** Residual $\chi(\alpha)$ as a function of α. The dashed horizontal line correspond to the noise floor $\sigma_E\sqrt{\tilde{m}}$ used in the BRD condition. **(b)** α_{opt} determined at various levels of SNR. As SNR increases, $P(1/T_1)$ is calculated with lower values of α_{opt}, and approaches $P_{model}(1/T_1)$.

Bibliography

[1] L. Balents, "Spin liquids in frustrated magnets", Nature **464**, 199–208 (2010).

[2] P. Anderson, "Resonating valence bonds: a new kind of insulator?", Materials Research Bulletin **8**, 153–160 (1973).

[3] A. P. Ramirez, "Strongly geometrically frustrated magnets", Annual Review of Materials Research **24**, 453–480 (1994).

[4] S. Blundell, *Magnetism in condensed matter*, Oxford Master Series in Condensed Matter Physics (OUP Oxford, 2001).

[5] D. A. Tennant, R. A. Cowley, S. E. Nagler, and A. M. Tsvelik, "Measurement of the spin-excitation continuum in one-dimensional $KCuF_3$ using neutron scattering", Phys. Rev. B **52**, 13368–13380 (1995).

[6] N. D. Mermin and H. Wagner, "Absence of ferromagnetism or antiferromagnetism in one- or two-dimensional isotropic heisenberg models", Phys. Rev. Lett. **17**, 1133–1136 (1966).

[7] P. A. Lee, "From high temperature superconductivity to quantum spin liquid: progress in strong correlation physics", Reports on Progress in Physics **71**, 012501 (2007).

[8] M. R. Norman, "Colloquium: herbertsmithite and the search for the quantum spin liquid", Rev. Mod. Phys. **88**, 041002 (2016).

[9] J. S. Gardner, M. J. P. Gingras, and J. E. Greedan, "Magnetic pyrochlore oxides", Rev. Mod. Phys. **82**, 53–107 (2010).

[10] Y. Shimizu, K. Miyagawa, K. Kanoda, M. Maesato, and G. Saito, "Spin liquid state in an organic mott insulator with a triangular lattice", Phys. Rev. Lett. **91**, 107001 (2003).

[11] K. M. Ranjith, C. Klein, A. A. Tsirlin, H. Rosner, C. Krellner, and M. Baenitz, "Magnetic resonance as a local probe for kagomé magnetism in barlowite Cu4(OH)6FBr", Scientific Reports **8**, 10851 (2018).

[12] K. Matan, T. Ono, Y. Fukumoto, T. J. Sato, J. Yamaura, M. Yano, K. Morita, and H. Tanaka, "Pinwheel valence-bond solid and triplet excitations in the two-dimensional deformed kagome lattice", Nature Physics **6**, 865–869 (2010).

[13] A. P. Ramirez, G. P. Espinosa, and A. S. Cooper, "Strong frustration and dilution-enhanced order in a quasi-2d spin glass", Phys. Rev. Lett. **64**, 2070–2073 (1990).

[14] T. Imai and Y. S. Lee, "Do quantum spin liquids exist?", Physics Today **69**, 30–36 (2016).

[15] R. W. Smaha, W. He, J. P. Sheckelton, J. Wen, and Y. S. Lee, "Synthesis-dependent properties of barlowite and zn-substituted barlowite", Journal of Solid State Chemistry **268**, 123–129 (2018).

[16] M. P. Shores, E. A. Nytko, B. M. Bartlett, and D. G. Nocera, "A structurally perfect s = 1/2 kagomé antiferromagnet", Journal of the American Chemical Society **127**, PMID: 16190686, 13462–13463 (2005).

[17] T. H. Han, J. S. Helton, S. Chu, A. Prodi, D. K. Singh, C. Mazzoli, P. Müller, D. G. Nocera, and Y. S. Lee, "Synthesis and characterization of single crystals of the spin-1/2 kagome-lattice antiferromagnets $Zn_xCu_{4-x}(OH)_6Cl_2$", Phys. Rev. B **83**, 100402 (2011).

[18] Z. Feng, Z. Li, X. Meng, W. Yi, Y. Wei, J. Zhang, Y.-C. Wang, W. Jiang, Z. Liu, S. Li, F. Liu, J. Luo, S. Li, G. qing Zheng, Z. Y. Meng, J.-W. Mei, and Y. Shi, "Gapped spin-1/2 spinon excitations in a new kagome quantum spin liquid compound Cu3Zn(OH)6FBr*", Chinese Physics Letters **34**, 077502 (2017).

[19] P. Mendels, F. Bert, M. A. de Vries, A. Olariu, A. Harrison, F. Duc, J. C. Trombe, J. S. Lord, A. Amato, and C. Baines, "Quantum magnetism in the paratacamite family: towards an ideal kagomé lattice", Phys. Rev. Lett. **98**, 077204 (2007).

[20] T.-H. Han, J. S. Helton, S. Chu, D. G. Nocera, J. A. Rodriguez-Rivera, C. Broholm, and Y. S. Lee, "Fractionalized excitations in the spin-liquid state of a kagome-lattice antiferromagnet", Nature **492**, 406–410 (2012).

[21] R. W. Smaha, W. He, J. M. Jiang, J. Wen, Y.-F. Jiang, J. P. Sheckelton, C. J. Titus, S. G. Wang, Y.-S. Chen, S. J. Teat, A. A. Aczel, Y. Zhao, G. Xu, J. W. Lynn, H.-C. Jiang, and Y. S. Lee, "Materializing rival ground states in the barlowite family of kagome magnets: quantum spin liquid, spin ordered, and valence bond crystal states", npj Quantum Materials **5**, 23 (2020).

[22] H. Kawamura and K. Uematsu, "Nature of the randomness-induced quantum spin liquids in two dimensions", Journal of Physics: Condensed Matter **31**, 504003 (2019).

[23] S. Sachdev, "Kagomé- and triangular-lattice heisenberg antiferromagnets: ordering from quantum fluctuations and quantum-disordered ground states with unconfined bosonic spinons", Phys. Rev. B **45**, 12377–12396 (1992).

[24] Y.-M. Lu, Y. Ran, and P. A. Lee, "Z_2 spin liquids in the s=1/2 heisenberg model on the kagome lattice: a projective symmetry-group study of schwinger fermion mean-field states", Phys. Rev. B **83**, 224413 (2011).

[25] S. Yan, D. A. Huse, and S. R. White, "Spin-liquid ground state of the $S = 1/2$ kagome heisenberg antiferromagnet", Science **332**, 1173–1176 (2011).

[26] Y. Ran, M. Hermele, P. A. Lee, and X.-G. Wen, "Projected-wave-function study of the spin-1/2 heisenberg model on the kagomé lattice", Phys. Rev. Lett. **98**, 117205 (2007).

[27] H. J. Liao, Z. Y. Xie, J. Chen, Z. Y. Liu, H. D. Xie, R. Z. Huang, B. Normand, and T. Xiang, "Gapless spin-liquid ground state in the s=1/2 kagome antiferromagnet", Phys. Rev. Lett. **118**, 137202 (2017).

[28] R. R. P. Singh and D. A. Huse, "Triplet and singlet excitations in the valence bond crystal phase of the kagome lattice heisenberg model", Phys. Rev. B **77**, 144415 (2008).

[29] T. Shimokawa, K. Watanabe, and H. Kawamura, "Static and dynamical spin correlations of the (S)=1/2 random-bond antiferromagnetic Heisenberg model on the triangular and kagome lattices", Phys. Rev. B **92**, 134407 (2015).

[30] R. R. P. Singh, "Valence bond glass phase in dilute kagome antiferromagnets", Phys. Rev. Lett. **104**, 177203 (2010).

[31] I. Kimchi, J. P. Sheckelton, T. M. McQueen, and P. A. Lee, "Scaling and data collapse from local moments in frustrated disordered quantum spin systems", Nature Communications **9**, 4367 (2018).

[32] A. Kitaev, "Anyons in an exactly solved model and beyond", Annals of Physics **321**, January Special Issue, 2–111 (2006).

[33] H. Takagi, T. Takayama, G. Jackeli, G. Khaliullin, and S. E. Nagler, "Concept and realization of kitaev quantum spin liquids", Nature Reviews Physics **1**, 264–280 (2019).

[34] G. Jackeli and G. Khaliullin, "Mott insulators in the strong spin-orbit coupling limit: from heisenberg to a quantum compass and kitaev models", Phys. Rev. Lett. **102**, 017205 (2009).

[35] Y. Singh and P. Gegenwart, "Antiferromagnetic mott insulating state in single crystals of the honeycomb lattice material Na_2IrO_3", Phys. Rev. B **82**, 064412 (2010).

[36] Y. Singh, S. Manni, J. Reuther, T. Berlijn, R. Thomale, W. Ku, S. Trebst, and P. Gegenwart, "Relevance of the heisenberg-kitaev model for the honeycomb lattice iridates A_2IrO_3", Phys. Rev. Lett. **108**, 127203 (2012).

[37] K. W. Plumb, J. P. Clancy, L. J. Sandilands, V. V. Shankar, Y. F. Hu, K. S. Burch, H.-Y. Kee, and Y.-J. Kim, "α-$RuCl_3$: a spin-orbit assisted mott insulator on a honeycomb lattice", Phys. Rev. B **90**, 041112 (2014).

[38] J. G. Rau, E. K.-H. Lee, and H.-Y. Kee, "Spin-orbit physics giving rise to novel phases in correlated systems: iridates and related materials", Annual Review of Condensed Matter Physics **7**, 195–221 (2016).

[39] F. Bahrami, M. Abramchuk, O. Lebedev, and F. Tafti, "Metastable kitaev magnets", Molecules **27**, 10.3390/molecules27030871 (2022).

[40] M. Abramchuk, C. Ozsoy-Keskinbora, J. W. Krizan, K. R. Metz, D. C. Bell, and F. Tafti, "Cu2iro3: a new magnetically frustrated honeycomb iridate", Journal of the American Chemical Society **139**, 15371–15376 (2017).

[41] K. Kitagawa, T. Takayama, Y. Matsumoto, A. Kato, R. Takano, Y. Kishimoto, S. Bette, R. Dinnebier, G. Jackeli, and H. Takagi, "A spin–orbital-entangled quantum liquid on a honeycomb lattice", Nature **554**, 341–345 (2018).

[42] F. Bahrami, W. Lafargue-Dit-Hauret, O. I. Lebedev, R. Movshovich, H.-Y. Yang, D. Broido, X. Rocquefelte, and F. Tafti, "Thermodynamic evidence of proximity to a kitaev spin liquid in $Ag_3LiIr_2O_6$", Phys. Rev. Lett. **123**, 237203 (2019).

[43] J. Knolle, R. Moessner, and N. B. Perkins, "Bond-disordered spin liquid and the honeycomb iridate $H_3LiIr_2O_6$: abundant low-energy density of states from random majorana hopping", Phys. Rev. Lett. **122**, 047202 (2019).

[44] P. A. Lee, "An end to the drought of quantum spin liquids", Science **321**, 1306–1307 (2008).

[45] J. Wang, W. Yuan, T. Imai, P. M. Singer, F. Bahrami, and F. Tafti, "NMR investigation on the honeycomb iridate $Ag_3LiIr_2O_6$", Phys. Rev. B **103**, 214405 (2021).

[46] J. Wang, W. Yuan, P. M. Singer, R. W. Smaha, W. He, J. Wen, Y. S. Lee, and T. Imai, "Emergence of spin singlets with inhomogeneous gaps in the kagome lattice heisenberg antiferromagnets Zn-barlowite and herbertsmithite", Nature Physics **17**, 1109–1113 (2021).

[47] W. Yuan, J. Wang, P. M. Singer, R. W. Smaha, J. Wen, Y. S. Lee, and T. Imai, "Emergence of the spin polarized domains in the kagome lattice heisenberg antiferromagnet Zn-barlowite $(Zn_{0.95}Cu_{0.05})Cu_3(OD)_6FBr$", npj Quantum Materials **7**, 120 (2022).

[48] C. P. Slichter, *Principles of magnetic resonance*, 3rd edition (Springer Berlin, Heidelberg, 1996).

[49] A. Abragam, *The principles of nuclear magnetism*, International series of monographs on physics (Oxford, Clarendon Press, 1961).

[50] N. Ashcroft and N. Mermin, *Solid state physics*, Science: Physics (Saunders College, 1976).

[51] K. R. Thurber, "^{17}O and ^{63}Cu NMR study of spin dynamics in low-dimensional spin 1/2 antiferromagnets", PhD thesis (Massachusetts Institute of Technology, 1999).

[52] E Fukushima and S. Roeder, *Experimental pulse nmr: a nuts and bolts approach* (Addison-Wesley Pub. Co., 1981).

[53] T. Shimizu, "On the electric field gradient at copper nuclei in oxides", Journal of the Physical Society of Japan **62**, 772–778 (1993).

[54] E. R. Andrew and D. P. Tunstall, "Spin-lattice relaxation in imperfect cubic crystals and in non-cubic crystals", Proceedings of the Physical Society **78**, 1 (1961).

[55] A. Narath, "Nuclear spin-lattice relaxation in hexagonal transition metals: titanium", Phys. Rev. **162**, 320–332 (1967).

[56] F. Mila and T. Rice, "Analysis of magnetic resonance experiments in $YBa_2Cu_3O_7$", Physica C: Superconductivity **157**, 561 –570 (1989).

[57] V. Jaccarino, *Theory of magnetism in transition metals*, Proc. of Int. School of Physics, Enrico Fermi, volume 37 (Academic Press, 1966).

[58] T. Moriya, "The effect of electron-electron interaction on the nuclear spin relaxation in metals", Journal of the Physical Society of Japan **18**, 516–520 (1963).

[59] R. White, *Quantum theory of magnetism*, Springer series in solid-state sciences (Springer-Verlag, New York, 1983).

[60] P. M. Singer, "^{17}O and ^{63}Cu nuclear resonance study of the inhomogeneous electronic state in $La_{2-x}Sr_xcuo_4$", PhD thesis (Massachusetts Institute of Technology, 2003).

[61] C. H. Pennington, "Single crystal nmr of high temperature superconductors", PhD thesis (University of Illinois, 1989).

[62] Y.-Q. Song, L. Venkataramanan, M. Hürlimann, M. Flaum, P. Frulla, and C. Straley, "T1–t2 correlation spectra obtained using a fast two-dimensional laplace inversion", Journal of Magnetic Resonance **154**, 261–268 (2002).

[63] L. Venkataramanan, Y.-Q. Song, and M. Hurlimann, "Solving fredholm integrals of the first kind with tensor product structure in 2 and 2.5 dimensions", IEEE Transactions on Signal Processing **50**, 1017–1026 (2002).

[64] B. Sun and K.-J. Dunn, "Two-dimensional nuclear magnetic resonance petrophysics", Magnetic Resonance Imaging **23**, Proceedings of the Seventh International Conference on Recent Advances in MR Applications to Porous Media, 259–262 (2005).

[65] J. Mitchell, T. Chandrasekera, and L. Gladden, "Numerical estimation of relaxation and diffusion distributions in two dimensions", Progress in Nuclear Magnetic Resonance Spectroscopy **62**, 34–50 (2012).

[66] P. M. Singer, D. Asthagiri, Z. Chen, A. Valiya Parambathu, G. J. Hirasaki, and W. G. Chapman, "Role of internal motions and molecular geometry on the NMR relaxation of hydrocarbons", The Journal of Chemical Physics **148**, 164507 (2018).

[67] P. M. Singer, D. Asthagiri, W. G. Chapman, and G. J. Hirasaki, "NMR spin-rotation relaxation and diffusion of methane", The Journal of Chemical Physics **148**, 204504 (2018).

[68] P. M. Singer, A. Arsenault, T. Imai, and M. Fujita, "^{139}La NMR investigation of the interplay between lattice, charge, and spin dynamics in the charge-ordered high-T_c cuprate $La_{1.875}Ba_{0.125}CuO_4$", Phys. Rev. B **101**, 174508 (2020).

[69] A. Arsenault, T. Imai, P. M. Singer, K. M. Suzuki, and M. Fujita, "Magnetic inhomogeneity in charge-ordered $La_{1.885}Sr_{0.115}CuO_4$ studied by NMR", Phys. Rev. B **101**, 184505 (2020).

[70] S. K. Takahashi, J. Wang, A. Arsenault, T. Imai, M. Abramchuk, F. Tafti, and P. M. Singer, "Spin excitations of a proximate kitaev quantum spin liquid realized in Cu_2IrO_3", Phys. Rev. X **9**, 031047 (2019).

[71] C. Lee, S. Lee, Y. Choi, C. Wang, H. Luetkens, T. Shiroka, Z. Jang, Y.-G. Yoon, and K.-Y. Choi, "Coexistence of random singlets and disordered kitaev spin liquid in $H_3LiIr_2O_6$", Phys. Rev. B **107**, 014424 (2023).

[72] D. C. Johnston, "Stretched exponential relaxation arising from a continuous sum of exponential decays", Phys. Rev. B **74**, 184430 (2006).

[73] T. Imai, T. Shimizu, T. Tsuda, H. Yasuoka, T. Takabatake, Y. Nakazawa, and M. Ishikawa, "Nuclear spin-lattice relaxation of ^{63,65}Cu at the Cu(2) sites of the high T_c superconductor $YBa_2Cu_3O_{7-\delta}$", Journal of the Physical Society of Japan **57**, 1771–1779 (1988).

[74] M. Itoh, H. Yasuoka, A. R. KING, and V. Jaccarino, "Decay of nuclear magnetization in the randomly diluted antiferromagnets $Fe_xZn_{1-x}F_2$ and $Mn_xZn_{1-x}F_2$", Journal of the Physical Society of Japan **55**, 964–972 (1986).

[75] D. C. Johnston, S.-H. Baek, X. Zong, F. Borsa, J. Schmalian, and S. Kondo, "Dynamics of magnetic defects in heavy fermion LiV_2O_4 from stretched exponential ^{7}Li nmr relaxation", Phys. Rev. Lett. **95**, 176408 (2005).

[76] P. Thayamballi and D. Hone, "Nuclear relaxation in a randomly diluted heisenberg paramagnet", Phys. Rev. B **21**, 1766–1773 (1980).

[77] A. Olariu, P. Mendels, F. Bert, F. Duc, J. C. Trombe, M. A. de Vries, and A. Harrison, "^{17}O nmr study of the intrinsic magnetic susceptibility and spin

dynamics of the quantum kagome antiferromagnet $ZnCu_3(OH)_6Cl_2$", Phys. Rev. Lett. **100**, 087202 (2008).

[78] T. Imai, T. Shimizu, H. Yasuoka, Y. Ueda, and K. Kosuge, "Anomalous temperature dependence of cu nuclear spin-lattice relaxation in $YBa_2Cu_3O_{6.91}$", Journal of the Physical Society of Japan **57**, 2280–2283 (1988).

[79] E. J. Fordham, L. Venkataramanan, J. Mitchell, and A. Valori, "What are, and what are not, inverse laplace transforms", Diffusion Fundam. **29**, 1 (2017).

[80] C. Lawson and R. Hanson, *Solving least squares problems* (Prentice-Hall, New York, 1974).

[81] A. Tikhonov and V. Arsenin, *Solutions of ill-posed problems*, Halsted Press book (Winston, 1977).

[82] E. Chouzenoux, S. Moussaoui, J. Idier, and F. c. Mariette, "Efficient maximum entropy reconstruction of nuclear magnetic resonance T1-T2 spectra", IEEE Transactions on Signal Processing **58**, 6040–6051 (2010).

[83] M. Prange and Y.-Q. Song, "Quantifying uncertainty in nmr t2 spectra using monte carlo inversion", Journal of Magnetic Resonance **196**, 54–60 (2009).

[84] L. Venkataramanan, F. K. Gruber, T. M. Habashy, and D. E. Freed, "Mellin transform of cpmg data", Journal of Magnetic Resonance **206**, 20–31 (2010).

[85] J. P. Butler, J. A. Reeds, and S. V. Dawson, "Estimating solutions of first kind integral equations with nonnegative constraints and optimal smoothing", SIAM Journal on Numerical Analysis **18**, 381–397 (1981).

[86] M. Hürlimann and L Venkataramanan, "Quantitative measurement of two-dimensional distribution functions of diffusion and relaxation in grossly inhomogeneous fields", Journal of Magnetic Resonance **157**, 31–42 (2002).

[87] F. Bahrami, E. M. Kenney, C. Wang, A. Berlie, O. I. Lebedev, M. J. Graf, and F. Tafti, "Effect of structural disorder on the kitaev magnet $Ag_3LiIr_2O_6$", Phys. Rev. B **103**, 094427 (2021).

[88] S.-H. Baek, S.-H. Do, K.-Y. Choi, Y. S. Kwon, A. U. B. Wolter, S. Nishimoto, J. van den Brink, and B. Büchner, "Evidence for a field-induced quantum spin liquid in α-$RuCl_3$", Phys. Rev. Lett. **119**, 037201 (2017).

[89] G. Simutis, N. Barbero, K. Rolfs, P. Leroy-Calatayud, K. Mehlawat, R. Khasanov, H. Luetkens, E. Pomjakushina, Y. Singh, H.-R. Ott, J. Mesot, A. Amato, and T. Shiroka, "Chemical and hydrostatic-pressure effects on the kitaev honeycomb material Na_2IrO_3", Phys. Rev. B **98**, 104421 (2018).

[90] P. Khuntia, S. Manni, F. R. Foronda, T. Lancaster, S. J. Blundell, P. Gegenwart, and M. Baenitz, "Local magnetism and spin dynamics of the frustrated honeycomb rhodate Li_2RhO_3", Phys. Rev. B **96**, 094432 (2017).

[91] M. Fu, T. Imai, T.-H. Han, and Y. S. Lee, "Evidence for a gapped spin-liquid ground state in a kagome heisenberg antiferromagnet", Science **350**, 655–658 (2015).

[92] P. Khuntia, M. Velazquez, Q. Barthélemy, F. Bert, E. Kermarrec, A. Legros, B. Bernu, L. Messio, A. Zorko, and P. Mendels, "Gapless ground state in the archetypal quantum kagome antiferromagnet $ZnCu_3(OH)_6Cl_2$", Nature Physics **16**, 469–474 (2020).

[93] D. E. Freedman, T. H. Han, A. Prodi, P. Müller, Q.-Z. Huang, Y.-S. Chen, S. M. Webb, Y. S. Lee, T. M. McQueen, and D. G. Nocera, "Site specific x-ray anomalous dispersion of the geometrically frustrated kagomé magnet, herbertsmithite, ZnCu3(OH)6Cl2", Journal of the American Chemical Society **132**, 16185–16190 (2010).

[94] R. W. Smaha, I. Boukahil, C. J. Titus, J. M. Jiang, J. P. Sheckelton, W. He, J. Wen, J. Vinson, S. G. Wang, Y.-S. Chen, S. J. Teat, T. P. Devereaux, C. Das Pemmaraju, and Y. S. Lee, "Site-specific structure at multiple length scales in kagome quantum spin liquid candidates", Phys. Rev. Mater. **4**, 124406 (2020).

[95] M. Azuma, Z. Hiroi, M. Takano, K. Ishida, and Y. Kitaoka, "Observation of a spin gap in $SrCu_2O_3$ comprising spin-1/2 quasi-1d two-leg ladders", Phys. Rev. Lett. **73**, 3463–3466 (1994).

[96] J. Kikuchi, H. Yasuoka, M. Hase, Y. Sasago, and K. Uchinokura, "Cu nuclear quadrupole resonance study of $CuGeO_3$", Journal of the Physical Society of Japan **63**, 872–875 (1994).

[97] T. Imai, K. R. Thurber, K. M. Shen, A. W. Hunt, and F. C. Chou, "^{17}o and ^{63}Cu nmr in undoped and hole doped Cu_2O_3 two-leg spin ladder $A_{14}Cu_{24}O_{41}$ (A_{14} = La_6Ca_8,Sr_{14},$Sr_{11}Ca_3$)", Phys. Rev. Lett. **81**, 220–223 (1998).

[98] H. Kageyama, K. Yoshimura, R. Stern, N. V. Mushnikov, K. Onizuka, M. Kato, K. Kosuge, C. P. Slichter, T. Goto, and Y. Ueda, "Exact dimer ground state and quantized magnetization plateaus in the two-dimensional spin system $SrCu_2(BO_3)_2$", Phys. Rev. Lett. **82**, 3168–3171 (1999).

[99] T. Moriya, "Nuclear Magnetic Relaxation in Antiferromagnetics, II", Progress of Theoretical Physics **16**, 641–657 (1956).

[100] Y. Itoh, T. Imai, T. Shimizu, T. Tsuda, H. Yasuoka, and Y. Ueda, "Nuclear quadrupole resonance of cu in the paramagnetic state of CuO", Journal of the Physical Society of Japan **59**, 1143–1146 (1990).

[101] T. Imai, E. A. Nytko, B. M. Bartlett, M. P. Shores, and D. G. Nocera, "^{63}Cu, ^{35}Cl, and ^{1}H NMR in the S=1/2 kagome lattice $ZnCu_3(OH)_6Cl_2$", Phys. Rev. Lett. **100**, 077203 (2008).

[102] T. Imai, M. Fu, T. H. Han, and Y. S. Lee, "Local spin susceptibility of the S=1/2 kagome lattice in $ZnCu_3(OH)_6Cl_2$", Phys. Rev. B **84**, 020411 (2011).

[103] J. Wang, W. Yuan, P. M. Singer, R. W. Smaha, W. He, J. Wen, Y. S. Lee, and T. Imai, "Freezing of the lattice in the kagome lattice heisenberg antiferromagnet zn-barlowite $ZnCu_3(OD)_6FBr$", Phys. Rev. Lett. **128**, 157202 (2022).

[104] R. N. Bhatt and P. A. Lee, "Scaling studies of highly disordered spin-$\frac{1}{2}$ antiferromagnetic systems", Phys. Rev. Lett. **48**, 344–347 (1982).

[105] T. Imai, J. Wang, R. W. Smaha, W. He, J. Wen, and Y. S. Lee, "Local probe investigation of the spin dynamics in the kagome and interlayers of orthorhombic barlowite $Cu_4(OD)_6FBr$: ^{79}Br and ^{63}Cu nqr study", Phys. Rev. Mater. **8**, 014403 (2024).

[106] E. Fordham, A. Sezginer, and L. Hall, "Imaging multiexponential relaxation in the (y, Log_eT_1) plane, with application to clay filtration in rock cores", Journal of Magnetic Resonance, Series A **113**, 139–150 (1995).